Tweaking the constant in the Linear Hadwiger Theorem

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Abstract

Norin and Steiner recently proved the Linear Hadwiger Conjecture. That is, there is an absolute constant C such that every graph G satisfies $\chi(G) \leqslant C\,\mathrm{had}(G)$, where $\chi(G)$ is the chromatic number and $\mathrm{had}(G)$ is the Hadwiger number of G. They estimate that their proof gives a non-optimised constant C of order 10^{100}. This paper uses AI assistance to optimise their argument, and shows that every graph G satisfies $\chi(G) \leqslant 1{,}118{,}430\,\mathrm{had}(G)$.

Contents

1 Introduction **2**

2 From Delcourt–Postle and Norin–Steiner to our bound **4**
2.1 The reduction of Delcourt and Postle . . . 5
2.2 The proof of Norin and Steiner . . . 5
2.3 The present paper . . . 6
2.4 Constants . . . 9

3 Preliminaries **11**
3.1 Definitions . . . 11
3.2 Connectivity, linkages and minors . . . 14
3.3 Extremal results . . . 16
3.4 Results of Norin and Steiner . . . 17
3.5 Paths to a large set . . . 17
3.6 A sharpened form of Mader's theorem . . . 20

4 An explicit chromatic-separable reduction **20**

5 Colouring graphs of order $O(a \log^2 a)$ **27**
5.1 The base case . . . 28
5.2 A bootstrapping step . . . 29
5.3 The schedule . . . 30
5.4 Small values of a . . . 32

6 Rooted minors via massed pairs **32**

7 A refined form of Lin's theorem **37**

8 The chromatic-inseparable case **41**

9 Proof of Theorem 1.2 **48**

A Postscript **53**

B Computer verification **55**

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1 Introduction

Hadwiger's Conjecture [15] suggests a deep relationship between complete graph minors and graph colourings, and is widely considered one of the most important open problems in graph theory. The conjecture asserts that $\chi(G) \leqslant \mathrm{had}(G)$ for every graph G. Here, the *chromatic number* $\chi(G)$ is the minimum number of colours in a colouring of G, where a *colouring* of G assigns one colour to each vertex so that adjacent vertices receive distinct colours. And the *Hadwiger number* $\mathrm{had}(G)$ is the largest integer t such that the complete graph K_t is a minor of G, where a graph H is a *minor* of a graph G if a graph isomorphic to H can be obtained from a subgraph of G by contracting edges.

The conjecture is trivial for $\mathrm{had}(G) \leqslant 2$. It was proved for $\mathrm{had}(G) = 3$ by Hadwiger [15] and Dirac [10]. For $\mathrm{had}(G) = 4$ it is equivalent to the Four Colour Theorem for planar graphs [50, 1, 2, 43, 16]. For $\mathrm{had}(G) = 5$ it was reduced to the Four Colour Theorem by Robertson, Seymour and Thomas [44]. It is open for $\mathrm{had}(G) \geqslant 6$. See the surveys of Seymour [45] and Norin [35] for background.

Much effort has gone into upper bounds on $\chi(G)$ as a function of $t = \mathrm{had}(G)$. Table 1 summarises the history.

Date	Upper bound on $\chi(G)$ when $\mathrm{had}(G) = t$	Reference
1964	2^{t-1}	Wagner [51]
1968	$O(t \log t)$	Mader [29]
1982/84	$O(t\sqrt{\log t})$	Kostochka [22, 23], Thomason [48]
2001	$(2\alpha + o(1))\, t\sqrt{\log t}$	Thomason [49]
2019	$O(t(\log t)^{\beta})$ for every $\beta > \frac{1}{4}$	Norin, Postle and Song [37]
2021	$O(t \log\log t)$	Delcourt and Postle [8]
Sep 2026	$O(t\sqrt{\log\log t})$	Liu and Luo [27], Lin [26]
Sep 2026	$O(t \log\log\log t)$	Liu and Luo [28]
Oct 2026	$O(t)$, with constant of order 10^{100}	Norin and Steiner [38]
Oct 2026	$1{,}118{,}430\, t$	this paper

Table 1: Upper bounds on the chromatic number of graphs with Hadwiger number t. Dates refer to the first appearance of each result. Here $\alpha = 0.3190863\ldots$ is Thomason's constant [49]. The order 10^{100} is the estimate of Norin and Steiner.

The bounds up to 2001 came from degeneracy. A graph is *d-degenerate* if every non-null subgraph has a vertex of degree at most d. A d-degenerate graph is $(d+1)$-colourable, by a greedy argument. Since $\mathrm{had}(H) \leqslant \mathrm{had}(G)$ for every subgraph H of G, an upper bound on the average degree of a graph in terms of its Hadwiger number gives an upper bound on its degeneracy, and hence on its chromatic number. Mader [29] showed that, for $t \geqslant 2$, every graph with no K_t minor has average degree $O(t \log t)$. Kostochka [22, 23] and Thomason [48] independently improved this to $O(t\sqrt{\log t})$. Thomason [49] later determined the extremal function asymptotically. Thomason defined an explicit constant $\alpha = 0.3190863\ldots$ and showed that every graph G with no K_t minor has at most $(\alpha + o(1))\, t\sqrt{\log t}\, v(G)$ edges, implying G has average degree and thus chromatic number at most $(2\alpha + o(1))\, t\sqrt{\log t}$.

The bound $O(t\sqrt{\log t})$ on the average degree is tight. Bollobás, Catlin and Erdős [5] proved that the Hadwiger number of the random graph $G(n, \frac{1}{2})$ is almost surely $(1 + o(1))\, n/\sqrt{\log_2 n}$. So for each t there is a graph with no K_t minor and average degree $\Omega(t\sqrt{\log t})$. Hence no argument based only on degeneracy gives a bound on $\chi(G)$ better than $\Theta(t\sqrt{\log t})$. This limitation is called the *degeneracy barrier*. Thomason [49] showed that the constant α is best possible, with random graphs as extremal examples, and observed that the extremal graphs are essentially disjoint unions of dense pseudo-random

graphs. Myers [33] made this precise.

However, these random graphs are far from counterexamples to Hadwiger's Conjecture. By a theorem of Bollobás [4], $G(n, \frac{1}{2})$ almost surely has chromatic number $(1 + o(1))\, n/(2 \log_2 n)$, which is much smaller than its Hadwiger number. Evidence for a linear bound comes from fractional colouring. Reed and Seymour [42] proved that $\chi_f(G) \leqslant 2\,\mathrm{had}(G)$ for every graph G, where χ_f is the fractional chromatic number. So the fractional relaxation of Hadwiger's Conjecture holds up to a factor of 2. In the same paper, Reed and Seymour asked whether there is an absolute constant C such that $\chi(G) \leqslant C\,\mathrm{had}(G)$ for every graph G. This question became known as the *Linear Hadwiger Conjecture*.

The degeneracy barrier stood for more than thirty years, until Norin, Postle and Song [37] broke it. They proved that graphs with no K_t minor are $O(t(\log t)^\beta)$-colourable for every $\beta > \frac{1}{4}$. Their approach shows that a graph with large chromatic number contains a small highly connected subgraph, and that many such subgraphs can be linked together to form a K_t minor. A key tool is their density increment theorem, which finds in a graph either a denser minor or a small subgraph of nearly the same density. A density bound of Norin and Postle [36] for unbalanced bipartite graphs with no K_t minor was used later by Delcourt and Postle [8] to find small dense subgraphs.

The next breakthrough came from Delcourt and Postle [8], who proved the following reduction. If every graph with no K_t minor and $O(t \log^4 t)$ vertices is $O(t)$-colourable, then the Linear Hadwiger Conjecture holds. So only graphs of order $t(\log t)^{O(1)}$ need to be considered. For such a graph G with n vertices, the Duchet–Meyniel theorem [11] gives an independent set of at least $n/(2t)$ vertices, and greedy colouring then gives $\chi(G) \in O(t \log(n/t)) \subseteq O(t \log\log t)$. Together with a more general form of their reduction, this gives the bound $O(t \log\log t)$ of Delcourt and Postle. Lin [25] later reduced the order of the graphs in this reduction to $O(t \log^2 t)$.

After the reduction of Delcourt and Postle, the problem was to colour small graphs with no K_t minor more efficiently. Graphs with $n \in O(t)$ vertices are trivially $O(t)$-colourable. So the difficulty lies with graphs whose order is more than linear in t but at most $t(\log t)^{O(1)}$. Liu and Luo [27] and, independently, Lin [26] improved the colouring of such graphs, and obtained the bound $O(t\sqrt{\log\log t})$. Liu and Luo [28] then improved this to $O(t \log\log\log t)$.

Shortly afterwards, Norin and Steiner [38] proved the Linear Hadwiger Conjecture. They colour graphs with at most $t(\log t)^{1/2-\varepsilon}$ vertices directly, using an argument based on the fractional colouring theorem of Reed and Seymour, a quantitative Pippenger–Frankl–Rödl theorem, and linear programming (see Section 2.2). They then bootstrap up to graphs with $t \log^5 t$ vertices, and finish with the reduction of Delcourt and Postle.

Theorem 1.1 (Norin and Steiner [38])**.** *There is an absolute constant C such that every graph G satisfies $\chi(G) \leqslant C\ \mathrm{had}(G)$.*

Because of Theorem 1.1, all that remains in the study of Hadwiger's Conjecture is to determine the best possible constant. Norin and Steiner [38, Section 4] estimate that their proof gives a non-optimised constant C of order 10^{100}. They write:

> "There is a lot of room for improvement, and we see no clear barriers to reducing the value of C below 10^{10}. Proving Theorem 2 with, say, $C = 100$ would likely require substantial new ideas."

Here their Theorem 2 is Theorem 1.1. Our main result confirms the first sentence of this quotation.

Theorem 1.2. *Every graph G satisfies $\chi(G) \leqslant 1{,}118{,}430\ \mathrm{had}(G)$.*

The author does not claim that Theorem 1.2 is a significant improvement on the result of Norin and Steiner (which is a real breakthrough). The present paper simply reports the outcomes of an experiment in using AI to optimise the constants.

Statement about AI use. I started by asking Claude Opus 5.5 to optimise the constant in the proof of Norin and Steiner [38]. Claude identified several areas for optimisation, worked out all the details, and produced a first draft of the paper. GPT-6 checked the proof and suggested further optimisations. I also identified optimisations, and instructed Claude how to improve the exposition. This process went through several iterations. I have read and revised the whole paper so that it is human-readable. I understand the proof at a high level, but have not checked all the technical and numerical details by hand. For these details I rely on automated verification.

Verification. Theorem 1.2 has been formally verified in Lean 4 [32] with the Mathlib library [31], building on the Lean library `LinearHadwiger`, which formalises the proof of Norin and Steiner (see Appendix B). The final Lean statement is exactly Theorem 1.2:

```
theorem HadwigerConstant.chromaticNumber_le_mul_hadwigerNumber
    (V : Type u) [Fintype V] (G : SimpleGraph V) :
    G.chromaticNumber ≤ 1118430 * hadwigerNumber G
```

Here `G.chromaticNumber` is the chromatic number of G as defined in Mathlib, and `hadwigerNumber G` is defined from standard Mathlib notions. The verification depends only on the standard axioms of Lean. All non-trivial numerical inequalities were checked by computer. The scripts and Lean code are available as ancillary files on arXiv.

Organisation. Section 2 explains how the present proof builds on the work of Delcourt and Postle and of Norin and Steiner, and where the constant in Theorem 1.2 comes from. Section 3 gives definitions and the results we use. Sections 4, 5 and 8 contain the three main ingredients: the chromatic-separable reduction, the colouring of graphs of order $O(a \log^2 a)$ by an additive bootstrap, and the chromatic-inseparable case. Section 6 shows, using Wollan's work, that dense highly connected graphs are woven (Lemma 6.4), and Section 7 proves the refined form of Lin's theorem. Both results are used in Section 8. Section 9 combines the three ingredients in a final separation step, and proves Theorem 1.2. Appendix A discusses how the constant might be reduced further. Appendix B gives details of the computer verification.

Constant naming conventions. Each constant or coefficient that is used outside the result introducing it is typically denoted by a letter with the number of that result in subscript. For example, $C_{6.4}$ is introduced in Lemma 6.4. Constants from other papers are named after the result of this paper in which they are restated. Exceptions include Thomason's constant α of Section 1, the constants C_β, C_{sep}, $C_{B.1}$, $C_{B.8}$ and C_{ins} of Section 2, which are those of Norin and Steiner and of Delcourt and Postle, and the constants C_{DP} and C_{Lin} of Delcourt and Postle and of Lin (Sections 3.3 and 7). We use C for constants in edge-density bounds, Λ for coefficients in lower bounds on the chromatic number that force a K_a minor, c for coefficients in upper bounds on the chromatic number, τ for a bound on the order of a subgraph, and ω for a proportion of density.

2 From Delcourt–Postle and Norin–Steiner to our bound

This section explains how the proof of Theorem 1.2 builds on the work of Delcourt and Postle [8] and of Norin and Steiner [38]. Section 2.1 describes the reduction of Delcourt and

Postle, Section 2.2 describes the proof of Norin and Steiner, Section 2.3 summarises how the present paper improves it, and Section 2.4 explains where the constant in Theorem 1.2 comes from.

2.1 The reduction of Delcourt and Postle

Delcourt and Postle [8] reduced the Linear Hadwiger Conjecture to colouring small graphs. Their ideas underlie both the proof of Norin and Steiner and the present paper. The following definition is a key ingredient. Let $s \geqslant 0$. A graph G is *s-chromatic-separable* if it has two vertex-disjoint subgraphs, each with chromatic number at least $\chi(G) - s$. Otherwise G is *s-chromatic-inseparable*. (Section 3 gives the precise definition.) To find a K_t minor in a graph of large chromatic number, Delcourt and Postle use three ingredients.

Small dense pieces. For every integer $k \geqslant t$, every graph with no K_t minor and with at least $C_{\mathrm{DP}}k$ edges per vertex has a k-connected subgraph with $O(t \log^3 t)$ vertices [8, Theorem 2.3]. This is the small-dense-subgraph theorem of Delcourt and Postle, stated in Section 3.3. Lin [25] later reduced the order to $O(t \log t)$.

The chromatic-inseparable case. Let G be a graph. Let Γ be the maximum chromatic number of a subgraph of G that has no K_t minor and $O(t \log^4 t)$ vertices. There is an absolute constant C_{ins} such that, if G is $C_{\mathrm{ins}}(t + \Gamma)$-chromatic-inseparable and $\chi(G) \geqslant 2C_{\mathrm{ins}}(t+\Gamma)$, then G has a K_t minor [8, Lemma 2.5]. (In their notation, $\Gamma = tg$.) The minor is built in stages. At each stage, small dense pieces supply new branch sets. Inseparability guarantees that, after the vertices already used are removed, a highly connected subgraph of large chromatic number remains for the next stage.

The chromatic-separable case. Here a K_a minor, for $a \leqslant t$, is built by a ternary recursion on a. A highly connected subgraph of large chromatic number is separated twice, giving three vertex-disjoint subgraphs of large chromatic number. A rooted $K_{2a/3}$ minor is built recursively in each of them. Each branch set of the K_a minor is then formed from branch sets of two of the three minors, so that any two of the new branch sets are adjacent. Delcourt and Postle combine the two cases in a single induction [8, Theorem 7.1], which gives their reduction [8, Theorem 1.6].

The present paper makes both cases explicit, and recombines them so that the constants add rather than multiply.

2.2 The proof of Norin and Steiner

We now describe the structure of the proof of Norin and Steiner [38]. Let $\mathrm{LH}(\beta)$ denote the following statement. There is a constant C_β such that, for all sufficiently large real t, every graph G with $\mathrm{had}(G) < t$ and $v(G) \leqslant t(\log t)^\beta$ satisfies $\chi(G) \leqslant C_\beta t$.

(I) *Small graphs.* Norin and Steiner [38, Theorem 4] prove that $\mathrm{LH}(\frac{1}{2} - \varepsilon)$ holds with constant $4 + \varepsilon$ for every $\varepsilon \in (0, \frac{1}{2})$. The proof uses three tools: the fractional colouring theorem of Reed and Seymour [42], a quantitative form of the Pippenger–Frankl–Rödl theorem on almost perfect matchings in hypergraphs, and a linear programming argument. Norin and Steiner [38, Theorem 9 and Appendix A] prove the quantitative form. An earlier draft [39] of their paper states the explicit bound $\chi(G) \leqslant 4\,\mathrm{had}(G) + \varepsilon v(G) + (100/\varepsilon)^{2000/\varepsilon^2}$, from which we derive our base case (Theorem 5.1 and Lemma 5.3).

(II) *Bootstrapping.* Norin and Steiner [38, Theorem 5] prove that $\mathrm{LH}(\beta)$ implies $\mathrm{LH}(\frac{4}{3}\beta)$, with new constant $C_{\frac{4}{3}\beta} = 4C_\beta + 2C_{\mathrm{sep}}(1 + C_\beta)$. Here C_{sep} is the constant in the

chromatic-separable reduction of Norin and Steiner [38, Theorem 15], which adapts the chromatic-separable case of Delcourt and Postle described above.

(III) *Large graphs.* By the reduction of Delcourt and Postle described above [8, Theorem 1.6], LH(5) implies Theorem 1.1. This step multiplies the constant C_5 by a further constant factor.

Norin and Steiner start from exponent $\frac{1}{3}$, and apply Part (II) ten times to reach exponent 5. In the notation of Norin and Steiner, $C_{\mathrm{sep}} = 39 \cdot 35 \cdot C_{B.1}^2 \cdot C_{B.8}$ in [38, Section B.3 and the proof of Lemma B.12]. The constant $C_{B.1}$ is the density constant of the small-dense-subgraph theorem of Delcourt and Postle (Section 3.3, where it is $C_{\mathrm{DP}} = 3{,}072{,}000$). Norin and Steiner need $C_{B.1} \geqslant 480$, and they need [38, Theorem B.1], which is this theorem, to hold with $C_{B.1}$ in place of C_{DP}. This is known for $C_{B.1} = C_{\mathrm{DP}}$, but not for $C_{B.1} = 480$. Also $C_{B.8}$ is the linkage constant of Bollobás and Thomason [6] in the form of [38, Theorem B.8], and Norin and Steiner only require $C_{B.8} \geqslant 1$. By the theorem of Thomas and Wollan [47, Corollary 1.3], we may take $C_{B.8} = 10$. Even with $C_{B.1} = 480$ and $C_{B.8} = 10$, we get $C_{\mathrm{sep}} = 39 \cdot 35 \cdot 480^2 \cdot 10 \approx 3.1 \cdot 10^9$. Then ten iterations of $C \mapsto 4C + 2C_{\mathrm{sep}}(1 + C)$ starting from 4 give a constant of at least $4.8 \cdot 10^{98}$. Here 4 is less than their starting constant $4 + \frac{1}{6}$, which is the constant of the small-graph bound of Norin and Steiner [38, Theorem 4] (Section 3.4) with $\varepsilon = \frac{1}{6}$. With $C_{B.1} = C_{\mathrm{DP}}$ in place of 480, the same computation gives a constant of at least $6.4 \cdot 10^{174}$. So the constant of Norin and Steiner is large for three reasons: C_{sep} is large, the constant is multiplied by more than $2C_{\mathrm{sep}}$ at each of the ten iterations of Part (II), and Part (III) multiplies the result by a further constant.

2.3 The present paper

We now summarise how the present paper improves the proof of Norin and Steiner [38]. Figure 1 shows how the main results depend on each other.

1. An explicit chromatic-separable reduction (Section 4). We reprove, with explicit constants, the chromatic-separable reduction of Norin and Steiner [38, Theorem 15], which is used in each step of Part (II). We use a theorem of Nguyen [34] on highly connected subgraphs with large chromatic number (Theorem 3.2) in place of the theorem of Girão and Narayanan [13] used by Norin and Steiner. We also use the linkage theorem of Thomas and Wollan [47]. The reduction must join some prescribed vertices, called *roots*, to the recursively built minors. It must also join other prescribed vertices, called *terminals*, to each other in pairs. The reduction does this through a *hub*, which is an induced subgraph with two properties. First, for a prescribed integer m, the hub is m-linked, so that any m pairs formed by $2m$ distinct vertices of the hub can be joined by disjoint paths inside the hub. Second, the hub has chromatic number at most $\lceil 25m/2 \rceil + 1$, so that deleting the hub costs few colours. We obtain the hub (Lemma 4.1) from a smallest induced subgraph whose chromatic number is at least a prescribed value. This is a vertex-critical induced subgraph, from which a sharpened form of Mader's theorem (Lemma 3.17) extracts a dense highly connected subgraph. The density form of the Thomas–Wollan theorem then makes the hub linked (see the remark after Lemma 4.1 for the saving over the connectivity form). Obtaining the hub in this way avoids the small-dense-subgraph theorem of Delcourt and Postle (Section 3.3). After the roots and terminals are joined through the hub, the proof of Lemma 4.2 reduces to showing that a $21a$-connected subgraph G_3 of large chromatic number is $(a, 0)$-woven (see Section 3.1). If G_3 has a K_{2a} minor, then G_3 is $(a, 0)$-woven by the rooting lemma of Kawarabayashi (Lemma 3.9). This is Step 7a of the proof of Lemma 4.2, and the idea is due to Delcourt and Postle [8, Section 7.1]. So in Step 7b we

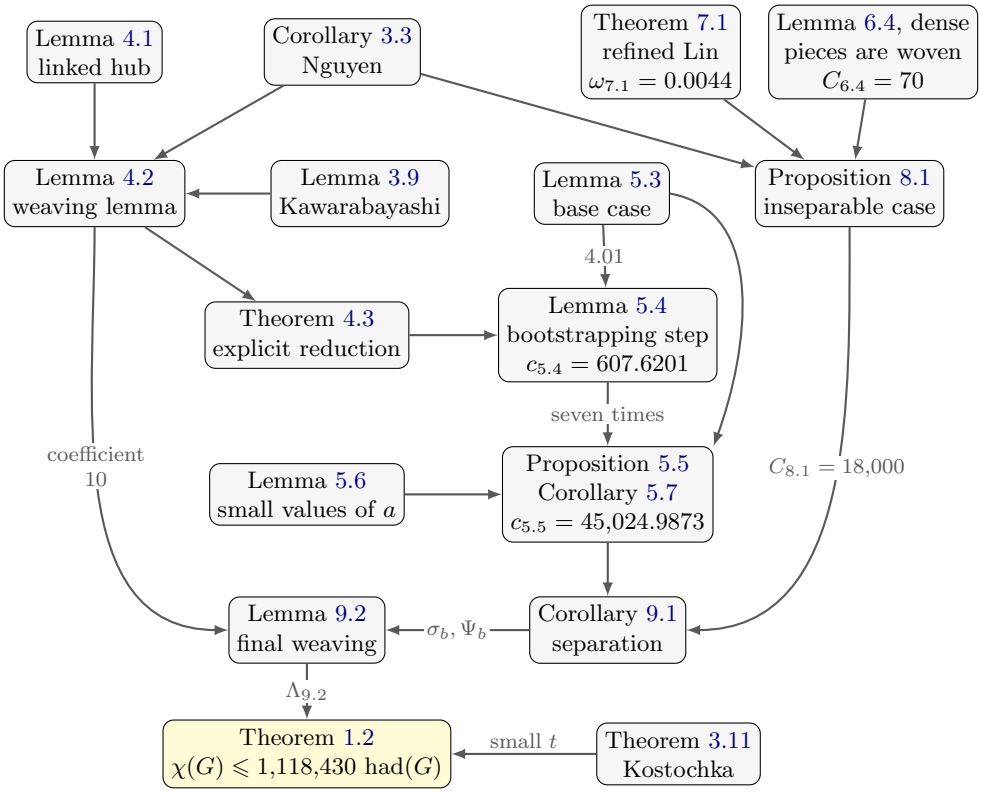


Figure 1: The structure of the proof of Theorem 1.2. An arrow means that the second result uses the first, or uses a constant from it. Most labels are the main constants passed along the arrows (see Table 2). Here 4.01 is the constant of Lemma 5.3, and 10 is the coefficient of Φ in Λ_U and of $78 + 2c_{5.5}$ in $\Lambda_{9.2}$.

may assume that G_3 has no K_{2a} minor. As a result, the chromatic separations in Step 7b are made in graphs with no K_{2a} minor, rather than no K_{14a} minor as in [8, 38]. We also allow an uneven distribution of the branch sets among the three subgraphs used in the recursion. The result is Theorem 4.3, with the cost $c_{4.3}(\Phi)$ given in Table 2, where Φ is the parameter in the hypothesis $(\mathcal{H}_{\Phi,a_0})$ of Section 3. This cost is 42 plus the minimum of two terms. The second comes from the uneven distribution, and is the smaller one when $\Phi > 190$.

2. An additive bootstrap (Section 5). Each bootstrapping step of Norin and Steiner splits the graph into two parts, G_1 and G_2. The vertex set of G_1 is partitioned into the vertex sets of large connected induced bipartite subgraphs. The part G_2 has no such subgraph. The chromatic-separable reduction of Norin and Steiner [38, Theorem 15], with constant C_{sep}, is applied to G_2. The chromatic-inseparable subgraphs of G_2 are handled by a Gyárfás path argument, which requires a colouring bound for small graphs, called *Gyárfás graphs* (see Lemma 3.15). Norin and Steiner colour the Gyárfás graphs with the bound obtained at the previous bootstrapping step. We colour them with a fixed earlier bound instead. At every step except the final one, this is the base case (Lemma 5.3, an explicit form of Part (I)). At the final step, it is the bound given by an extra bootstrapping step, called the *side step*, which is used only for this purpose. As explained after Lemma 5.4, colouring the Gyárfás graphs with a fixed earlier bound makes each step map a constant c to $2c$ plus a constant independent of c. Five main steps, the side step and a final step suffice (Proposition 5.5). This gives $c_{5.5} = 45{,}024.9873$, compared with about 77,681.0 for seven steps that all use the base case (see the remark

after Proposition 5.5).

3. The chromatic-inseparable case (Section 8). Let H be a graph with no K_t minor. Let Γ be an upper bound on the chromatic number of the small subgraphs of H. For the lemma of Delcourt and Postle, these are the subgraphs with $O(t \log^4 t)$ vertices. For Proposition 8.1, they are the subgraphs with at most $N_t = O(t \log^2 t)$ vertices. By the lemma of Delcourt and Postle in Section 2.1, if H is $C_{\text{ins}}(t+\Gamma)$-chromatic-inseparable, then $\chi(H) < 2C_{\text{ins}}(t+\Gamma)$. So the large constant C_{ins} multiplies Γ, both in the separation cost $C_{\text{ins}}(t+\Gamma)$ and in the chromatic threshold $2C_{\text{ins}}(t+\Gamma)$ (see Section 3). We rework the proof of Delcourt and Postle as follows, to keep the large constant C_{ins} out of the separation cost. Proposition 8.1 shows that, if $\log t \geqslant 1.6 \cdot 10^6$, then every graph with no K_t minor and chromatic number at least $(32 + 2C_{8.1})t + 2\Gamma + 2$ is $(78t + 2\Gamma)$-chromatic-separable. Here $C_{8.1} = 18{,}000$ is the constant in the density threshold $d = C_{8.1}t$ of the chromatic-inseparable case. Thus Γ has coefficient only 2, and the large constant $C_{8.1}$ enters only the chromatic threshold, not the separation cost. Section 9 explains how this affects the final constant.

4. Rooted minors via massed pairs (Section 6). In the chromatic-inseparable case, each piece must be woven, that is, for every choice of roots and terminal pairs it must have a rooted complete minor and a linkage that meet only in the roots that are also terminals (Section 3). Delcourt and Postle obtain this from Wollan's extremal theorem for rooted minors [52, Theorem 1.1], applied to the graph $K_s + \ell K_2$, where s is the number of roots and $\ell \approx 10t$ is the number of terminal pairs. Their application of Wollan's theorem requires density about $5 \cdot 10^5\, t$. We use instead three structural theorems from Wollan's proof [52, Theorems 2.1–2.3], concerning a minimal counterexample (Theorem 6.1), together with the strategy of Wollan's proof of [52, Theorem 5.1] (see the outline after Lemma 6.3). Lemma 6.4 then requires only density $70t$. Delcourt and Postle delete the roots and apply Wollan's theorem to the rest of the piece. We instead apply Lemma 6.4 to the whole piece, roots included. This is possible for two reasons. First, in the proof of Lemma 6.2, each root that is not a terminal is a root of the rooted model, and each root that is a terminal is replaced by a neighbour, as in [8, Lemma 5.13]. Second, keeping the roots costs only a few edges per root in the density condition (M1) of Section 6. Lemma 6.4 needs pieces that are $9t$-connected and have density $C_{6.4}t$, and these two requirements give the condition $\omega_{7.1}C_{8.1} \geqslant C_{6.4} + 9$ on $C_{8.1}$ in Proposition 8.1 (see Step 1 of the proof of Proposition 8.1).

5. A refined form of Lin's theorem (Section 7). The chromatic-inseparable case uses a theorem that finds small highly connected subgraphs in dense graphs with no K_t minor. The theorem of Delcourt and Postle in Section 2.1 has density constant $C_{\text{DP}} \approx 3.07 \cdot 10^6$ (Section 3.3). Lin's improvement of it has density constant $C_{\text{Lin}} \approx 1.31 \cdot 10^9$ (Section 7). We rerun Lin's argument with different parameters, and obtain pieces that are dense, not only highly connected (Theorem 7.1). For $t \geqslant 10^{13}$, every graph with no K_t minor and density $d \geqslant 1.8 \cdot 10^4\, t$ contains a $(9t+1)$-connected piece with at most $\tau_{7.1}\, t \log t$ vertices and density more than $\omega_{7.1}d - 9t$, where $\omega_{7.1} = 0.0044$ and $\tau_{7.1} = 4.56 \cdot 10^{18}$. Section 7 explains the sources of the improvement. Because the pieces have $O(t \log t)$ vertices, the core built in the proof of Proposition 8.1 has $O(t \log^2 t)$ vertices, and so do the small graphs that must be coloured. The exponent 2 in this order bound determines the number of bootstrapping steps (see the remark after Proposition 5.5).

6. A more economical construction (Sections 6 and 8). In the chromatic-inseparable case, each piece must be $(2x, \ell)$-woven, where x is the number of branch

sets added at each stage and ℓ bounds the number of linkage pairs. Delcourt and Postle take $\ell = 10t$ and $x = \lceil t/\sqrt{\log t}\rceil$. We take $x = \lceil t/(M\sqrt{\log t})\rceil$ with $M = 10^5$. The reasons for this choice, and its cost, are explained before Lemma 6.3. At stage n of the construction the number of pairs is $(4n-3)x$. Since there are at most $M\sqrt{\log t}+1$ stages, $(4n-3)x \leqslant \lfloor 4.004t \rfloor$ when $\log t \geqslant 1.6 \cdot 10^6$. So we take $\ell = \lfloor 4.004t \rfloor$. Finally, as in the proof of Delcourt and Postle, paths from the pieces are joined inside one designated piece, which we call the *junction*, by knitting (Lemma 3.6). Lemma 3.6 uses the density form of the Thomas–Wollan linkage theorem. So the connectivity required by the knitting step is at most $7.001t$ (Step 4 of the proof of Proposition 8.1), instead of about $31t$ with the connectivity form of the Thomas–Wollan theorem.

7. Explicit size thresholds (Sections 5 and 9). Our asymptotic statements hold above explicit size thresholds. The thresholds T_{base} and T_{boot} are lower bounds on $\log\log a$ in Section 5, and T_{fin} is a lower bound on $\log\log t$. The other thresholds are lower bounds on $\log t$ or on t, and are implied by $\log\log t \geqslant T_{\text{fin}}$. Section 5.4 and Case 1 of the proof of Theorem 1.2 treat the values below T_{boot} and T_{fin}, respectively. For t below a size threshold t_0, the estimate of Norin and Steiner [38, Section 4] uses the Kostochka–Thomason degeneracy bound, which presumably contributes a factor of order $\sqrt{\log t_0}$ to the constant of Norin and Steiner. In the proof of Theorem 1.2, where G has no K_t minor, we colour with the bound of Theorem 3.11 only when $\log\log t < 18.4$. Then this bound gives $\chi(G) < 600{,}000\,t + 1$.

Most of the ingredients above re-optimise arguments of Delcourt and Postle [8], Norin and Steiner [38], Lin [25] and Wollan [52]. The new ingredients are as follows:

- The hub is built in a new way (Lemma 4.1 and item 1 above).
- Step 7a of the proof of Lemma 4.2 uses an observation of Delcourt and Postle inside the recursion, so that Step 7b separates graphs with no K_{2a} minor (item 1 above).
- The branch sets in the proof of Lemma 4.2 are distributed unevenly among the three subgraphs (item 1 above).
- The bootstrap is additive, with a new schedule (Section 5).
- Lemma 6.3 adapts Wollan's massed-pair argument to rooted models of $K_s + \ell K_2$ with s small, in which only the clique part needs a minor argument.
- In Step 5 of the proof of Proposition 8.1, the first stage of the chromatic-inseparable case is treated separately. At later stages, the construction, which follows Delcourt and Postle, makes the new branch sets pairwise adjacent through a rooted clique model in a piece. This piece exists only from the second stage onwards.

2.4 Constants

The constant in Theorem 1.2 is

$$c_{1.2} := 1829.6 + 2C_{8.1} + 24\,c_{5.5} < 1{,}118{,}430.$$

Here $c_{5.5} = 45{,}024.9873$ is a constant such that every graph with no K_a minor and at most $N_a = 24\tau_{7.1}M^2\,a\log^2 a$ vertices is $c_{5.5}a$-colourable (Proposition 5.5 and Corollary 5.7), and $C_{8.1} = 18{,}000$ is the density constant of item 3 above (Proposition 8.1). The term 1829.6 collects the lower-order terms (see the proof of Theorem 1.2). The term $24c_{5.5}$ accounts for about 97% of $c_{1.2}$.

The terms of $c_{1.2}$ come from three changes to the argument of Norin and Steiner, which remove the three reasons for their large constant listed after Part (III) in Section 2.2. First, the reduction to chromatic-inseparable subgraphs costs $c_{4.3}(\Phi)$ in place of $C_{\text{sep}}(1+\Phi)$, up to lower-order terms (item 1 above). Second, the Gyárfás graphs are coloured with a fixed

Constant	Value	Role	Where
Φ	4.01 or 615.6401 ($78+2c_{5.5}$ plays the role of Φ in $\Lambda_{9.2}$)	in $(\mathcal{H}_{\Phi,a_0})$, s-chromatic-inseparable and had $< b$ give $s \leqslant \Phi b$	Sec. 3
Λ_{E}, Λ_{U}	$(1+\eta)(517.5+12\Phi)$, $(1+\eta)(897.5+10\Phi)$	coefficient of a in the weaving lemma, in the even and uneven schemes	Lem. 4.2
η	$5/(a_0-4)$, $a_0 = t/\log t$	slack in Λ_{E}, Λ_{U} and $\Lambda_{9.2}$	Lem. 4.2, 5.4, 9.2
$c_{4.3}(\Phi)$	$42+\min\{517.5+12\Phi,$ $897.5+10\Phi\}$	term added by a bootstrapping step whose Gyárfás graphs are Φa-colourable	Thm. 4.3
ξ	10^{-4}	error term of a bootstrapping step (which requires $0 \leqslant \Phi \leqslant 10^4$)	Lem. 5.4
$c_{5.4}$	607.6201	each main bootstrapping step maps c to $2c+c_{5.4}$	Lem. 5.4
$c_{5.5}$	45,024.9873	colouring graphs with at most N_a vertices	Prop. 5.5
N_a	$24\tau_{7.1}M^2\, a \log^2 a$	order bound for the small graphs	Prop. 5.5
T_{base}, T_{boot}	1160, 1161	size thresholds on $\log\log a$ in Section 5	Sec. 5.1, Prop. 5.5
M	10^5	each stage adds $x = \lceil t/(M\sqrt{\log t})\rceil$ branch sets	Lem. 6.4
$C_{6.4}$	70	density that makes a $9t$-connected graph $(2x, \lfloor 4.004t\rfloor)$-woven	Lem. 6.4
$\omega_{7.1}$	0.0044	proportion of density kept by the refined Lin theorem	Thm. 7.1
$\tau_{7.1}$	$4.56\cdot 10^{18}$	pieces have at most $\tau_{7.1}\, t\log t$ vertices	Thm. 7.1
$C_{8.1}$	18,000	density threshold $d = C_{8.1}t$, with $\omega_{7.1}C_{8.1} \geqslant C_{6.4}+9$	Prop. 8.1
k, d	$12t$, $C_{8.1}t$	connectivity and density threshold in the chromatic-inseparable case	Prop. 8.1
σ_b, Ψ_b	$(78+2c_{5.5})b$, $(32+2C_{8.1}+2c_{5.5})b+2$	separation cost and chromatic threshold for graphs with no K_b minor	Cor. 9.1
T_{fin}	18.4	size threshold on $\log\log t$ in Section 9	Lem. 9.2
$\Lambda_{9.2}$	$(1+\eta)(897.5+10(78+$ $2c_{5.5})) \approx 9.022\cdot 10^5$	coefficient of a in the final weaving lemma	Lem. 9.2
$\Lambda'_{9.2}$	$110+2C_{8.1}+4c_{5.5}+\frac{2}{t} \approx$ $2.162\cdot 10^5$	coefficient of t in the final weaving lemma, which pays for two successive separations	Lem. 9.2
$c_{1.2}$	$<$ 1,118,430	$1829.6+2C_{8.1}+24c_{5.5}$	Sec. 2.4

Table 2: Constants, parameters and thresholds used in more than one place. The parameters of the bootstrap schedule are listed in Table 4.

earlier bound, and the additive bootstrap of item 2 above gives $c_{5.5}$. Third, Part (III) is replaced by a final separation step (Corollary 9.1 and Lemma 9.2). This step is one more round of the chromatic-separable reduction. Each of its chromatic separations is supplied by the chromatic-inseparable case (Proposition 8.1 and items 3–6 above), with $c_{5.5}$ as the bound for the small graphs. In a graph with no K_b minor, such a separation costs $(78+2c_{5.5})b$ (Corollary 9.1). In Lemma 9.2, which builds a K_a minor, these separations are made in graphs with no K_{2a} minor. So $78+2c_{5.5}$ plays the role of Φ in Λ_{U}. The coefficient 10 of Φ in Λ_{U} therefore gives the term $20c_{5.5}$ of $c_{1.2}$. Two successive separations account for a further $4c_{5.5}$. Of this $4c_{5.5}$, the chromatic threshold contributes $2c_{5.5}$. The cost of the first separation contributes the other $2c_{5.5}$, and this extra $2c_{5.5}$ ensures that each part of the first separation can be separated again. The chromatic threshold also contributes $2C_{8.1}$ (see Section 9). These and the other main constants are listed in Table 2.

3 Preliminaries

3.1 Definitions

All graphs are finite and simple. For an integer $n \geqslant 0$, let $[n] := \{1, \dots, n\}$, so $[0] = \emptyset$. All logarithms are natural. For $k \geqslant 1$, we write $\log^k t := (\log t)^k$. For a graph G, let $v(G)$ and $e(G)$ denote the numbers of vertices and edges of G. The *order* of G is $v(G)$. The *density* of a non-null graph G is $e(G)/v(G)$, which is half its average degree. For $v \in V(G)$, let $N_G(v)$ be the set of neighbours of v in G, and let $\deg_G(v) := |N_G(v)|$. Let $\delta(G)$ and $\Delta(G)$ denote the minimum and maximum degree. For an integer $k \geqslant 0$, a graph is *k-connected* if it has more than k vertices and remains connected after deleting any set of fewer than k vertices. For a non-null graph G, the *connectivity* $\kappa(G)$ is the largest k such that G is k-connected. So G is k-connected if and only if $\kappa(G) \geqslant k$, and $\kappa(G) \leqslant \delta(G)$. A subgraph H of G is *spanning* if $V(H) = V(G)$. For $X \subseteq V(G)$, let $G[X]$ denote the subgraph induced by X. Let $G - X := G[V(G) \setminus X]$. For disjoint $A, B \subseteq V(G)$, let $e_G(A, B)$ be the number of edges of G with one end in A and the other in B, and let $G(A, B)$ be the bipartite subgraph of G with vertex set $A \cup B$ whose edges are these edges. For graphs G_1, G_2, the graph $G_1 \cup G_2$ has vertex set $V(G_1) \cup V(G_2)$ and edge set $E(G_1) \cup E(G_2)$. The *disjoint union* $G_1 + G_2$ is the union of vertex-disjoint copies of G_1 and G_2. For an integer $\ell \geqslant 0$, let ℓG denote the disjoint union of ℓ copies of G. The *null graph* has no vertices.

Contraction. Let uv be an edge of G. *Contracting* uv means deleting u and v and adding a new vertex adjacent to every vertex of $(N_G(u) \cup N_G(v)) \setminus \{u, v\}$. The resulting graph is simple. Now let $A \subseteq V(G)$ be non-empty with $G[A]$ connected, and let T be a spanning tree of $G[A]$. The graph G/A is obtained from G by contracting the edges of T one after another. The vertices of A are replaced by a single vertex x_A, which is adjacent to exactly those vertices of $V(G) \setminus A$ that have a neighbour in A. In particular, G/A does not depend on the choice of T or on the order of the contractions. If $|A| = 1$, then $G/A = G$ and x_A is the vertex of A. We say that G/A is obtained by *contracting* A. Contracting a connected subgraph H means contracting $V(H)$. Finally, let $A_1, \dots, A_m \subseteq V(G)$ be pairwise disjoint and non-empty, with each $G[A_i]$ connected. Contracting $A_1, \dots, A_m$ means contracting A_1, then A_2, and so on, up to A_m. This is well defined, since each A_i still induces a connected subgraph after $A_1, \dots, A_{i-1}$ are contracted. The result is obtained from $G - (A_1 \cup \dots \cup A_m)$ by adding vertices $x_{A_1}, \dots, x_{A_m}$, where x_{A_i} is adjacent to every remaining vertex with a neighbour in A_i, and x_{A_i} is adjacent to x_{A_j} if and only if G has an edge between A_i and A_j. So the result does not depend on the order.

Colourings. A colouring of G is a map $c : V(G) \to \mathbb{N}$ with $c(u) \neq c(v)$ for every edge uv, and $\chi(G)$ is the minimum number of colours used by a colouring. An *independent set* is a set of pairwise non-adjacent vertices. The *independence number* $\alpha(G)$ is the maximum size of an independent set. The *fractional chromatic number* $\chi_f(G)$ is the minimum of $\sum_I x_I$ over all non-negative weights x_I on the independent sets I of G such that $\sum_{I \ni v} x_I \geqslant 1$ for every vertex v.

Minors and models. We set $\mathrm{had}(G) := 0$ if G is null. A *model* of a graph H in G is a family $\{B_v : v \in V(H)\}$ of pairwise disjoint non-empty subsets of $V(G)$ such that each $G[B_v]$ is connected and there is an edge of G between B_u and B_v for every edge uv of H. The sets B_v are called *branch sets*. A *K_s model* is a model of K_s, that is, a family of s pairwise adjacent branch sets. A graph G has an H minor if and only if it contains a model of H. Let $X = \{x_v : v \in V(H)\}$ be a set of $|V(H)|$ distinct vertices of G. The model is *rooted at X* (according to the bijection $x_v \mapsto v$) if $x_v \in B_v$ for every $v \in V(H)$. For a K_a model and a set R of a vertices, *rooted at R* means rooted at R according to some bijection between R and $V(K_a)$. Equivalently, each branch set contains exactly one vertex of R. If $\mathcal{A}$ is a model, then $V(\mathcal{A})$ denotes the union of its branch sets. Similarly, for

a family $\mathcal{P}$ of paths, $V(\mathcal{P}) := \bigcup_{P\in\mathcal{P}} V(P)$. A subgraph or family is *disjoint from* a set X if its vertex set is disjoint from X, and two subgraphs are *disjoint* if they are vertex-disjoint.

Separations. A *separation* of G is a pair (A, B) of subsets of $V(G)$ with $A \cup B = V(G)$ and no edge between $A \setminus B$ and $B \setminus A$. Its *order* is $|A \cap B|$. It is *trivial* if $A \subseteq B$ or $B \subseteq A$.

Paths and linkages. A *path* $P = v_0v_1 \dots v_m$ in G is a sequence of distinct vertices with $v_{i-1}v_i \in E(G)$ for each $i \in [m]$. We identify P with the subgraph it forms. Its *ends* are its *first* vertex v_0 and its *last* vertex v_m. The other vertices are *internal*. For $X \subseteq V(G)$, a family $\mathcal{P}$ of paths in G is *pairwise disjoint outside* X if $V(P) \cap V(Q) \subseteq X$ for all distinct $P, Q \in \mathcal{P}$. A path P is *induced* in G if $G[V(P)] = P$. Every subpath of an induced path is induced. If P is an induced path of G, then $\chi(G[V(P)]) \leqslant 2$. Given a path P in G with ends u and v, a shortest u–v path in $G[V(P)]$ is an induced path of G with the same ends and with vertex set contained in $V(P)$. For $A, B \subseteq V(G)$, an *A–B path* is a path whose first vertex is its only vertex in A and whose last vertex is its only vertex in B. Let $s_1, \dots, s_\ell, t_1, \dots, t_\ell$ be vertices such that $s_i \neq s_j$ and $t_i \neq t_j$ for $i \neq j$, and such that if $s_i = t_j$ then $i = j$. An *$\{(s_i, t_i)\}_{i\in[\ell]}$-linkage* is a family of pairwise vertex-disjoint paths $P_1, \dots, P_\ell$ such that P_i has ends s_i and t_i. A graph is *ℓ-linked* [17, 24] if it has an $\{(s_i, t_i)\}_{i\in[\ell]}$-linkage for every choice of 2ℓ distinct vertices $s_1, \dots, s_\ell, t_1, \dots, t_\ell$. Unlike Thomas and Wollan [47], we do not require an ℓ-linked graph to have at least 2ℓ vertices. This makes no difference below, since every graph that we show to be ℓ-linked has more than 2ℓ vertices. Every linkage used with Theorem 3.5 below has 2ℓ distinct ends.

Woven graphs. The following notion is due to Delcourt and Postle [8, Definition 5.12], in a form that allows any number of terminal pairs up to b. Let $b' \in \{0, \dots, b\}$, and let $R = \{r_1, \dots, r_a\}$, $S = \{s_1, \dots, s_{b'}\}$ and $T = \{t_1, \dots, t_{b'}\}$ be sets of vertices of a graph G, where $r_1, \dots, r_a$ are distinct and $s_1, \dots, s_{b'}, t_1, \dots, t_{b'}$ satisfy the conditions in the definition of a linkage. The sets R, S and T need not be disjoint. The graph G is *(a, b)-woven* if, for all such b', R, S and T, there is a K_a model $\mathcal{M}$ rooted at R and an $\{(s_i, t_i)\}_{i\in[b']}$-linkage $\mathcal{P}$ with $V(\mathcal{M}) \cap V(\mathcal{P}) = R \cap (S \cup T)$. As illustrated in Figure 2, this definition means that the only intersections between the model and the linkage are unavoidable. Since b' ranges over $\{0, \dots, b\}$, every (a, b)-woven graph is (a, c)-woven for every integer c with $0 \leqslant c \leqslant b$. Lemma 3.1 shows that the number of roots can also be decreased. Taking $b' = 0$ and any set R of a vertices shows that every (a, b)-woven graph with at least a vertices has a K_a minor.

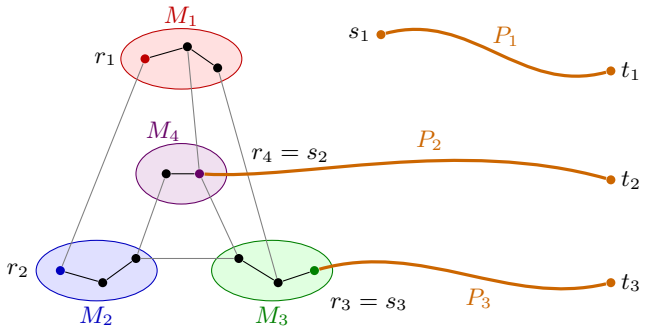


Figure 2: The definition of (a, b)-woven with $a = 4$ and $b = 3$. The shaded branch sets $M_1, \dots, M_4$ form a K_4 model rooted at $R = \{r_1, \dots, r_4\}$, and the paths P_1, P_2, P_3 form an $\{(s_i, t_i)\}_{i\in[3]}$-linkage. Since $s_2 = r_4$ and $s_3 = r_3$, the model and the linkage meet exactly in $R \cap (S \cup T) = \{r_3, r_4\}$.

Lemma 3.1. *Let $a \geqslant a' \geqslant 0$ and $b \geqslant 0$ be integers. If G is (a, b)-woven and $v(G) \geqslant a+2b$, then G is (a', b)-woven.*

Proof. Let $b' \in \{0, \dots, b\}$, and let $R' = \{r_1, \dots, r_{a'}\}$, $S = \{s_1, \dots, s_{b'}\}$ and $T = \{t_1, \dots, t_{b'}\}$ be as in the definition of (a', b)-woven. Since $|R' \cup S \cup T| \leqslant a' + 2b'$ and $v(G) \geqslant a + 2b$, there are $a - a'$ distinct vertices of G outside $R' \cup S \cup T$. Call these *padding* vertices $r_{a'+1}, \dots, r_a$, and let $R := \{r_1, \dots, r_a\}$. Since G is (a, b)-woven, there is a K_a model $\mathcal{M} = \{M_1, \dots, M_a\}$ with $r_i \in M_i$ and an $\{(s_i, t_i)\}_{i \in [b']}$-linkage $\mathcal{P}$ with $V(\mathcal{M}) \cap V(\mathcal{P}) = R \cap (S \cup T)$. Let $\mathcal{M}' := \{M_1, \dots, M_{a'}\}$, which is a $K_{a'}$ model rooted at R'. The padding vertices lie outside $S \cup T$, so $R \cap (S \cup T) = R' \cap (S \cup T)$, and hence $V(\mathcal{M}') \cap V(\mathcal{P}) \subseteq R' \cap (S \cup T)$. Conversely, each $r_i \in R' \cap (S \cup T)$ lies in $M_i \subseteq V(\mathcal{M}')$ and is an end of a path of $\mathcal{P}$. Hence $V(\mathcal{M}') \cap V(\mathcal{P}) = R' \cap (S \cup T)$. □

Knit graphs. The following notion is due to Bollobás and Thomason [6]. We use the form of Delcourt and Postle [8, Definition 5.7], without the requirement $\ell \leqslant k \leqslant v(G)$. A graph G is *(k, ℓ)-knit* if, for every set S of k vertices and every partition of S into $p \geqslant \ell$ non-empty parts $S_1, \dots, S_p$, there are pairwise disjoint sets $R_1, \dots, R_p$ with $S_i \subseteq R_i$ and each $G[R_i]$ connected.

Chromatic separability. The following notion is due to Delcourt and Postle [8, Definition 2.4]. Let s be real. A graph G is *s-chromatic-separable* if it has vertex-disjoint subgraphs H_1, H_2, possibly null, with $\chi(H_i) \geqslant \chi(G) - s$ for $i = 1, 2$. Otherwise G is *s-chromatic-inseparable*. Delcourt and Postle take s to be an integer, and do not allow null subgraphs. We allow real s and null subgraphs, which changes nothing when $\chi(G) > s$. Taking $H_1 = G$ and H_2 null shows that every s-chromatic-inseparable graph satisfies $\chi(G) > s$. If G is s-chromatic-inseparable and $s' \leqslant s$, then G is s'-chromatic-inseparable. If H_1, H_2 are as above, then $G[V(H_1)]$ and $G[V(H_2)]$ are vertex-disjoint induced subgraphs of G with $\chi(G[V(H_i)]) \geqslant \chi(H_i) \geqslant \chi(G) - s$. So G is s-chromatic-separable if and only if it has two vertex-disjoint induced subgraphs, possibly null, each with chromatic number at least $\chi(G) - s$.

Chromatic threshold and separation cost. Several of our results state that, for a class of graphs and reals $\Psi, \sigma \geqslant 0$, every graph H in the class with $\chi(H) \geqslant \Psi$ is σ-chromatic-separable. Here Ψ is the *chromatic threshold* and σ is the *separation cost*. The separation cost bounds the loss in chromatic number in passing from H to each of the two vertex-disjoint subgraphs. Such a statement implies that every s-chromatic-inseparable graph H in the class satisfies $s < \sigma$ or $\chi(H) < \Psi$. In particular, if $\sigma \leqslant \Psi$, then every such H satisfies $s < \Psi$, since $s < \chi(H)$. It will be convenient to introduce the following hypothesis. Let $\Phi \geqslant 0$ and $a_0 \geqslant 1$ be real. A graph G satisfies $(\mathcal{H}_{\Phi, a_0})$ if for all reals $s \geqslant 0$ and $b \geqslant a_0$ and for every induced subgraph H of G, if H is s-chromatic-inseparable and $\mathrm{had}(H) < b$, then $s \leqslant \Phi b$. This property is inherited by induced subgraphs. It is adapted from the hypothesis on the function f in a theorem of Norin and Steiner [38, Theorem 15]. The differences are that the lower bound a_0 on b is a free parameter, and that b is not bounded above. The corresponding theorem of Delcourt and Postle [8, Theorem 7.1] uses a different hypothesis, namely an upper bound on $\chi(H)/a$ over small subgraphs H with no K_a minor.

Tangency and cores. The following notions are due to Delcourt and Postle [8, Definition 6.3]. A subgraph J of G is *tangent* to a model $\mathcal{A} = \{A_1, \dots, A_s\}$ if $|V(J) \cap A_i| = 1$ for every $i \in [s]$. A set $S \subseteq V(G)$ is a *core* of $\mathcal{A}$ if, for all distinct i, j, some edge of G has one end in $A_i \cap S$ and the other in $A_j \cap S$. Such edges are called *core edges*.

Bipartite pieces. Following Norin and Steiner [38, Section 3], where it is denoted $b(G)$, let $\mathrm{bip}(G)$ denote the maximum number of vertices in a connected induced bipartite subgraph of G. For an integer $k \geqslant 1$, we have $\mathrm{bip}(G) < k$ if and only if G has no connected induced bipartite subgraph with exactly k vertices. Indeed, deleting a leaf of a spanning tree of a connected induced bipartite subgraph with more than k vertices leaves a connected induced bipartite subgraph with one vertex fewer.

The extremal function. For a positive integer s, let $D(s)$ denote the maximum chro-

matic number of a graph with no K_s minor. For real $y \geqslant 1$, let $D(y) := D(\lfloor y \rfloor)$.

3.2 Connectivity, linkages and minors

This subsection collects the results on connectivity, linkages and rooted minors that we use, starting with highly connected subgraphs of graphs with large chromatic number.

Theorem 3.2 (Nguyen [34, Theorem 1.1]). *Let $k \geqslant 1$ be an integer. Every graph F with $\chi(F) \geqslant (3 + \frac{1}{16})k$ contains a $(k+1)$-connected induced subgraph with more than $\chi(F) - k$ vertices and chromatic number at least $\chi(F) - 2k + 2$.*

Nguyen states Theorem 3.2 for subgraphs, not necessarily induced. The proof gives an induced subgraph, and the induced form is the one formalised. We use Theorem 3.2 in the following form.

Corollary 3.3. *Let $k \geqslant 2$ be an integer and let F be a graph with $\chi(F) \geqslant \lceil 3.0625(k-1) \rceil$. Then F contains a k-connected induced subgraph F' with $\chi(F') \geqslant \chi(F) - 2k$.*

Proof. Apply Theorem 3.2 with $k - 1 \geqslant 1$ in place of k. Since $\chi(F)$ is an integer, $\chi(F) \geqslant \lceil 3.0625(k-1) \rceil$ is equivalent to $\chi(F) \geqslant (3 + \frac{1}{16})(k-1)$. This gives a k-connected induced subgraph F' of F with $\chi(F') \geqslant \chi(F) - 2(k-1) + 2 = \chi(F) - 2k + 4 \geqslant \chi(F) - 2k$. □

We use Menger's theorem in the following form, which is the form formalised in `LinearHadwiger`. For $A, B \subseteq V(G)$, a *path from A to B* is a path whose first vertex lies in A and whose last vertex lies in B. It may have further vertices in A or in B.

Theorem 3.4 (Menger). *Let G be a graph, let $A, B \subseteq V(G)$, and let $k \geqslant 0$ be an integer. Suppose that every set of vertices of G that meets every path from A to B in G has at least k vertices. Then G contains k pairwise vertex-disjoint paths from A to B. In particular, this holds if G is k-connected and $|A|, |B| \geqslant k$.*

The last sentence holds since, if $Q \subseteq V(G)$ and $|Q| < k$, then $G - Q$ is connected and contains a vertex $a \in A$ and a vertex $b \in B$, so some path from A to B avoids Q. Let P be a path from A to B. Let a be the last vertex of P in A, and let b be the first vertex in B of the subpath of P from a to the last vertex of P. The subpath of P from a to b is an A–B path, and we say that it is obtained from P by *trimming*. Trimming each path of a family of pairwise vertex-disjoint paths from A to B gives a family of pairwise vertex-disjoint A–B paths.

Theorem 3.5 (Thomas and Wollan [47, Corollary 1.2]). *Let $\ell \geqslant 1$ be an integer. Every 2ℓ-connected graph G with at least $5\ell\, v(G)$ edges is ℓ-linked.*

We call Theorem 3.5 the *density form* of the Thomas–Wollan theorem. Its *connectivity form* [47, Corollary 1.3] states that every 10ℓ-connected graph G is ℓ-linked, which follows immediately from Theorem 3.5.

Lemma 3.6 (cf. Delcourt and Postle [8, Theorem 5.8]). *Let ℓ and k be integers with $1 \leqslant \ell \leqslant k$. Every graph G with $\kappa(G) \geqslant 2k - \ell$ and $e(G) \geqslant 5k\, v(G)$ is (k, ℓ)-knit.*

Proof. Let $|S| = k$, and let $S_1, \ldots, S_p$ be a partition of S into non-empty parts, where $p \geqslant \ell$, with $S_i = \{v_{i,1}, \ldots, v_{i,n_i}\}$. As illustrated in Figure 3, we choose for each $v_{i,j}$ with $j \geqslant 2$ a private neighbour $u_{i,j}$ of $v_{i,1}$, and then link $v_{i,j}$ to $u_{i,j}$. Since $\delta(G) \geqslant \kappa(G) \geqslant 2k - \ell \geqslant 2k - p$, we can choose pairwise distinct vertices $u_{i,j} \notin S$, for $1 \leqslant i \leqslant p$ and $2 \leqslant j \leqslant n_i$, such that $u_{i,j}$ is adjacent to $v_{i,1}$. Indeed, there are $k - p$ such pairs (i, j), and each $v_{i,1}$ has at least $2k - p$ neighbours, and hence at least $k - p$ neighbours outside S. So such a choice exists

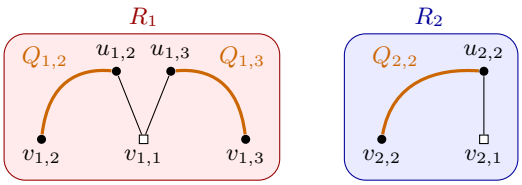


Figure 3: The proof of Lemma 3.6 with $p = 2$, $S_1 = \{v_{1,1}, v_{1,2}, v_{1,3}\}$ and $S_2 = \{v_{2,1}, v_{2,2}\}$. Each $u_{i,j}$ is a neighbour of $v_{i,1}$ outside S, and the path $Q_{i,j}$ in $G - L$ joins $v_{i,j}$ to $u_{i,j}$. The vertices of $L = \{v_{1,1}, v_{2,1}\}$ (squares) lie on no path, and R_1 and R_2 are shaded.

by Hall's theorem. Let $L := \{v_{i,1} : i \in [p]\}$. There are $k - p$ pairs $(v_{i,j}, u_{i,j})$, for $i \in [p]$ and $2 \leqslant j \leqslant n_i$. These pairs have pairwise distinct ends and avoid L. The graph $G - L$ is $(\kappa(G) - p)$-connected, and $\kappa(G) - p \geqslant 2k - \ell - p \geqslant 2(k - p)$. At most $p\,v(G)$ edges of G have an end in L, so $e(G - L) \geqslant (5k - p)\,v(G) \geqslant 5(k - p)\,v(G - L)$. If $p < k$, then $G - L$ is $(k - p)$-linked by Theorem 3.5. So $G - L$ has a linkage $\{Q_{i,j} : i \in [p],\ 2 \leqslant j \leqslant n_i\}$, where $Q_{i,j}$ is a path from $v_{i,j}$ to $u_{i,j}$. If $p = k$, then there are no pairs, and this linkage is empty. The paths $Q_{i,j}$ avoid L. For each $i \in [p]$, let $R_i := \{v_{i,1}\} \cup \bigcup_{j=2}^{n_i} V(Q_{i,j})$. Then $G[R_i]$ is connected, since $Q_{i,j}$ joins $v_{i,j}$ to $u_{i,j}$ and $u_{i,j}v_{i,1} \in E(G)$ for $2 \leqslant j \leqslant n_i$. Also $S_i \subseteq R_i$, and the sets R_i are pairwise disjoint, since the paths $Q_{i,j}$ are pairwise disjoint and avoid L. □

Delcourt and Postle [8, Theorem 5.8] state a result of this kind for Ck-connected graphs, with an unspecified constant C. They state that it is an easy corollary of their Theorem 5.6 [8], and remark that the Thomas–Wollan theorem gives the value 10 for the linkage constant in their Theorem 5.6. Since Lemma 3.6 assumes $e(G) \geqslant 5k\,v(G)$, its proof can use Theorem 3.5. To make $G - L$ $(k - p)$-linked, the density form needs connectivity $2(k - p)$, whereas the connectivity form needs $10(k - p)$.

For a graph G and $W, B \subseteq V(G)$, a *double fan* from W to B in G is a family $\{P_{w,j} : w \in W,\ j \in \{1, 2\}\}$ of paths in G that are pairwise disjoint outside W, such that each $P_{w,j}$ has first vertex w, has its last vertex in B, and has no vertex in W other than w.

Lemma 3.7 (Delcourt and Postle [8, Lemma 5.3])**.** *Let G be a graph, and let $A_1, A_2, B \subseteq V(G)$ with $A_1 \cap A_2 = \emptyset$. Suppose that, for each $i \in \{1, 2\}$, the graph $G - A_{3-i}$ contains a double fan from A_i to B. Then G contains $|A_1| + |A_2|$ pairwise vertex-disjoint paths from $A_1 \cup A_2$ to B.*

Delcourt and Postle assume in addition that A_1, A_2 and B are disjoint and non-empty, and that the paths in the double fans are A_i–B paths. Their argument proves the form above, which is the one in `LinearHadwiger`: if a set Q meets every path from $A := A_1 \cup A_2$ to B, then the two paths of the double fan starting at each $a \in A_i \setminus Q$ meet Q in distinct vertices outside A, so $|Q \setminus A| \geqslant 2|A_i \setminus Q|$ for $i \in \{1, 2\}$. Hence $|Q| \geqslant |A|$, and Theorem 3.4 applies.

Lemma 3.8 (Delcourt and Postle [8, Lemma 5.15], see also [38, Lemma B.11])**.** *Let $a, b \geqslant 0$ be integers. Let $\mathcal{P} = \{P_1, \ldots, P_b\}$ be an $\{(s_i, t_i)\}_{i \in [b]}$-linkage in a graph G, where P_i has ends s_i and t_i, and let $S := \{s_1, \ldots, s_b\}$ and $T := \{t_1, \ldots, t_b\}$. Let H be a subgraph of G such that $G[V(H)]$ is (a, c)-woven for some integer $c \geqslant b$, and let $R \subseteq V(H)$ with $|R| = a$. Then G has an $\{(s_i, t_i)\}_{i \in [b]}$-linkage $\mathcal{P}'$ and a K_a model $\mathcal{M}$ in $G[V(H)]$ rooted at R such that $V(\mathcal{P}') \subseteq V(H) \cup V(\mathcal{P})$ and $R \cap (S \cup T) \subseteq V(\mathcal{P}') \cap V(\mathcal{M}) \subseteq R \cap V(\mathcal{P})$.*

If H is (a, c)-woven, then so is $G[V(H)]$, since adding edges keeps a graph woven. Delcourt and Postle [8] assume that H is (a, c)-woven, $a \geqslant 1$ and $s_i \neq t_i$, and state that $\mathcal{M}$ is a model in G. Their proof gives the form above, which is the one in `LinearHadwiger`.

It applies the wovenness of $G[V(H)]$ to the pairs formed by the first and last vertices in $V(H)$ of the paths P_i that meet $V(H)$, and replaces the part of each such P_i between these two vertices by the resulting path. These two vertices may coincide, so the condition $s_i \neq t_i$ is not needed. There are at most $b \leqslant c$ such pairs, so no lower bound on $v(H)$ is needed.

Lemma 3.9 (Kawarabayashi [18, statement (3.1)], see also [8, Lemma 5.14])**.** *Let $a \geqslant 1$ be an integer and let G be a graph. If $\kappa(G) \geqslant a$ and G has a K_{2a} minor, then G is $(a, 0)$-woven.*

Kawarabayashi states (3.1) for graphs with no separation of order less than a that separates a vertex from the set of roots. This condition holds when $\kappa(G) \geqslant a$.

Lemma 3.10 (Delcourt and Postle [8, Lemma 6.2])**.** *Let G be a connected graph and $\emptyset \neq S \subseteq V(G)$. Then there is an induced connected subgraph H of G and a set S' with $S \subseteq S' \subseteq V(H)$, $|S'| \leqslant 3|S|$ and $\chi(H - S') \leqslant 2$.*

3.3 Extremal results

This subsection collects extremal results for graphs with no K_t minor, namely edge-density bounds for graphs and for bipartite graphs, small highly connected subgraphs of dense graphs, and a lower bound on the independence number, together with its consequence for colouring.

Theorem 3.11 (Kostochka [22, 23] and Thomason [48], see also [8, Theorem 4.3])**.** *Let $s \geqslant 2$ be an integer. Every non-null graph G with $e(G) \geqslant 30\, s\sqrt{\log s}\, v(G)$ has a K_s minor. Consequently, every graph G with no K_s minor satisfies $e(G) \leqslant 30\, s\sqrt{\log s}\, v(G)$ and $\chi(G) < 60\, s\sqrt{\log s} + 1$. In particular, $D(s) < 60\, s\sqrt{\log s} + 1$.*

Delcourt and Postle [8, Theorem 4.3] state Theorem 3.11 with the constant 30, and note that it follows from the results of Kostochka [22]. The explicit bound of Kostochka [23, Theorem 1] holds for all s, but with a larger constant (about 270). The constant 30 affects the size thresholds, and also the constant in Theorem 1.2. Indeed, in Case 1 of the proof of Theorem 1.2, where $\log\log t < 18.4$, Theorem 3.11 gives $\chi(G) < 600{,}000\, t + 1$, which is below 1,118,430 had(G). With the constant 270 in place of 30, this bound would be about $5.3 \cdot 10^6\, t$.

Theorem 3.12 (Norin and Postle [36, Theorem 3.2], see also [8, Theorem 4.1])**.** *Let $C_{3.12} := 6400$ and let $t \geqslant 3$ be an integer. Let B be a bipartite graph with no K_t minor and with bipartition (V_1, V_2). Then*

$$e(B) \;\leqslant\; C_{3.12}\, t\sqrt{\log t}\, \sqrt{|V_1||V_2|} + (t-2)(|V_1| + |V_2|).$$

Norin and Postle show in the proof of [36, Theorem 3.2] that the constant 6400 suffices. Theorem 3.12 is formalised in `LinearHadwiger` with this constant.

Delcourt and Postle [8, Theorem 2.3] proved the following *small-dense-subgraph theorem*. Let $C_{\mathrm{DP}} := 480 \cdot 6400 = 3{,}072{,}000$. For all integers $t \geqslant 3$ and $k \geqslant t$, every non-null graph G with no K_t minor and $e(G) \geqslant C_{\mathrm{DP}} k\, v(G)$ has a k-connected subgraph with at most ${C_{\mathrm{DP}}}^2\, t \log^3 t$ vertices. Delcourt and Postle state this theorem with $480\, C_{3.12}$ in place of C_{DP}, where $C_{3.12}$ is the constant of the bipartite density bound in Theorem 3.12, which they state as their Theorem 4.1 [8]. We do not use the small-dense-subgraph theorem, although it is formalised in `LinearHadwiger`, with a k-connected induced subgraph. Instead, Section 7 proves a refined form of Lin's theorem, which finds pieces that are both dense and highly connected.

Theorem 3.13 (Reed and Seymour [42], see also [38, Theorem 6 and Corollary 17]). *Every graph J satisfies $\chi_f(J) \leqslant 2\,\mathrm{had}(J)$, and hence $v(J) \leqslant 2\alpha(J)\,\mathrm{had}(J)$. In particular, if $a \geqslant 2$ is an integer and J has no K_a minor, then $\alpha(J) \geqslant v(J)/(2a-2)$.*

Duchet and Meyniel [11] proved a slightly stronger form of the last statement of Theorem 3.13 by a direct argument, and Delcourt and Postle [8, Theorem 3.3] use their result. (We use the theorem of Reed and Seymour here, instead of the theorem of Duchet and Meyniel, so that the proof matches the Lean formalisation.)

Results of Delcourt and Postle [8, Lemma 3.1 and Corollary 3.2] imply that every non-null graph J such that $\alpha(J') \geqslant v(J')/p$ for every non-null subgraph J' of J satisfies $\chi(J) \leqslant \big(2+\max\{\log(v(J)/p), 0\}\big)p$. By Theorem 3.13, this condition holds with $p = 2a-2$ for every graph with no K_a minor. We use the result in the following form.

Lemma 3.14 (Delcourt and Postle [8, Lemma 3.1 and Corollary 3.2]). *Let $a \geqslant 2$ be an integer and let $p := 2a-2$. Every non-null graph J with no K_a minor satisfies $\chi(J) \leqslant \big(2+\max\{\log(v(J)/p), 0\}\big)p$.*

3.4 Results of Norin and Steiner

Norin and Steiner [38, Theorem 4] proved that, for every $\varepsilon \in (0, \frac{1}{2})$, there is $t_0 = t_0(\varepsilon)$ such that, for every real $t \geqslant t_0$, every graph G with $\mathrm{had}(G) < t$ and $v(G) \leqslant t(\log t)^{1/2-\varepsilon}$ satisfies $\chi(G) \leqslant (4+\varepsilon)t$. We call this result the *small-graph bound* of Norin and Steiner. We do not use it, and it is not formalised. Lemma 5.3 is an explicit instance of it, derived from a quantitative form (Theorem 5.1).

Lemma 3.15 (cf. Norin and Steiner [38, Corollary 17]). *Let s, b and Γ be real with $\Gamma \geqslant 0$, and let $k \geqslant 2$ be an integer. Suppose that every graph J with $\mathrm{had}(J) < b$ and $v(J) \leqslant 2bk(k-1)$ satisfies $\chi(J) \leqslant \Gamma$. If H is an s-chromatic-inseparable graph with $\mathrm{had}(H) < b$ and $\mathrm{bip}(H) < k$, then $s < \Gamma$.*

In applications of Lemma 3.15, we call the graphs J in its hypothesis *Gyárfás graphs*. As noted in Section 3.1, $\mathrm{bip}(H) < k$ means that H has no connected induced bipartite subgraph with exactly k vertices. The proof of Lemma 3.15 is that of Norin and Steiner [38, Corollary 17]. It uses the following form of the Gyárfás path argument [14], due to Norin and Steiner [39, Lemma 13] (see also [38, Lemma 16]). Let $q \geqslant 1$ and $k \geqslant 2$ be integers, and let G be a graph with $\mathrm{bip}(G) < k$ and $\chi(G) \geqslant q$ that has no two vertex-disjoint induced subgraphs with chromatic number at least q. Then there is a set $X \subseteq V(G)$ with $\alpha(G[X]) \leqslant k(k-1)$ and $\chi(G-X) \leqslant q-1$. Norin and Steiner assume only $\mathrm{bip}(G) \leqslant k$. The form above is the one formalised in `LinearHadwiger`.

Proof of Lemma 3.15. If $s < 0$, then $s < 0 \leqslant \Gamma$. So let $s \geqslant 0$. If $\chi(H) \leqslant s$, then two null subgraphs show that H is s-chromatic-separable. So $\chi(H) > s$. Let $q := \lceil \chi(H) - s \rceil$. Then q is an integer with $1 \leqslant q \leqslant \chi(H)$ and $q < \chi(H) - s + 1$. Since $q \geqslant \chi(H) - s$ and H is s-chromatic-inseparable, H has no two vertex-disjoint induced subgraphs with chromatic number at least q. So the Gyárfás path argument gives a set $X \subseteq V(H)$ with $\alpha(H[X]) \leqslant k(k-1)$ and $\chi(H-X) \leqslant q-1$. Hence $\chi(H[X]) \geqslant \chi(H) - \chi(H-X) \geqslant \chi(H) - q + 1 > s$. Since $\mathrm{had}(H[X]) \leqslant \mathrm{had}(H) < b$, Theorem 3.13 gives $v(H[X]) \leqslant 2\alpha(H[X])\,\mathrm{had}(H[X]) \leqslant 2bk(k-1)$. So $\chi(H[X]) \leqslant \Gamma$ by hypothesis, and hence $s < \chi(H[X]) \leqslant \Gamma$. □

3.5 Paths to a large set

The proofs of Lemma 4.2 and Proposition 8.1 use the following two constructions of paths. Both are taken from the proofs of Delcourt and Postle [8, Theorem 7.1 and Lemma 6.4]. Part (a) is their application of Menger's theorem, which they state without details. We

obtain it by replacing each vertex of the set W by two non-adjacent copies. Parts (b) and (c) adapt the identification argument in their proof of [8, Lemma 6.4], combined with their Lemma 5.3 (Lemma 3.7). Instead of identifying the two vertices of each pair, we add an auxiliary vertex adjacent to both, and work in the union of the given paths. Parts (b) and (c) are stated in these forms in `LinearHadwiger`.

Lemma 3.16. *Let G be a graph, and let $W, B \subseteq V(G)$.*

(a) Suppose that $W \cap B = \emptyset$, that G is r-connected for some integer $r \geqslant 2|W|$, and that $|B| \geqslant 2|W|$. Then G contains a double fan $\mathcal{P}$ from W to B with $\chi(G[V(\mathcal{P})]) \leqslant 4|W|$.

(b) Let $\mathcal{P}$ be a double fan from W to B in G. Let $m \geqslant 0$ be an integer, and let $U \subseteq V(G)$ be disjoint from $B \cup V(\mathcal{P})$. Let $\{Q_{i,j} : i \in [m],\ j \in \{1,2\}\}$ be a family $\mathcal{Q}$ of $2m$ pairwise disjoint paths in $G - W$, each with its first vertex in U, no other vertex in U, and its last vertex in B. Then G contains $|W| + m$ pairwise disjoint paths P'_w ($w \in W$) and Q'_i ($i \in [m]$) with the following properties. Each P'_w has first vertex w. Each Q'_i has as its first vertex the first vertex of $Q_{i,1}$ or of $Q_{i,2}$. Each of these paths has its last vertex in B, meets $W \cup U \cup B$ only in its ends, and has all its vertices in $V(\mathcal{P}) \cup V(\mathcal{Q})$.

(c) Let $\mathcal{P}$, m, U and $\mathcal{Q}$ be as in (b), except that the paths of $\mathcal{Q}$ may have further vertices in U. Then G contains $|W|+m$ pairwise disjoint paths P'_w ($w \in W$) and Q'_i ($i \in [m]$) with the following properties. Each P'_w has first vertex w. Each Q'_i has as its first vertex a vertex of U on $Q_{i,1}$ or on $Q_{i,2}$. Each of these paths has its last vertex in B and meets $W \cup U \cup B$ only in its ends.

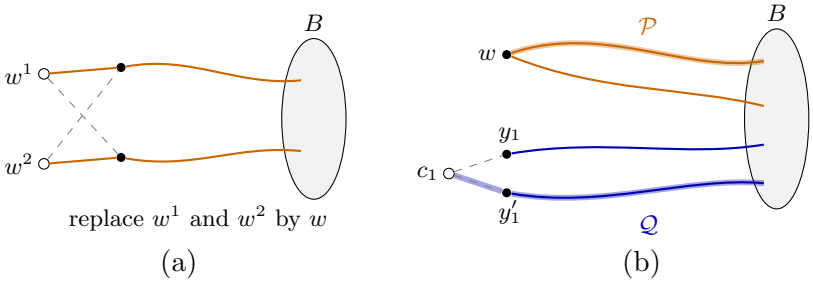


Figure 4: The proof of Lemma 3.16. (a) The vertex w is replaced by two non-adjacent copies w^1 and w^2 (white), each adjacent to the neighbours of w outside W. Replacing w^1 and w^2 by w in the Menger paths from w^1 and w^2 gives two paths from w to B that meet only in w. (b) The proof of part (b). Here $W = \{w\}$ and $m = 1$. An auxiliary vertex c_1 (white), adjacent only to y_1 and y'_1, is added, and Lemma 3.7 gives disjoint paths from w and from c_1 in the union of the paths. In the highlighted outcome, the path from c_1 continues through y'_1, and deleting c_1 gives a path from y'_1.

Proof. (a) As illustrated in Figure 4(a), we replace each vertex of W by two copies and apply Menger's theorem. Let $G^\times$ be obtained from $G - W$ by adding, for each $w \in W$, two new non-adjacent vertices w^1 and w^2, each adjacent to every vertex of $N_G(w) \setminus W$. Let $W^\times := \{w^1, w^2 : w \in W\}$, so $|W^\times| = 2|W|$. We claim that every set $Q \subseteq V(G^\times)$ that meets every path from $W^\times$ to B in $G^\times$ has at least $2|W|$ vertices. Suppose not. Let $O := Q \setminus W^\times$ and $W_0 := \{w \in W : w^1, w^2 \in Q\}$, and let $D := O \cup W_0 \subseteq V(G)$. Then $|D| \leqslant |O| + 2|W_0| \leqslant |Q| < 2|W| \leqslant r$, so $G - D$ is connected. Since $|Q| < |W^\times|$, some w^i is not in Q, and then $w \notin D$. Since $|O| < 2|W| \leqslant |B|$, some $b \in B$ is not in O, and $b \notin D$ since $b \notin W$. Let P_0 be a w–b path in $G - D$, let z be the last vertex of P_0 in W, and let P_1 be the subpath of P_0 from z to b. Since $z \notin W_0$, some z^j is not in Q. The second vertex of P_1 lies in $N_G(z) \setminus W$, so replacing z by z^j in P_1 gives a path in $G^\times$ from z^j to

b. This path avoids Q, since its vertices other than z^j lie in $V(G) \setminus (W \cup D)$ and hence not in O. This contradiction proves the claim. By Theorem 3.4, $G^\times$ has $2|W|$ pairwise vertex-disjoint paths from $W^\times$ to B. Since $|W^\times| = 2|W|$, each vertex of $W^\times$ is the first vertex of exactly one of them, and hence is the only vertex of $W^\times$ on this path. Replacing the first vertex w^i of each of these paths by w gives a path in G from w to B with no other vertex in W, since all its other vertices lie in $V(G) \setminus W$ and its second vertex is a neighbour of w. The two paths obtained from w^1 and w^2 start at w, and any two of the resulting paths are disjoint outside W. So they form a double fan from W to B. Finally, replace each path P of this double fan by a shortest path between its ends in $G[V(P)]$. This is an induced path of G with the same ends and with vertex set contained in $V(P)$, so the resulting family $\mathcal{P}$ is again a double fan from W to B. Each path of $\mathcal{P}$ is induced in G, and so has a proper 2-colouring. Assign each vertex of $V(\mathcal{P})$ to one path of $\mathcal{P}$ containing it, and colour the vertex with the pair consisting of this path and the colour of the vertex in the 2-colouring of this path. If two adjacent vertices are assigned to the same path, then they are adjacent on this induced path, and so have distinct colours in its 2-colouring. So adjacent vertices receive distinct pairs, and $\chi(G[V(\mathcal{P})]) \leqslant 2 \cdot 2|W| = 4|W|$. Moreover, if $R \subseteq V(G)$ contains $W \cup B$ and $G[R]$ satisfies the hypotheses of (a), then (a) applied to $G[R]$ gives a double fan $\mathcal{P}$ from W to B in G with $V(\mathcal{P}) \subseteq R$ and $\chi(G[V(\mathcal{P})]) \leqslant 4|W|$, since $G[R][V(\mathcal{P})] = G[V(\mathcal{P})]$.

(b) As illustrated in Figure 4(b), we add an auxiliary vertex for each pair of paths $Q_{i,1}, Q_{i,2}$ and apply Lemma 3.7. For $i \in [m]$, let y_i and y'_i be the first vertices of $Q_{i,1}$ and $Q_{i,2}$, and let $Y := \{y_1, y'_1, \dots, y_m, y'_m\}$. Let $C := \{c_1, \dots, c_m\}$ be a set of m new vertices. Let G^+ be the graph with vertex set $V(\mathcal{P}) \cup V(\mathcal{Q}) \cup C$ whose edges are the edges of the paths in $\mathcal{P} \cup \mathcal{Q}$ together with the edges $c_i y_i$ and $c_i y'_i$ for $i \in [m]$. The paths of $\mathcal{P}$ avoid U, and each path of $\mathcal{Q}$ meets U only in its first vertex, so $V(G^+) \cap U = Y$. Each $y \in \{y_i, y'_i\}$ is not in B, since $U \cap B = \emptyset$. So the path of $\mathcal{Q}$ starting at y has a second vertex, and y has exactly two neighbours in G^+, namely c_i and this second vertex. In G^+ the hypotheses of Lemma 3.7 hold with $A_1 = W$, $A_2 = C$ and the same B, since:

- The paths of $\mathcal{P}$ form a double fan from W to B in $G^+ - C$.
- Adding c_i at the start of $Q_{i,1}$ and of $Q_{i,2}$, for each $i \in [m]$, gives a double fan from C to B in $G^+ - W$, since the paths of $\mathcal{Q}$ are pairwise disjoint and lie in $G - W$.

Lemma 3.7 gives $|W| + m$ pairwise disjoint paths from $W \cup C$ to B in G^+. Since $|W \cup C| = |W| + m$, exactly one of them starts at each vertex of $W \cup C$, and so each of them meets $W \cup C$ only in its first vertex. Replace each of these paths by its subpath from its first vertex to its first vertex in B. Let L be one of the resulting paths, and let $y \in Y \cap V(L)$, with $y \in \{y_i, y'_i\}$. Since $y \notin W \cup C \cup B$, the vertex y is an internal vertex of L, so both its neighbours in G^+ are its neighbours on L. In particular $c_i \in V(L)$. Since L meets $W \cup C$ only in its first vertex, L starts at c_i, and y is the second vertex of L. Hence the paths starting in W avoid Y, and, for each $i \in [m]$, the path starting at c_i meets Y only in its second vertex, which lies in $\{y_i, y'_i\}$. For each $i \in [m]$, delete c_i from the path starting at c_i. All edges of G^+ not incident with C are edges of G, so this gives $|W| + m$ pairwise disjoint paths in G with all their vertices in $V(\mathcal{P}) \cup V(\mathcal{Q})$. The path P'_w starts at w, and the path Q'_i starts at y_i or at y'_i. Each of them meets W only in its first vertex, meets U only in its first vertex since $V(G^+) \cap U = Y$, and meets B only in its last vertex.

(c) For each path $Q_{i,j}$ of $\mathcal{Q}$, let $Q^*_{i,j}$ be the subpath of $Q_{i,j}$ from its last vertex in U to its last vertex. The paths $Q^*_{i,j}$ are pairwise disjoint paths in $G - W$, each with its first vertex in U, no other vertex in U, and its last vertex in B. Applying (b) to them gives the required paths, since the first vertex of each $Q^*_{i,j}$ is a vertex of U on $Q_{i,j}$. □

3.6 A sharpened form of Mader's theorem

The proofs of Lemmas 4.1 and 6.3 and of Theorem 7.1 use the following form of Mader's theorem.

Lemma 3.17 (Mader [30], see [9, Theorem 1.4.3]). *Let $k \geqslant 1$ be an integer, and let G be a non-null graph with $e(G) \geqslant 2k\, v(G)$. Let $\gamma := e(G)/v(G)$. Let $\gamma' := \frac{1}{2}\bigl(\gamma + \sqrt{\gamma^2 - 2\gamma k}\bigr)$. Then G has a $(k+1)$-connected induced subgraph H with $\delta(H) > \gamma$, $v(H) > 2\gamma'$ and $e(H) > \gamma'\, v(H) \geqslant (\gamma - k)\, v(H)$.*

The bounds $e(H) > \gamma' v(H)$ and $v(H) > 2\gamma'$ are used in the proof of Lemma 4.1.

Proof. The bounds $\delta(H) > \gamma$ and $e(H) > (\gamma - k)v(H)$ are in the statement and proof of [9, Theorem 1.4.3]. The bounds $v(H) > 2\gamma'$ and $e(H) > \gamma' v(H)$ are a slight sharpening that follows from the same proof. We give that proof, and record these bounds along the way. Here $\gamma \geqslant 2k$. Call an induced subgraph G' of G *good* if $v(G') \geqslant 2k$ and $e(G') > \gamma\,(v(G') - k)$. The graph G itself is good, since $e(G) = \gamma\, v(G)$ and $v(G) > 2\gamma \geqslant 4k$ (the average degree 2γ is less than $v(G)$). Let H be a good subgraph with as few vertices as possible. No good graph has exactly $2k$ vertices, since it would have more than $\gamma k \geqslant 2k^2 > \binom{2k}{2}$ edges. So $v(H) > 2k$. If H had a vertex v with $\deg_H(v) \leqslant \gamma$, then $H - v$ would have $v(H) - 1 \geqslant 2k$ vertices and $e(H - v) > \gamma(v(H) - k) - \gamma = \gamma(v(H - v) - k)$ edges. So $H - v$ would be good, contradicting the choice of H. Hence $\delta(H) > \gamma$. In particular $v(H) > \gamma$. Let $n := v(H)$ and $r := \sqrt{\gamma^2 - 2\gamma k}$, which is real since $\gamma \geqslant 2k$. Since H is good and $e(H) \leqslant n(n-1)/2 < n^2/2$, we have $n^2/2 > \gamma(n - k)$, that is, $n^2 - 2\gamma n + 2\gamma k > 0$. The roots of this quadratic are $\gamma \pm r$, and $n > \gamma \geqslant \gamma - r$, so $n > \gamma + r = 2\gamma'$. Hence

$$\frac{e(H)}{n} \;>\; \gamma\Bigl(1 - \frac{k}{n}\Bigr) \;>\; \gamma - \frac{\gamma k}{\gamma + r} \;=\; \gamma - \frac{\gamma - r}{2} \;=\; \gamma',$$

using $(\gamma + r)(\gamma - r) = 2\gamma k$. Finally, $\gamma' \geqslant \gamma - k$ is equivalent to $r \geqslant \gamma - 2k$. Since $\gamma - 2k \geqslant 0$, squaring shows that this is equivalent to $2\gamma k \geqslant 4k^2$, which holds since $\gamma \geqslant 2k$.

It remains to show that H is $(k+1)$-connected. We have $v(H) > 2k \geqslant k + 1$. Suppose that H has a separation (U_1, U_2) with $|U_1 \cap U_2| \leqslant k$ and both $U_1 \setminus U_2$ and $U_2 \setminus U_1$ non-empty. Let $H_i := H[U_i]$. A vertex of $U_1 \setminus U_2$ has all its neighbours in U_1, so $v(H_1) > \delta(H) > \gamma \geqslant 2k$. Similarly $v(H_2) > 2k$. Each H_i has fewer vertices than H, so it is not good, and hence $e(H_i) \leqslant \gamma(v(H_i) - k)$. Every edge of H lies in H_1 or H_2, and $v(H_1) + v(H_2) \leqslant v(H) + k$. Therefore

$$e(H) \;\leqslant\; e(H_1) + e(H_2) \;\leqslant\; \gamma\bigl(v(H_1) + v(H_2) - 2k\bigr) \;\leqslant\; \gamma\,(v(H) - k),$$

a contradiction, since H is good. So H is $(k+1)$-connected. □

4 An explicit chromatic-separable reduction

This section proves Theorem 4.3, the engine of each bootstrapping step. Roughly, it states that if $\mathrm{had}(G) < t$ and, for every real $b \geqslant a_0$, every s-chromatic-inseparable induced subgraph H of G with $\mathrm{had}(H) < b$ satisfies $s \leqslant \Phi b$, then $\chi(G)$ is at most about $c_{4.3}(\Phi)\, t$, where $c_{4.3}(\Phi)$ is defined in Theorem 4.3. The section makes explicit a lemma and a theorem of Norin and Steiner [38, Lemma B.12 and Theorem 15]. These are adaptations of results of Delcourt and Postle [8, Theorems 7.1 and 1.6]. Lemma 4.1 constructs the hub used in the proof of Lemma 4.2. Theorem 4.3 follows from Lemma 4.2, and is used in Lemma 5.4. Lemma 4.2 is also adapted in Section 9, as Lemma 9.2.

Lemma 4.1. *Let $m \geqslant 1$ be an integer and let $q := \lceil 25m/2 \rceil + 1$. Every graph F with $\chi(F) \geqslant q$ has an m-linked induced subgraph H_0 with $v(H_0) > 10m$ and $\chi(H_0) \leqslant q$.*

Proof. Let S be a smallest set of vertices of F with $\chi(F[S]) \geqslant q$, and let $F_0 := F[S]$. Then $\chi(F_0 - v) < q$ for every vertex v of F_0. Since $q \geqslant 1$, F_0 has a vertex, and deleting a vertex lowers the chromatic number by at most 1, so $\chi(F_0) = q$. If some vertex v of F_0 had $\deg_{F_0}(v) < q-1$, then a $(q-1)$-colouring of $F_0 - v$ would extend to F_0, a contradiction. Hence $\delta(F_0) \geqslant q-1$, a classical property of critical graphs (see Dirac [10]). Thus $\delta(F_0) \geqslant 25m/2$, so $\gamma := e(F_0)/v(F_0) \geqslant 25m/4 \geqslant 4m$. Apply Lemma 3.17 to F_0 with $2m$ in place of k, and let $\gamma' := \frac{1}{2}(\gamma + \sqrt{\gamma^2 - 4m\gamma})$. Then $\gamma' \geqslant 5m$. Indeed, this is clear if $\gamma \geqslant 10m$. Otherwise $0 < 10m - \gamma$, and $(10m-\gamma)^2 \leqslant \gamma^2 - 4m\gamma$ since this is equivalent to $\gamma \geqslant 25m/4$. So $\sqrt{\gamma^2 - 4m\gamma} \geqslant 10m - \gamma$, and hence $\gamma' \geqslant 5m$. So F_0 has a $(2m+1)$-connected induced subgraph H_0 with $e(H_0) > 5m\, v(H_0)$ and $v(H_0) > 10m$. By Theorem 3.5, H_0 is m-linked. Since F_0 is an induced subgraph of F, so is H_0. Finally, $\chi(H_0) \leqslant \chi(F_0) = q$. □

Lemma 4.1 needs only a $(2m+1)$-connected subgraph, because it uses Theorem 3.5. The connectivity form of the Thomas–Wollan theorem would need a $10m$-connected subgraph, and extracting one by Corollary 3.3 needs $\chi(F) \geqslant \lceil 3.0625(10m-1) \rceil \approx 30.6m$, instead of about $12.5m$. The density form needs the sharper bound $e(H) > \gamma' v(H)$ of Lemma 3.17. The weaker bound $e(H) > (\gamma - k)v(H)$ gives only $e(H) > 4.25m\, v(H)$, which is less than the $5m\, v(H)$ that Theorem 3.5 requires.

Lemma 4.2. *Let $\eta \in (0,1]$, $a_0 \geqslant 4 + 5/\eta$ and $\Phi \geqslant 0$ be real. Let*

$$\Lambda_{\mathrm{E}} := (1+\eta)\big(517.5 + 12\Phi\big), \qquad \Lambda_{\mathrm{U}} := (1+\eta)\big(897.5 + 10\Phi\big),$$

and let $\Lambda := \min\{\Lambda_{\mathrm{E}}, \Lambda_{\mathrm{U}}\}$. Let $a \geqslant 1$ be an integer, and let G be a $21a$-connected graph that satisfies $(\mathcal{H}_{\Phi,a_0})$ (defined in Section 3.1) and

$$\chi(G) \;\geqslant\; D\big(\tfrac{46}{3}a_0 + 4\big) + 1 + \Lambda a.$$

Then G is $(a, 3a)$-woven if $\Lambda = \Lambda_{\mathrm{E}}$, and $(a, \lceil 10a/3 \rceil)$-woven otherwise.

We first give the idea of the proof, which uses induction on a. The roots and terminals are joined by paths to a linked subgraph H_0, called the hub. Through the hub, the terminals are joined in pairs, and the roots are joined to a $21a$-connected subgraph G_3 of large chromatic number. This reduces the required wovenness of G to the $(a,0)$-wovenness of G_3 (Steps 1–6). If G_3 has a K_{2a} minor, then Lemma 3.9 finishes the proof (Step 7a). Otherwise two chromatic separations in a subgraph of G_3 give three disjoint $21a$-connected subgraphs of large chromatic number. These subgraphs are woven by induction (Steps 7b and 8). Each branch set of the required K_a model in G_3 is assembled from branch sets in two of these three subgraphs (Step 9).

Step 9 distributes the a branch sets among the three subgraphs. In the *uneven scheme*, the subgraph J_3', which is obtained by one chromatic separation rather than two (Step 7b), receives at most $\lceil 4a/5 \rceil$ of them. Each of the other two subgraphs J_1' and J_2' then receives only $\lceil 3a/5 \rceil$ branch sets. Each of J_1' and J_2' must still be woven with respect to all $2a$ paths of the linkage in Step 9. So the lemma must give $(a, \lceil 10a/3 \rceil)$-wovenness rather than $(a, 3a)$-wovenness. Indeed, Step 8 applies it by induction with $\lceil 3a/5 \rceil$ in place of a, and Step 9 uses that $\lceil 10\lceil 3a/5 \rceil / 3 \rceil \geqslant 2a$. In the *even scheme*, each subgraph receives at most $\lceil 2a/3 \rceil$ branch sets, and $3\lceil 2a/3 \rceil \geqslant 2a$, so $(a, 3a)$-wovenness suffices. The uneven scheme reduces the coefficient of Φ from 12 to 10, at the cost of increasing the constant term from 517.5 to 897.5. The even scheme gives the smaller value of Λ when $\Phi < 190$, and the two values coincide at $\Phi = 190$. So Lemma 4.2 uses the even scheme if and only if $\Phi \leqslant 190$. The even scheme is therefore used in the bootstrapping steps whose Gyárfás graphs are coloured by the base case, where $\Phi = 4.01$ (Proposition 5.5). The uneven scheme is used in

the final step of Proposition 5.5, where $\Phi = 615.6401$ is the constant of the side step, and in the final application (Lemma 9.2), where the role of Φ is played by $78+2c_{5.5} \approx 9.0 \cdot 10^4$. The scheme is determined by Φ, so it is the same at every level of the induction below. We give the proof for the uneven scheme ($\Lambda = \Lambda_{\mathrm{U}} < \Lambda_{\mathrm{E}}$), and then list the changes for the even scheme. Figure 5 illustrates Steps 3–6 and Step 9.

Proof. We proceed by induction on a. Every subgraph extracted below by Corollary 3.3 is induced, and the subgraphs produced by the chromatic separations in Step 7b are chosen induced. Hence every graph to which the induction hypothesis is applied is an induced subgraph of G and satisfies $(\mathcal{H}_{\Phi,a_0})$. The proof uses the following subgraphs, which are defined in Steps 3, 5 and 7b. Each of them is an induced subgraph of G.

$$G \supseteq G_1 \supseteq H_0 \qquad \text{and} \qquad G_1 \supseteq G_2 \supseteq G_3 \supseteq G_4 \supseteq J_i \supseteq J_i' \quad (i \in \{1,2,3\}).$$

For the uneven scheme, let $\ell_0 := \lceil 10a/3 \rceil$ and $h := 2a + 4\ell_0$. We show that G is (a, ℓ_0)-woven. In Step 1, $\ell \in \{0, \dots, \ell_0\}$ denotes the number of terminal pairs. Then

$$\ell \leqslant \ell_0 \leqslant \tfrac{10a+2}{3}, \qquad a + \ell \leqslant \tfrac{13}{3}a + 1, \qquad a + 2\ell \leqslant \tfrac{23}{3}a + 2, \qquad h \leqslant \tfrac{46}{3}a + 4.$$

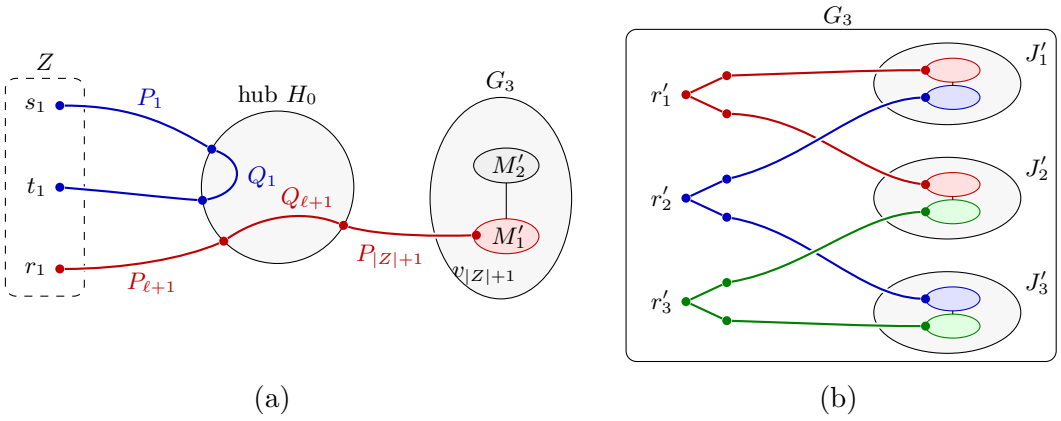


Figure 5: The proof of Lemma 4.2. (a) Steps 3–6. Paths through the $(a+\ell)$-linked hub H_0 join the terminals s_1 and t_1 (blue), and join the root r_1 to the branch set M_1' of a K_a model $\{M_1', M_2', \dots\}$ in G_3 (red). (b) Step 9. Each root r_i' with $i \in I_{jk}$ is joined through its neighbours s_i' and s_{a+i}' to branch sets of $\mathcal{M}_j$ and $\mathcal{M}_k$. Here r_1', r_2' and r_3' (red, blue and green) lie in I_{12}, I_{13} and I_{23}. Any two of these index pairs intersect, so any two new branch sets are adjacent. Only the branch sets containing terminals are shown.

Step 1. Reduction to disjoint terminals. Let $\ell \in \{0, \dots, \ell_0\}$, and let $R = \{r_1, \dots, r_a\}$, $S = \{s_1, \dots, s_\ell\}$ and $T = \{t_1, \dots, t_\ell\}$ be as in the definition of (a, ℓ_0)-woven. As illustrated in Figure 6, we modify the roots and terminals, using that $\delta(G) \geqslant 21a$. The list $r_1, \dots, r_a, s_1, \dots, s_\ell, t_1, \dots, t_\ell$ has $a+2\ell$ entries, which need not be distinct vertices. Since $2(a+2\ell) \leqslant \frac{46}{3}a+4 \leqslant 21a$, each entry has at least $\delta(G) - |R \cup S \cup T| \geqslant 21a - (a+2\ell) \geqslant a+2\ell$ neighbours outside $R \cup S \cup T$. So, by Hall's theorem, we can assign to each entry a neighbour of it outside $R \cup S \cup T$, such that the $a+2\ell$ assigned vertices are distinct. For each i with $s_i = t_i$, let t_i^* be the vertex assigned to the entry t_i. For each i with $r_i \in S \cup T$, let r_i^* be the vertex assigned to the entry r_i. Let $t_i^* := t_i$ for each i with $s_i \neq t_i$, and let $r_i^* := r_i$ for each i with $r_i \notin S \cup T$. Let $R^* := \{r_1^*, \dots, r_a^*\}$ and $T^* := \{t_1^*, \dots, t_\ell^*\}$. The vertices r_i^* are distinct, the vertices t_i^* are distinct, and R^*, S and T^* are pairwise disjoint. So (R^*, S, T^*) is as in the definition of (a, ℓ_0)-woven.

Suppose that $\mathcal{M}^* = \{M_1^*, \dots, M_a^*\}$ is a K_a model with $r_i^* \in M_i^*$ for each $i \in [a]$, and that $\mathcal{P}^*$ is an $\{(s_i, t_i^*)\}$-linkage with $V(\mathcal{M}^*) \cap V(\mathcal{P}^*) = \emptyset$. For each i with $s_i = t_i$, replace the path of $\mathcal{P}^*$ from s_i to t_i^* by the one-vertex path s_i. For each i with $r_i \neq r_i^*$, add r_i to M_i^*. This keeps $G[M_i^*]$ connected, since $r_i r_i^* \in E(G)$. Each such r_i lies in $S \cup T$. If $r_i = t_j$ and $s_j = t_j$, then $r_i = s_j \in S$. Otherwise $r_i \in S$ or $r_i = t_j = t_j^* \in T^*$. So $r_i \in S \cup T^* \subseteq V(\mathcal{P}^*)$, and hence $r_i \notin V(\mathcal{M}^*)$. Hence the new branch sets are pairwise disjoint. Each contains exactly one vertex of R, since every r_j either equals $r_j^* \in M_j^*$ or lies in $V(\mathcal{P}^*)$. Let $\mathcal{M}$ and $\mathcal{P}$ be the resulting families. Then $\mathcal{M}$ is a K_a model rooted at R, and $\mathcal{P}$ is an $\{(s_i, t_i)\}$-linkage. Since $V(\mathcal{P}) \subseteq V(\mathcal{P}^*)$ and $V(\mathcal{M}^*) \cap V(\mathcal{P}^*) = \emptyset$, every vertex $x \in V(\mathcal{M}) \cap V(\mathcal{P})$ is one of the added vertices, so $x = r_i$ for some i with $r_i \in S \cup T$. Conversely, each $r_i \in S \cup T$ lies in the ith branch set of $\mathcal{M}$ and in $\{s_j, t_j\}$ for some j, so r_i is an end of the path of $\mathcal{P}$ from s_j to t_j. Hence $V(\mathcal{M}) \cap V(\mathcal{P}) = R \cap (S \cup T)$. So it suffices to treat pairwise disjoint sets R, S, T, which we assume from now on.

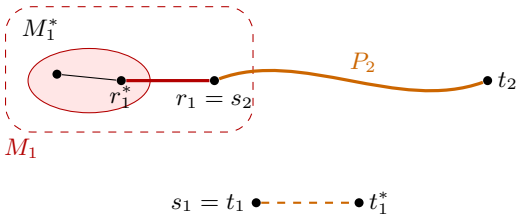


Figure 6: Step 1 of the proof of Lemma 4.2. The root $r_1 = s_2$ is replaced in R by a neighbour r_1^*, and the pair with $s_1 = t_1$ gets a new end t_1^*. Given a model $\mathcal{M}^*$ rooted at R^* and a linkage $\mathcal{P}^*$ disjoint from it, we add r_1 to M_1^* to obtain M_1 (dashed), and replace the path from s_1 to t_1^* (dashed) by the one-vertex path s_1. The model and the linkage then meet exactly in r_1.

Let $Z := R \cup S \cup T$, so $|Z| = a + 2\ell$. Index Z as $z_i := s_i$ $(1 \leqslant i \leqslant \ell)$, $z_{\ell+i} := r_i$ $(1 \leqslant i \leqslant a)$ and $z_{a+\ell+i} := t_i$ $(1 \leqslant i \leqslant \ell)$.

Step 2. Two easy cases. Suppose G has a K_h minor. Since $h \geqslant 2|Z|$, the graph G has a $K_{2|Z|}$ minor. Since $\kappa(G) \geqslant 21a \geqslant |Z|$, Lemma 3.9 shows that G is $(|Z|, 0)$-woven, so G has a $K_{|Z|}$ model $\{B_z : z \in Z\}$ rooted at Z. The sets B_{r_i} form a K_a model rooted at R. For each $i \leqslant \ell$, the sets B_{s_i} and B_{t_i} are adjacent and induce connected subgraphs, so their union contains an s_i–t_i path. These paths are pairwise disjoint and disjoint from the sets B_{r_i}. So G is (a, ℓ_0)-woven. We may therefore assume that G has no K_h minor. If $a \leqslant a_0$, then $h \leqslant \frac{46}{3}a_0 + 4$. Since h is an integer, it follows that $h \leqslant \lfloor \frac{46}{3}a_0 + 4 \rfloor$. The function D is non-decreasing, since a graph with no K_s minor has no K_{s+1} minor. So $\chi(G) \leqslant D(h) \leqslant D(\frac{46}{3}a_0 + 4)$, contrary to the hypothesis. Hence $a > a_0 \geqslant 4 + 5/\eta \geqslant 9$, and so $a \geqslant 10$, since a is an integer.

Step 3. The hub. Let $G_1 := G - Z$. Then $\kappa(G_1) \geqslant 21a - |Z| \geqslant 2a$. This bound is used only in Step 6. Also $\chi(G_1) \geqslant \chi(G) - |Z| \geqslant \lceil 12.5(a+\ell) \rceil + 1$, since $\chi(G) - |Z|$ is an integer, $\chi(G) \geqslant 1 + \Lambda a \geqslant 1 + 897.5a \geqslant 62a + 16$ and $|Z| + 12.5(a+\ell) + 1 \leqslant 62a + 16$. By Lemma 4.1 with $a + \ell$ in place of m, G_1 has an $(a+\ell)$-linked induced subgraph H_0, which we call the *hub*, with $v(H_0) > 10(a + \ell)$ and $\chi(H_0) \leqslant \lceil 12.5(a + \ell) \rceil + 1 \leqslant 12.5(a + \ell) + 2$.

Step 4. Paths to the hub. We have $\kappa(G) \geqslant 21a$ and $2|Z| \leqslant \frac{46}{3}a + 4 \leqslant 21a$. The set $V(H_0)$ is disjoint from Z, since $H_0 \subseteq G_1$, and $v(H_0) > 10(a + \ell) \geqslant 2|Z|$. So Lemma 3.16(a), with $W = Z$, $B = V(H_0)$ and $r = 21a$, gives a double fan $\mathcal{P}_Z$ from Z to $V(H_0)$ in G with $\chi(G[V(\mathcal{P}_Z)]) \leqslant 4|Z|$.

Step 5. The next high-chromatic subgraph. Let $G_2 := G - (Z \cup V(\mathcal{P}_Z) \cup V(H_0))$. Deleting Z, $V(H_0)$ and $V(\mathcal{P}_Z)$ lowers the chromatic number by at most $|Z|$, $\chi(H_0)$ and $4|Z|$,

respectively. So

$$\chi(G_2) \geqslant \chi(G)-5|Z|-12.5(a+\ell)-2 = \chi(G)-(17.5a+22.5\ell+2) \geqslant \chi(G)-(92.5a+17),$$

since $3\ell \leqslant 10a+2$. Since $\chi(G) \geqslant 1+897.5a$, the right side is at least $805a-16 \geqslant \lceil 3.0625(21a-1)\rceil$. By Corollary 3.3, G_2 has a $21a$-connected induced subgraph G_3 with $\chi(G_3) \geqslant \chi(G_2)-42a$.

Step 6. Reduction to G_3. We show that if G_3 is $(a,0)$-woven, then G is (a,ℓ_0)-woven. Since G_1 is $2a$-connected by Step 3, and $V(G_3)$ and $V(H_0)$ are subsets of $V(G_1)$, each with more than $2a$ vertices, Theorem 3.4 gives a family $\mathcal{P}_Y = \{Q_{i,j} : i \in [a],\ j \in \{1,2\}\}$ of $2a$ pairwise disjoint paths from $V(G_3)$ to $V(H_0)$ in G_1. Since $G_3 \subseteq G_2 = G-(Z \cup V(\mathcal{P}_Z) \cup V(H_0))$, the set $V(G_3)$ is disjoint from $V(H_0) \cup V(\mathcal{P}_Z)$. The paths of $\mathcal{P}_Y$ lie in $G_1 = G-Z$. So Lemma 3.16(c) applies, with $\mathcal{P}_Z$ and $\mathcal{P}_Y$ in place of $\mathcal{P}$ and $\mathcal{Q}$, and with $W = Z$, $B = V(H_0)$, $U = V(G_3)$ and $m = a$. It gives $|Z|+a = 2(a+\ell)$ pairwise disjoint paths $P_1,\dots,P_{2(a+\ell)}$ in G, each with its last vertex in $V(H_0)$. For $i \leqslant |Z|$, the path P_i starts at z_i. For $i \leqslant a$, the path $P_{|Z|+i}$ starts at a vertex $v_{|Z|+i}$ of $V(G_3)$. Let $u_i \in V(H_0)$ be the last vertex of P_i. Each P_i meets $Z \cup V(G_3) \cup V(H_0)$ only in its ends. So each P_i meets $V(H_0)$ only in u_i, and meets $Z \cup V(G_3)$ only in its first vertex.

Since G_3 is $(a,0)$-woven, it has a K_a model $\{M'_1,\dots,M'_a\}$ with $v_{|Z|+i} \in M'_i$. Since H_0 is $(a+\ell)$-linked, it contains pairwise disjoint paths $Q_1,\dots,Q_{a+\ell}$ with Q_i joining u_i to $u_{a+\ell+i}$. For $i \leqslant a$, let

$$M_i := V(P_{\ell+i}) \cup V(Q_{\ell+i}) \cup V(P_{|Z|+i}) \cup M'_i.$$

Here $Q_{\ell+i}$ joins $u_{\ell+i}$ to $u_{a+2\ell+i} = u_{|Z|+i}$. The paths $P_1,\dots,P_{2(a+\ell)}$ are pairwise disjoint, and so are the paths $Q_1,\dots,Q_{a+\ell}$. Each Q_j lies in H_0, and each P_k meets $V(H_0)$ only in u_k. Each vertex u_k is an end of exactly one path Q_j. So each Q_j meets $V(P_1) \cup \cdots \cup V(P_{2(a+\ell)})$ only in its ends. Also $V(G_3) \cap V(H_0) = \emptyset$, and each P_k meets $Z \cup V(G_3)$ only in its first vertex. Hence $G[M_i]$ is connected and contains $z_{\ell+i} = r_i$, and the sets M_i are pairwise disjoint. Any two of them are adjacent, because the sets M'_i are. So $\{M_i\}$ is a K_a model rooted at R. For $i \leqslant \ell$, the path P_i joins s_i to u_i, the path Q_i joins u_i to $u_{a+\ell+i}$, and the path $P_{a+\ell+i}$ joins $z_{a+\ell+i} = t_i$ to $u_{a+\ell+i}$. Hence $P_i \cup Q_i \cup P_{a+\ell+i}$ is an s_i–t_i path. These paths are pairwise disjoint and disjoint from every M_j. Hence G is (a,ℓ_0)-woven. It remains to show that G_3 is $(a,0)$-woven.

Step 7a. Using Lemma 3.9 in G_3. If G_3 has a K_{2a} minor, then, since $\kappa(G_3) \geqslant 21a \geqslant a$, Lemma 3.9 shows that G_3 is $(a,0)$-woven, and we are done by Step 6. So we may assume that $\mathrm{had}(G_3) < 2a$.

Step 7b. Three disjoint high-chromatic subgraphs. To show that G_3 is $(a,0)$-woven, fix an arbitrary set $R' = \{r'_1,\dots,r'_a\}$ of a vertices of G_3. The graphs G_4, J_1, J_2, J_3 and J'_1, J'_2, J'_3 defined below depend on R'. Since $\delta(G_3) \geqslant 21a$, we can choose pairwise distinct vertices $s'_1,\dots,s'_{2a} \in V(G_3) \setminus R'$ such that s'_i and s'_{a+i} are neighbours of r'_i. Let $S' := \{s'_1,\dots,s'_{2a}\}$ and $G_4 := G_3-(R' \cup S')$. Then $\chi(G_4) \geqslant \chi(G_3)-3a$, G_4 is an induced subgraph of G, and $\mathrm{had}(G_4) \leqslant \mathrm{had}(G_3) < 2a$ by Step 7a. Let $\sigma := 2a\Phi$. By Step 2, $2a > a > a_0$. Let $\sigma' := \frac{1}{2}(\sigma + \lfloor\sigma\rfloor + 1)$, so that $\sigma < \sigma' < \lfloor\sigma\rfloor + 1$. If G_4 were σ'-chromatic-inseparable, then $(\mathcal{H}_{\Phi,a_0})$, with σ', $2a$ and G_4 in place of s, b and H, would give $\sigma' \leqslant 2a\Phi = \sigma$, which is false. So G_4 has vertex-disjoint induced subgraphs F_1, F_2 with $\chi(F_i) \geqslant \chi(G_4)-\sigma' > \chi(G_4)-\lfloor\sigma\rfloor-1$. Since chromatic numbers are integers, $\chi(F_i) \geqslant \chi(G_4)-\lfloor\sigma\rfloor \geqslant \chi(G_4)-\sigma$. The same argument applied to F_1 gives vertex-disjoint induced subgraphs $F_{1,1}, F_{1,2}$ of F_1 with $\chi(F_{1,j}) \geqslant \chi(F_1)-\sigma$. Let $J_1 := F_{1,1}$, $J_2 := F_{1,2}$ and $J_3 := F_2$. Then $\chi(J_i) \geqslant \chi(G_4)-2\sigma$ for $i \in \{1,2\}$, and $\chi(J_3) \geqslant \chi(G_4)-\sigma$. By Steps 3–5 and the bounds above,

$$\chi(J_i) \geqslant \chi(G)-(92.5a+17)-42a-3a-2\sigma = \chi(G)-(137.5a+17)-4\Phi a$$

for each $i \in \{1,2,3\}$, since $\chi(J_3) \geqslant \chi(G_4) - \sigma \geqslant \chi(G_4) - 2\sigma$. Since $\chi(G) \geqslant 1 + \big(897.5 + 10\Phi\big)a$, the right side is at least $700a \geqslant \lceil 3.0625(21a-1)\rceil$. In particular, these subgraphs are non-null. By Corollary 3.3, each J_i contains a $21a$-connected induced subgraph J_i' with $\chi(J_i') \geqslant \chi(J_i) - 42a$.

Step 8. Accounting. Table 3 lists the losses in chromatic number from G to J_i', for both schemes. Collecting the losses for the uneven scheme, with $\sigma = 2\Phi a$, we obtain

$$\chi(J_i') \;\geqslant\; \chi(G) - L_i, \qquad \text{where} \quad L_i := \begin{cases} 179.5a + 17 + 4\Phi a & \text{if } i \in \{1,2\}, \\ 179.5a + 17 + 2\Phi a & \text{if } i = 3. \end{cases}$$

Let $a_1' = a_2' := \lceil 3a/5 \rceil$ and $a_3' := \lceil 4a/5 \rceil$. Then $a_i' < a$ (as $a \geqslant 10$), $a - a_1' = \lfloor 2a/5 \rfloor \geqslant (2a-4)/5$ and $a - a_3' = \lfloor a/5 \rfloor \geqslant (a-4)/5$. Since $a > a_0 \geqslant 4 + 5/\eta$ by Step 2, we have $\eta(a-4) \geqslant 5$. Hence $(1+\eta)(a-4) \geqslant a+1$ and $(1+\eta)(a-2) \geqslant (1+\eta)(a-4)+2 \geqslant a+3$. Since $\Lambda = \Lambda_{\mathrm{U}} = 5(1+\eta)\big(179.5 + 2\Phi\big)$,

$$\Lambda(a - a_3') \;\geqslant\; (1+\eta)(a-4)\big(179.5+2\Phi\big) \;\geqslant\; (a+1)\big(179.5+2\Phi\big) \;\geqslant\; L_3$$

and

$$\Lambda(a - a_1') \;\geqslant\; 2(1+\eta)(a-2)\big(179.5+2\Phi\big) \;\geqslant\; 2(a+3)\big(179.5+2\Phi\big) \;\geqslant\; L_1 = L_2.$$

Therefore $\chi(J_i') \geqslant D(\frac{46}{3}a_0 + 4) + 1 + \Lambda a_i'$ for each $i \in \{1,2,3\}$. Since J_i' is $21a$-connected, it is $21a_i'$-connected. Also J_i' is an induced subgraph of G, so it satisfies $(\mathcal{H}_{\Phi,a_0})$. So by induction each J_i' is $(a_i', \lceil 10a_i'/3 \rceil)$-woven.

Step 9. Weaving. Let $p := \lfloor 2a/5 \rfloor$, and partition $[a] = I_{12} \cup I_{13} \cup I_{23}$ with $|I_{13}| = |I_{23}| = p$ and $|I_{12}| = a - 2p$. Then

$$|I_{12}| + |I_{13}| \;=\; |I_{12}| + |I_{23}| \;=\; a - p \;=\; a_1' \;=\; a_2' \qquad \text{and} \qquad |I_{13}| + |I_{23}| \;=\; 2p \;\leqslant\; a_3'.$$

For $j \in \{1,2,3\}$, let X_j be a set of a_j' vertices of J_j', which exists since $v(J_j') > 21a$. For each $i \in I_{jk}$ with $j < k$, choose terminals $x_i \in X_j$ and $x_i' \in X_k$, all distinct. This is possible, since the numbers of terminals needed in X_1, X_2 and X_3 are $|I_{12}| + |I_{13}|$, $|I_{12}| + |I_{23}|$ and $|I_{13}| + |I_{23}|$, which are at most a_1', a_2' and a_3' by the display above.

Since G_3 is $21a$-connected and $|R'| = a$, the graph $G_3 - R'$ is $20a$-connected. So $G_3 - R'$ has minimum degree at least $20a$, and hence at least $10a\, v(G_3 - R')$ edges. Hence $G_3 - R'$ is $2a$-linked by Theorem 3.5 with $\ell = 2a$. The $4a$ vertices $s_1', \dots, s_{2a}'$ and x_i, x_i' $(i \in [a])$ are distinct, since $S' \cap V(J_j') = \emptyset$ for each $j \in \{1,2,3\}$, as $J_j' \subseteq G_4$. So $G_3 - R'$ contains a linkage $\mathcal{P}^{(0)}$ consisting of a path from s_i' to x_i and a path from s_{a+i}' to x_i', for each $i \in [a]$. For $j = 1, 2, 3$ in turn, apply Lemma 3.8 in the graph G_3 with J_j', a_j', X_j and $\mathcal{P}^{(j-1)}$ in place of H, a, R and $\mathcal{P}$. Since J_j' is an induced subgraph of G_3, we have $G_3[V(J_j')] = J_j'$, which is $(a_j', \lceil 10a_j'/3 \rceil)$-woven by Step 8. Since $a_j' \geqslant 3a/5$, we have $\lceil 10a_j'/3 \rceil \geqslant 2a$, which is the number of pairs of $\mathcal{P}^{(j-1)}$. So Lemma 3.8 applies, with $c = \lceil 10a_j'/3 \rceil$. We obtain a linkage $\mathcal{P}^{(j)}$ for the same pairs and a $K_{a_j'}$ model $\mathcal{M}_j$ in J_j' rooted at X_j, with

$$V(\mathcal{P}^{(j)}) \cap V(\mathcal{M}_j) \subseteq X_j \qquad \text{and} \qquad V(\mathcal{P}^{(j)}) \subseteq V(J_j') \cup V(\mathcal{P}^{(j-1)}).$$

Let $\mathcal{P}'' := \mathcal{P}^{(3)}$. For $j < j'$, induction on j' gives $V(\mathcal{P}^{(j')}) \subseteq V(\mathcal{P}^{(j)}) \cup V(J_{j+1}') \cup \cdots \cup V(J_{j'}')$, and the graphs $J_{j+1}', \dots, J_{j'}'$ are disjoint from $\mathcal{M}_j$. Hence $V(\mathcal{P}'') \cap V(\mathcal{M}_j) \subseteq X_j$ for each $j \in \{1,2,3\}$. The case $j = 0$ gives $V(\mathcal{P}'') \subseteq V(\mathcal{P}^{(0)}) \cup V(J_1') \cup V(J_2') \cup V(J_3')$. The linkage $\mathcal{P}^{(0)}$ lies in $G_3 - R'$, and each J_j' is a subgraph of $G_4 = G_3 - (R' \cup S')$. So $r_i' \notin V(\mathcal{P}'')$ for each $i \in [a]$.

For $i \in [a]$, let P''_i and P''_{a+i} be the paths of $\mathcal{P}''$ from s'_i to x_i and from s'_{a+i} to x'_i. For $i \in I_{jk}$, let C_i be the branch set of $\mathcal{M}_j$ containing x_i, let C'_i be the branch set of $\mathcal{M}_k$ containing x'_i, and let $B_i := \{r'_i\} \cup V(P''_i) \cup V(P''_{a+i}) \cup C_i \cup C'_i$. Each $G[B_i]$ is connected, because r'_i is adjacent to s'_i and s'_{a+i}. The sets B_i are pairwise disjoint since:

- The paths of $\mathcal{P}''$ are pairwise disjoint.
- The models $\mathcal{M}_j$ lie in the disjoint graphs J'_j. Distinct vertices of X_j lie in distinct branch sets of $\mathcal{M}_j$, since $\mathcal{M}_j$ is rooted at X_j.
- The paths meet the models only in vertices of the sets X_j. Each branch set of $\mathcal{M}_j$ contains exactly one vertex of X_j. So, for each $i' \in [a]$, every path of $\mathcal{P}''$ meets $C_{i'}$ only in $x_{i'}$, and meets $C'_{i'}$ only in $x'_{i'}$. Since the paths of $\mathcal{P}''$ are pairwise disjoint, $P''_{i'}$ is the only path of $\mathcal{P}''$ that meets $C_{i'}$, and $P''_{a+i'}$ is the only path of $\mathcal{P}''$ that meets $C'_{i'}$.
- $r'_i \notin V(\mathcal{P}'')$, and $r'_i \notin V(J'_1) \cup V(J'_2) \cup V(J'_3)$, since each J'_j is a subgraph of $G_4 = G_3 - (R' \cup S')$.

Any two of the pairs $\{1,2\},\{1,3\},\{2,3\}$ intersect. So any two of the sets B_i contain distinct branch sets of a common model $\mathcal{M}_j$, and are therefore adjacent. Hence $\{B_i\}$ is a K_a model rooted at R', and G_3 is $(a,0)$-woven. This completes the proof for the uneven scheme.

The even scheme. Suppose now that $\Lambda = \Lambda_{\mathrm{E}}$. The proof is the same, with $\ell_0 := 3a$ and $h := 14a$, so that $\ell \leqslant 3a$, and with the following changes in the numbers. In Step 1, $|Z| \leqslant 7a$, so $2(a+2\ell) \leqslant 14a \leqslant 21a$. In Step 2, if $a \leqslant a_0$ then $\chi(G) \leqslant D(14a) \leqslant D(\frac{46}{3}a_0+4)$, and so $a > a_0 \geqslant 4 + 5/\eta \geqslant 9$ and $a \geqslant 10$. In Step 3, $\chi(G_1) \geqslant \chi(G) - 7a \geqslant 50a+1$, since $\chi(G) \geqslant 1 + \Lambda a \geqslant 1 + 517.5a$. The hub H_0 is $(a+\ell)$-linked, where $a+\ell \leqslant 4a$, with $v(H_0) > 10(a+\ell)$ and $\chi(H_0) \leqslant 12.5(a+\ell)+2 \leqslant 50a+2$. In Step 4, $2|Z| \leqslant 14a$, and the colouring bound of Lemma 3.16(a) is $4|Z| \leqslant 28a$. In Step 5, $\chi(G_2) \geqslant \chi(G) - (85a+2) \geqslant 432.5a - 1 \geqslant \lceil 3.0625(21a-1) \rceil$. In Step 7b, $\sigma = 2a\Phi$ as before, and each of J_1, J_2, J_3 satisfies

$$\chi(J_i) \;\geqslant\; \chi(G_4) - 2\sigma \;\geqslant\; \chi(G) - (130a+2) - 4\Phi a \;\geqslant\; 300a \;\geqslant\; \lceil 3.0625(21a-1) \rceil.$$

In Step 8, $\chi(J'_i) \geqslant \chi(G) - (172a+2) - 4\Phi a$ for each $i \in \{1,2,3\}$, and we take $a'_1 = a'_2 = a'_3 := \lceil 2a/3 \rceil$. Then $a - a'_i \geqslant (a-2)/3$. As in the uneven scheme, $(1+\eta)(a-2) \geqslant a+3$, and so

$$\Lambda_{\mathrm{E}}(a-a'_i) \;\geqslant\; (1+\eta)(a-2)\big(172.5+4\Phi\big) \;\geqslant\; (a+3)\big(172.5+4\Phi\big) \;\geqslant\; 172a+2+4\Phi a.$$

Hence $\chi(J'_i) \geqslant D(\frac{46}{3}a_0+4) + 1 + \Lambda_{\mathrm{E}} a'_i$, and by induction each J'_i is $(a'_i, 3a'_i)$-woven. In Step 9, let I_{12}, I_{13}, I_{23} have sizes (n,n,n), $(n+1,n,n)$ or $(n+1,n+1,n)$ according as $a = 3n$, $3n+1$ or $3n+2$. Then each of the three sums $|I_{12}| + |I_{13}|$, $|I_{12}| + |I_{23}|$ and $|I_{13}| + |I_{23}|$ is at most $\lceil 2a/3 \rceil$. Lemma 3.8 applies to each J'_j with $c = 3a'_j$, since J'_j is $(a'_j, 3a'_j)$-woven and $3a'_j \geqslant 2a$. The rest of the proof is unchanged. □

Comparison with earlier proofs. The proof of Lemma 4.2 follows the proof of a theorem of Delcourt and Postle [8, Theorem 7.1], and its adaptation by Norin and Steiner [38, Lemma B.12]. Hypothesis $(\mathcal{H}_{\Phi,a_0})$ replaces the bound of Delcourt and Postle on small subgraphs and the hypothesis of Norin and Steiner on chromatic inseparability. The main differences are as follows. First, we construct the hub by Lemma 4.1, whereas Delcourt and Postle obtain it from their Theorem 2.3, and Norin and Steiner from [38, Theorem B.1] and the theorem of Girão and Narayanan [38, Theorem B.3]. Second, Step 7a applies Lemma 3.9 to G_3, which Delcourt and Postle [8, Section 7.1] observe would suffice if only $(a,0)$-wovenness were needed. So Step 7b separates graphs with no K_{2a} minor, whereas these proofs separate graphs with no K_{14a} minor. Third, we use Corollary 3.3 in place

of the theorem of Girão and Narayanan in Steps 5 and 7b. Fourth, the uneven scheme is new, and the even scheme is the distribution of Delcourt and Postle. Finally, Step 6 groups the paths into pairs $\{Q_{i,1}, Q_{i,2}\}$, since Lemma 3.7 requires two paths at each start vertex.

Operation	Uneven scheme	Even scheme
Delete Z and the paths $\mathcal{P}_Z$ (Steps 3 and 5)	$5\lvert Z\rvert = 5a + 10\ell$	$35a$
Delete the hub H_0 (Steps 3 and 5)	$12.5(a+\ell)+2$	$50a+2$
Subtotal (Step 5)	$17.5a + 22.5\ell + 2 \leqslant 92.5a + 17$	$85a+2$
Extract G_3 (Step 5)	$42a$	$42a$
Delete $R' \cup S'$ (Step 7b)	$3a$	$3a$
Chromatic separations (Step 7b)	2σ, 2σ, σ for J_1, J_2, J_3	2σ for each J_i
Extract J_i' (Step 7b)	$42a$	$42a$
Total loss (Step 8)	$179.5a + 17 + 2\sigma$ $(i = 1, 2)$ $179.5a + 17 + \sigma$ $(i = 3)$	$172a + 2 + 2\sigma$
where	$\sigma = 2a\Phi$	$\sigma = 2a\Phi$
Sizes a_1', a_2', a_3'	$\lceil 3a/5\rceil, \lceil 3a/5\rceil, \lceil 4a/5\rceil$	$\lceil 2a/3\rceil$ each

Table 3: Upper bounds on the losses in chromatic number in the proof of Lemma 4.2, where ℓ is the number of terminal pairs, with $\ell \leqslant \lceil 10a/3\rceil \leqslant (10a+2)/3$ in the uneven scheme and $\ell \leqslant 3a$ in the even scheme, and $\lvert Z\rvert = a + 2\ell$.

Given the chromatic separations of Step 7b, the uneven scheme minimises the coefficient of Φ. Suppose that J_1', J_2', J_3' host fractions f_1, f_2, f_3 of the a branch sets in Step 9. Each branch set uses two of these subgraphs, so $f_1 + f_2 + f_3 = 2$. Ignoring rounding, the coefficient of Φ in $\Lambda/(1+\eta)$ is $2\max\{2/(1-f_1), 2/(1-f_2), 1/(1-f_3)\}$, since J_1', J_2' are obtained by two separations of cost $2a\Phi$ and J_3' by one. This is minimised, with value 10, at $f_1 = f_2 = \frac{3}{5}$ and $f_3 = \frac{4}{5}$. Delcourt and Postle use $f_i = \frac{2}{3}$, which gives 12.

Theorem 4.3. *Let $t \geqslant 2$ be an integer, and let $\eta \in (0,1]$, $a_0 \geqslant 4 + 5/\eta$ and $\Phi \geqslant 0$ be real. Let $c_{4.3}(\Phi) := 42 + \min\{517.5 + 12\Phi,\ 897.5 + 10\Phi\}$. Let $\Lambda = \min\{\Lambda_{\mathrm{E}}, \Lambda_{\mathrm{U}}\}$ be as in Lemma 4.2. If G satisfies $\mathrm{had}(G) < t$ and $(\mathcal{H}_{\Phi,a_0})$, then*

$$\chi(G) \ < \ D\big(\tfrac{46}{3}a_0 + 4\big) + 1 + (42 + \Lambda)\, t,$$

where $42 + \Lambda = c_{4.3}(\Phi) + \eta\big(c_{4.3}(\Phi) - 42\big)$.

Theorem 4.3 parallels a theorem of Norin and Steiner [38, Theorem 15 and Section B.3], and the derivation by Delcourt and Postle of their Theorem 1.6 from their Theorem 7.1 [8].

Proof. Suppose for a contradiction that $\chi(G)$ is at least the right side of the displayed bound. Then $\chi(G) \geqslant 42t + \Lambda t \geqslant \lceil 3.0625(21t-1)\rceil$, since $D(\frac{46}{3}a_0 + 4) \geqslant 0$ and $\Lambda \geqslant 517.5$. By Corollary 3.3, G has a $21t$-connected induced subgraph G' with $\chi(G') \geqslant \chi(G) - 42t \geqslant D(\frac{46}{3}a_0 + 4) + 1 + \Lambda t$. The graph G' satisfies $(\mathcal{H}_{\Phi,a_0})$, since it is an induced subgraph of G. By Lemma 4.2 with $a = t$, G' is $(t,0)$-woven. Since $v(G') > 21t \geqslant t$, we may take any t vertices of G' as roots, so G' has a K_t minor, contradicting $\mathrm{had}(G) < t$. □

5 Colouring graphs of order $O(a \log^2 a)$

This section proves Corollary 5.7, which states that, for every integer $a \geqslant 3$, every graph with no K_a minor and at most N_a vertices is $c_{5.5}a$-colourable, where $N_a \in O(a \log^2 a)$ is defined in Proposition 5.5. It is used in Section 9. As in Parts (I) and (II) of the proof of Norin and Steiner (Section 2.2), we start from an explicit base case (Section 5.1)

and apply bootstrapping steps (Sections 5.2 and 5.3). Unlike in Part (II), the Gyárfás graphs are coloured with a fixed earlier result. So each step maps a constant c to $2c$ plus a term that does not depend on c (see the remark after Lemma 5.4). We call such a step *additive*, since the cost $c_{4.3}(\Phi)$ of the chromatic-separable reduction is added to $2c$, rather than multiplied by the previous constant as in Part (II). The base case (Lemma 5.3) is an explicit instance of the small-graph bound of Norin and Steiner (Section 3.4), with base constant 4.01 and exponent 0.49, derived from a quantitative theorem of Norin and Steiner (Theorem 5.1). Each bootstrapping step adds an error term $\xi = 10^{-4}$ to the constant (see Lemma 5.4). Section 5.4 treats the values of a with $\log\log a < T_{\mathrm{boot}}$, where the threshold T_{boot} is defined in Proposition 5.5.

5.1 The base case

This subsection makes the small-graph bound of Norin and Steiner (Section 3.4) explicit, for the constant 4.01 and the exponent 0.49 used below. We derive it from the following quantitative theorem. It is Theorem 2 of an earlier draft [39] of the paper of Norin and Steiner [38], and it is formalised in `LinearHadwiger` (see Appendix B). Let $T_{\mathrm{base}} := 1160$.

Theorem 5.1 (Norin and Steiner [39, Theorem 2])**.** *For every graph G and every real $\varepsilon \in (0, 1]$,*

$$\chi(G) \;\leqslant\; 4\,\mathrm{had}(G) + \varepsilon\, v(G) + A_\varepsilon, \qquad \text{where } A_\varepsilon := (100/\varepsilon)^{2000/\varepsilon^2}.$$

The next lemma bounds the additive term A_ε for the value of ε used in Lemma 5.3.

Lemma 5.2. *Let a be real with $y := \log\log a \geqslant T_{\mathrm{base}}$, and let $\varepsilon := 0.0099\,(\log a)^{-0.49}$. Then $0 < \varepsilon \leqslant 1$ and $A_\varepsilon \leqslant 10^{-4}a$.*

Proof. Since $\log a = e^y \geqslant 1$, we have $0 < \varepsilon \leqslant 0.0099$. Let $K := 2000/0.0099^2$ and $U := e^{0.98y} \geqslant 1$. Then $2000/\varepsilon^2 = KU$, $\log(100/\varepsilon) = \log(100/0.0099) + 0.49y$ and $\log a = e^y = Ue^{0.02y}$. Since $100/0.0099$ and 10^4 are less than $2.7^{10} < e^{10}$, we have $\log(100/0.0099) < 10$ and $\log 10^4 < 10$. So $\log A_\varepsilon = KU\log(100/\varepsilon) < KU(10 + 0.49y)$. Since $\log 10^4 < 10 \leqslant 10U$, it suffices to show that $KU(10 + 0.49y) + 10U \leqslant Ue^{0.02y}$, that is,

$$K(10 + 0.49y) + 10 \;\leqslant\; e^{0.02y}.$$

Since $e^z \geqslant 1 + z$ for every real z, we have $e^{0.02y} = e^{23.2}e^{0.02y-23.2} \geqslant e^{23.2}(0.02y - 22.2)$, where $0.02y - 22.2 \geqslant 1$. Since $e > 2.7182818283$ and $e^{0.2} \geqslant 1 + 0.2 + 0.2^2/2 = 1.22$, we have $e^{23.2} = e^{23}e^{0.2} \geqslant 2.7182818283^{23} \cdot 1.22 > 1.188 \cdot 10^{10}$. So it suffices to show that $f(y) := 1.188 \cdot 10^{10}(0.02y - 22.2) - K(10 + 0.49y) - 10$ is non-negative for $y \geqslant 1160$. Since $1.188 \cdot 10^{10}$ exceeds both $24.5K$ and $578.4K + 10$, the function f is linear with positive slope $0.02 \cdot 1.188 \cdot 10^{10} - 0.49K$, and $f(1160) = 1.188 \cdot 10^{10} - 578.4K - 10 > 0$. So the inequality holds for every $y \geqslant T_{\mathrm{base}}$. Hence $\log A_\varepsilon < \log a - \log 10^4$, that is, $A_\varepsilon < 10^{-4}a$. □

Lemma 5.3. *For every real a with $\log\log a \geqslant T_{\mathrm{base}}$, every graph J with $\mathrm{had}(J) < a$ and $v(J) \leqslant a(\log a)^{0.49}$ satisfies $\chi(J) \leqslant 4.01\,a$.*

Proof. Let $\varepsilon := 0.0099\,(\log a)^{-0.49}$. By Lemma 5.2, $0 < \varepsilon \leqslant 1$ and $A_\varepsilon \leqslant 10^{-4}a$. Also $4\,\mathrm{had}(J) < 4a$ and $\varepsilon\, v(J) \leqslant 0.0099\,a$. So Theorem 5.1 gives $\chi(J) \leqslant 4a + 0.0099\,a + 10^{-4}a = 4.01\,a$. □

With $\varepsilon = 0.0099\,(\log a)^{-0.49}$, the bound $A_\varepsilon \leqslant 10^{-4}a$ holds if and only if $\log\log a$ is at least about 1159.49. For $\log\log a$ below about 1158.5, no choice of ε makes $4a + \varepsilon a(\log a)^{0.49} + A_\varepsilon$ at most $4.01\,a$. So this argument cannot lower T_{base} by much.

5.2 A bootstrapping step

The following lemma is an explicit form of a theorem of Norin and Steiner [38, Theorem 5]. Hypothesis (a) is the bound that the lemma extends from exponent β to exponent $\beta+\zeta$, and hypothesis (b) is the bound used for the Gyárfás graphs. The conditions $(\beta_{\mathrm{G}}-2\zeta)T \geqslant 1$, $\zeta T \geqslant 5$ and $\beta_{\mathrm{G}} \leqslant 1$ ensure that the Gyárfás graphs are small enough for (b), and the conditions $T \geqslant 1000.9$ and $\Phi \leqslant 10^4$ ensure that the error terms are at most ξt. The proof needs only $T \geqslant 100$. We state the lemma with $T \geqslant 1000.9$ because this is the value in the formalisation, and every application of the lemma has $T \geqslant 1160.9$.

Lemma 5.4. *Let $\xi := 10^{-4}$. Let β be real, and let $T \geqslant 1000.9$, $\zeta > 0$, $\beta_{\mathrm{G}} \leqslant 1$, $c \geqslant 0$ and $0 \leqslant \Phi \leqslant 10^4$ be real such that $(\beta_{\mathrm{G}}-2\zeta)T \geqslant 1$ and $\zeta T \geqslant 5$. Suppose that:*

(a) For every real t with $\log\log t \geqslant T$, every graph G with $\mathrm{had}(G) < t$ and $v(G) \leqslant t(\log t)^{\beta}$ satisfies $\chi(G) \leqslant ct$.

(b) For every real a with $\log\log a \geqslant T - 0.1$, every graph J with $\mathrm{had}(J) < a$ and $v(J) \leqslant a(\log a)^{\beta_{\mathrm{G}}}$ satisfies $\chi(J) \leqslant \Phi a$.

Then, for every real t with $\log\log t \geqslant T$, every graph G with $\mathrm{had}(G) < t$ and $v(G) \leqslant t(\log t)^{\beta+\zeta}$ satisfies $\chi(G) \leqslant (2c + c_{4.3}(\Phi) + \xi)t$. In particular, if $\Phi = 4.01$, then $\chi(G) \leqslant (2c + c_{5.4})t$, where $c_{5.4} := c_{4.3}(4.01) + \xi = 42 + 517.5 + 12 \cdot 4.01 + 10^{-4} = 607.6201$.

In the corresponding step of Norin and Steiner (Part (II) in Section 2.2), which deduces $\mathrm{LH}(\frac{4}{3}\beta)$ from $\mathrm{LH}(\beta)$, the Gyárfás graphs are coloured using the hypothesis $\mathrm{LH}(\beta)$ itself. That is, they are coloured with the constant C_β obtained in the previous bootstrapping step. The constant C_β then enters the hypothesis of the chromatic-separable reduction, and the new constant $4C_\beta + 2C_{\mathrm{sep}}(1 + C_\beta)$ of Part (II) contains the product $C_{\mathrm{sep}}C_\beta$. In Lemma 5.4 the Gyárfás graphs are coloured by hypothesis (b), with a fixed constant Φ such as the constant 4.01 of Lemma 5.3. So the term $c_{4.3}(\Phi)$ does not depend on c. The factor 2 in $2c + c_{4.3}(\Phi)$ comes from colouring G_1 through its contraction G_1' in the proof of Lemma 5.4.

Proof. We follow the proof of Theorem 5 by Norin and Steiner [38]. Apart from the change explained above, we verify $(\mathcal{H}_{\Phi,a_0})$ with $a_0 = t/\log t$, that is, for every $b \geqslant t/\log t$ and not only for $b \leqslant t$. We also use Theorem 4.3 in place of their Theorem 15, split G as in [39, Lemma 12], and pass to real t by rounding t up. Let t be real with $y := \log\log t \geqslant T$, let G be a graph with $\mathrm{had}(G) < t$ and $v(G) \leqslant t(\log t)^{\beta+\zeta}$, let $p := (\log t)^{\zeta}$ and let $k := \lceil p \rceil$. Let $L := \log t = e^y$ and $a_0 := t/L$. Since $T > 0$ and $(\beta_{\mathrm{G}} - 2\zeta)T \geqslant 1$, we have $\beta_{\mathrm{G}} > 2\zeta > 0$. The condition $T \geqslant 1000.9$ is used only through its consequence $y \geqslant 100$, which gives $L \geqslant e^{100} \geqslant 2^{100} \geqslant 10^{30}$, and $11y \leqslant e^y = L$, since $e^y \geqslant y^2/2 \geqslant 11y$ for $y \geqslant 22$. Since $t = e^L \geqslant L^3/6$ and $L^2 \geqslant 10^{60}$, we also have $t \geqslant a_0 \geqslant L^2/6 \geqslant 10^{59}$. Since $\zeta y \geqslant \zeta T \geqslant 5$, we have $p = e^{\zeta y} \geqslant e^5 \geqslant 32$. So $k \geqslant 2$ and $k(k-1) < (p+1)p \leqslant \frac{33}{32}p^2$. Choose a maximal family $H_1, \ldots, H_m$ of pairwise vertex-disjoint connected induced bipartite subgraphs of G, each with exactly k vertices. Let $G_1 := G[V(H_1) \cup \cdots \cup V(H_m)]$ and $G_2 := G - V(G_1)$. By maximality, G_2 has no connected induced bipartite subgraph with exactly k vertices, so $\mathrm{bip}(G_2) < k$ (see Section 3.1).

(i) *Colouring G_1.* The sets $V(H_i)$ partition $V(G_1)$ into sets inducing connected subgraphs of chromatic number at most 2. Let G_1' be obtained from G_1 by contracting each H_i to a vertex. Then G_1' is a minor of G, so $\mathrm{had}(G_1') < t$. Also $v(G_1') = m \leqslant \lfloor v(G)/k \rfloor \leqslant v(G)/p \leqslant t(\log t)^{\beta}$. By (a), G_1' has a colouring ψ with at most ct colours. For each i, let ϕ_i be a proper 2-colouring of H_i. For each i, let z_i be the vertex of G_1' obtained by contracting $V(H_i)$, and give each vertex $v \in V(H_i)$ the pair $(\psi(z_i), \phi_i(v))$. Adjacent vertices in the same H_i differ in the second coordinate. If $v \in V(H_i)$ and $v' \in V(H_j)$ are adjacent with $i \neq j$, then $z_iz_j \in E(G_1')$, so $\psi(z_i) \neq \psi(z_j)$, and the pairs of v and v' differ in the first coordinate. Hence

$\chi(G_1) \leqslant 2\chi(G_1') \leqslant 2ct$. The colouring is a special case of a lemma of Norin and Steiner [38, Lemma 13], who credit the argument to Reed and Seymour [42] and to Steiner [46].

(ii) *The Gyárfás graphs.* Let H be an s-chromatic-inseparable induced subgraph of G_2 with $\mathrm{had}(H) < b$ for some real $b \geqslant a_0$, and let $b' := \min\{b, t\}$. Then $\mathrm{had}(H) < b'$, as $\mathrm{had}(H) \leqslant \mathrm{had}(G) < t$. Since $b' \geqslant a_0$ and $11y \leqslant L$, we have $\log b' \geqslant \log a_0 = L - y \geqslant \frac{10}{11}L$. As $\log \frac{11}{10} < 0.1$, it follows that $\log\log b' \geqslant y - 0.1 \geqslant T - 0.1$. So (b) applies with b' in place of a. Since $y \geqslant T$ and $\beta_G - 2\zeta > 0$, we have $L^{\beta_G - 2\zeta} = e^{(\beta_G - 2\zeta)y} \geqslant e^{(\beta_G - 2\zeta)T} \geqslant e > 2.7$, that is, $L^{\beta_G} \geqslant 2.7\, p^2$. Since $0 < \beta_G \leqslant 1$, we have $(\frac{10}{11})^{\beta_G} \geqslant \frac{10}{11}$, and so

$$(\log b')^{\beta_G} \;\geqslant\; \big(\tfrac{10}{11}L\big)^{\beta_G} \;\geqslant\; \tfrac{10}{11} \cdot 2.7\, p^2 \;>\; 2 \cdot \tfrac{33}{32}\, p^2 \;>\; 2k(k-1).$$

Consider Lemma 3.15 with b' and $\Phi b' \geqslant 0$ in place of b and Γ. Every Gyárfás graph J in its hypothesis, that is, every graph J with $\mathrm{had}(J) < b'$ and $v(J) \leqslant 2b'k(k-1)$, satisfies $v(J) \leqslant b'(\log b')^{\beta_G}$, and so $\chi(J) \leqslant \Phi b'$ by (b) with b' in place of a. Since H is an induced subgraph of G_2, we have $\mathrm{bip}(H) \leqslant \mathrm{bip}(G_2) < k$. Also H is s-chromatic-inseparable and $\mathrm{had}(H) < b'$. So Lemma 3.15 gives $s < \Phi b' \leqslant \Phi b$. So G_2 satisfies $(\mathcal{H}_{\Phi,a_0})$.

(iii) *The chromatic-separable reduction.* Apply Theorem 4.3 to G_2, with $t' := \lceil t \rceil$ in place of t, $a_0 = t/\log t$ and $\eta := 5/(a_0 - 4)$. Then $t' \geqslant 2$, $a_0 = 4 + 5/\eta$ and $0 < \eta \leqslant 10/a_0 \leqslant 10^{-58}$, as $a_0 \geqslant 10^{59}$. This gives $\chi(G_2) < D(\frac{46}{3}a_0 + 4) + 1 + (42 + \Lambda)t'$, where $42 + \Lambda = c_{4.3}(\Phi) + \eta c_0$ with $c_0 := c_{4.3}(\Phi) - 42 \geqslant 0$. Since $t' \leqslant t + 1$, it follows that $\chi(G_2) < c_{4.3}(\Phi)\, t + R$, where

$$R \;:=\; D\big(\tfrac{46}{3}a_0 + 4\big) + 1 + c_{4.3}(\Phi) + \eta c_0(t+1).$$

Let $s_0 := \lfloor \frac{46}{3}a_0 + 4 \rfloor \leqslant 16a_0$. By Theorem 3.11, $D(\frac{46}{3}a_0 + 4) + 1 \leqslant 60\, s_0\sqrt{\log s_0} + 2 \leqslant 960\, a_0\sqrt{\log(16t)} + 2$. Since $\log(16t) = L + \log 16 \leqslant 2L$, $\sqrt{2L} \leqslant 10^{-12}L$ (as $L \geqslant 10^{30}$) and $a_0 L = t$, this is at most $960 \cdot 10^{-12}\, t + 2 \leqslant 10^{-7}t$, as $t \geqslant 10^{59}$. Since $0 \leqslant \Phi \leqslant 10^4$, we have $c_{4.3}(\Phi) \leqslant 939.5 + 10\Phi < 10^6$ and $c_0 \leqslant 897.5 + 10\Phi < 10^6$. Hence, as $\eta \leqslant 10^{-58}$ and $t \geqslant 10^{59}$,

$$R \;\leqslant\; 10^{-7}t + 10^6 + 10^{-52}(t+1) \;\leqslant\; 10^{-7}t + 2 \cdot 10^{-52}\, t \;\leqslant\; \xi t.$$

So $\chi(G_2) < (c_{4.3}(\Phi) + \xi)t$.

Hence $\chi(G) \leqslant \chi(G_1) + \chi(G_2) \leqslant (2c + c_{4.3}(\Phi) + \xi)t$. □

5.3 The schedule

We now combine Lemmas 5.3 and 5.4.

Proposition 5.5. *Let* $c_{5.5} := 45{,}024.9873$. *Let* $N_a := 24\tau_{7.1}M^2\, a \log^2 a$, *where* $\tau_{7.1} = 4.56 \cdot 10^{18}$ *and* $M = 10^5$ *are as in Theorem 7.1 and Lemma 6.4 below. Let* $T_{\mathrm{boot}} := 1161$. *For every real* a *with* $\log\log a \geqslant T_{\mathrm{boot}}$, *every graph* J *with* $\mathrm{had}(J) < a$ *and* $v(J) \leqslant N_a$ *satisfies* $\chi(J) \leqslant c_{5.5}a$.

Proof. Every application of Lemma 5.4 in this proof has $T \in \{T_{\mathrm{boot}} - 0.1, T_{\mathrm{boot}}\}$, so $T \geqslant T_{\mathrm{boot}} - 0.1 \geqslant 1000.9$, and $\Phi \in \{4.01, c_{\mathrm{side}}\}$, so $0 \leqslant \Phi \leqslant 10^4$.

Let $\beta_0 := 0.49$, the exponent of Lemma 5.3, and let $\vartheta := 0.0005$. Let $\zeta := \beta_0/2 - \vartheta = 0.2445$ and $\beta_j := \beta_0 + j\zeta$ for $j \geqslant 0$, so that $\beta_1 = 0.7345$ and $\beta_5 = 1.7125$. Let $\zeta_{\mathrm{f}} := \beta_1/2 - 1.1\vartheta = 0.3667$ and $\beta_{\mathrm{fin}} := \beta_5 + \zeta_{\mathrm{f}} = 4.25\beta_0 - 6.6\vartheta = 2.0792$. In each application below, (ζ, β_G) is $(0.2445, 0.49)$ or $(0.3667, 0.7345)$. In both cases $\beta_G \leqslant 1$, and

$\beta_G - 2\zeta$ is $2\vartheta = 0.001$ or $2.2\vartheta = 0.0011$, respectively. Since $T \geqslant 1000.9$, it follows that $(\beta_G - 2\zeta)T \geqslant 0.001 \cdot 1000.9 > 1$ and $\zeta T \geqslant 0.2445 \cdot 1000.9 > 5$. Every constant c below is positive. So the conditions of Lemma 5.4 on T, ζ, β_G, c and Φ hold in every application.

The side step. Apply Lemma 5.4 with $T = T_{\text{boot}} - 0.1 = 1160.9$, $\beta = \beta_G = \beta_0$, $c = \Phi = 4.01$ and $\zeta = 0.2445$. Hypotheses (a) and (b) hold by Lemma 5.3, since $T - 0.1 = 1160.8 \geqslant T_{\text{base}}$. Lemma 5.4 gives the constant $c_{\text{side}} := 2 \cdot 4.01 + c_{5.4} = 615.6401$ for exponent $\beta_1 = 0.7345$. That is, for every real a with $\log\log a \geqslant T_{\text{boot}} - 0.1$, every graph J with $\text{had}(J) < a$ and $v(J) \leqslant a(\log a)^{0.7345}$ satisfies $\chi(J) \leqslant c_{\text{side}} a$. The constant c_{side} equals the constant $\bar{c}_1$ of the first main step below. The side step differs from the first main step only in its size threshold. It is needed because hypothesis (b) of the final step requires the bound for every real a with $\log\log a \geqslant T_{\text{boot}} - 0.1$, whereas the main steps give it only when $\log\log a \geqslant T_{\text{boot}}$.

The main steps. Let $\bar{c}_0 := 4.01$ and $\bar{c}_j := 2\bar{c}_{j-1} + c_{5.4}$ for $j \geqslant 1$. For $j = 1, \dots, 5$ in turn, apply Lemma 5.4 with $T = T_{\text{boot}}$, $\beta = \beta_{j-1}$, $\beta_G = \beta_0$, $c = \bar{c}_{j-1}$, $\Phi = 4.01$ and $\zeta = 0.2445$. Hypothesis (a) holds by Lemma 5.3 for $j = 1$ and by the previous application for $j \geqslant 2$, and (b) holds by Lemma 5.3. This gives the constant $\bar{c}_5 = 2^5 \cdot 4.01 + 31\, c_{5.4} = 18{,}964.5431$ for exponent $\beta_5 = 1.7125$ and size threshold T_{boot}.

The final step. Apply Lemma 5.4 with $T = T_{\text{boot}}$, $\beta = \beta_5$, $\zeta = \zeta_f$, $\beta_G = \beta_1$, $c = \bar{c}_5$ and $\Phi = c_{\text{side}}$. Hypothesis (a) is the result of the fifth main step, and (b) is the result of the side step, since $T - 0.1 = T_{\text{boot}} - 0.1$. Since $c_{\text{side}} > 190$, we have $c_{4.3}(c_{\text{side}}) = 42 + 897.5 + 10 c_{\text{side}} = 7095.901$. So the constant obtained is $2\bar{c}_5 + c_{4.3}(c_{\text{side}}) + 10^{-4} = c_{5.5}$, for exponent $\beta_{\text{fin}} = 2.0792$. Hence every graph J with $\text{had}(J) < a$ and $v(J) \leqslant a(\log a)^{2.0792}$ satisfies $\chi(J) \leqslant c_{5.5} a$, whenever $\log\log a \geqslant T_{\text{boot}}$.

Finally, $N_a \leqslant a(\log a)^{2.0792}$ whenever $(\log a)^{0.0792} \geqslant 24\tau_{7.1} M^2$. Since $24\tau_{7.1} M^2 \leqslant 2.7^{79} < e^{79}$, this holds whenever $0.0792 \log\log a \geqslant 79$, and $0.0792\, T_{\text{boot}} \geqslant 79$. So the proposition follows. □

Step	Exponent	Gyárfás graphs	Scheme	Constant	Threshold
Base case	0.49	–	–	4.01	$T_{\text{base}} = 1160$
Side step	$0.49 \to 0.7345$	base case	even	$c_{\text{side}} = 615.6401$	1160.9
Main step 1	$0.49 \to 0.7345$	base case	even	$\bar{c}_1 = 615.6401$	$T_{\text{boot}} = 1161$
Main step 2	$0.7345 \to 0.979$	base case	even	$\bar{c}_2 = 1838.9003$	T_{boot}
Main step 3	$0.979 \to 1.2235$	base case	even	$\bar{c}_3 = 4285.4207$	T_{boot}
Main step 4	$1.2235 \to 1.468$	base case	even	$\bar{c}_4 = 9178.4615$	T_{boot}
Main step 5	$1.468 \to 1.7125$	base case	even	$\bar{c}_5 = 18{,}964.5431$	T_{boot}
Final step	$1.7125 \to 2.0792$	side step	uneven	$c_{5.5} = 45{,}024.9873$	T_{boot}

Table 4: The schedule in the proof of Proposition 5.5.

Table 4 summarises the schedule. For each step, it lists the exponents before and after the step, the result used to colour the Gyárfás graphs, the scheme of Lemma 4.2 (which is even if and only if $\Phi \leqslant 190$), the constant obtained, and the size threshold, which is a lower bound on $\log\log a$. The final exponent β_{fin} must exceed 2, since the inequality $N_a \leqslant a(\log a)^{\beta_{\text{fin}}}$ requires $(\log a)^{\beta_{\text{fin}}-2} \geqslant 24\tau_{7.1} M^2$, and $24\tau_{7.1} M^2 > 1$. Since $\beta_{\text{fin}} = 4.25\beta_0 - 6.6\vartheta$, this schedule requires $\beta_0 > 8/17$, that is, a base case with exponent $\frac{1}{2} - \delta$ for some $\delta < 1/34$. The value of ϑ balances two thresholds. A smaller ϑ raises the threshold on T imposed by the condition $(\beta_G - 2\zeta)T \geqslant 1$ of Lemma 5.4, which is $1/(2\vartheta)$ for the side and main steps and $1/(2.2\vartheta)$ for the final step. A larger ϑ lowers β_{fin}, and so raises the threshold $\log(24\tau_{7.1} M^2)/(\beta_{\text{fin}} - 2)$. With the chosen ϑ, these thresholds are at most 1000 and less than 873.4, respectively. (The proof of Proposition 5.5 uses the cruder bound $24\tau_{7.1} M^2 < e^{79}$, which needs only $\log\log a \geqslant 79/0.0792$, and $79/0.0792 < 998$.) Both are

below the threshold 1000.9 of Lemma 5.4, and hence below T_{base}. So T_{boot} is determined by T_{base}. The threshold 2 for β_{fin} comes from the order bound N_t in Proposition 8.1, which is in $O(t \log^2 t)$ because the pieces of Theorem 7.1 have $O(t \log t)$ vertices and the core S built in the at most $M\sqrt{\log t}+1$ stages of the proof of Proposition 8.1 has $O(M^2 t \log^2 t)$ vertices. With pieces of order $O(t \log^3 t)$, as in the small-dense-subgraph theorem of Delcourt and Postle, the order bound would be in $O(t \log^4 t)$, and more steps would be needed. If the Gyárfás graphs are coloured with Lemma 5.3 at every step, then each step raises the exponent by less than $0.49/2$, since the condition $(\beta_{\text{G}} - 2\zeta)T \geqslant 1$ of Lemma 5.4 forces $2\zeta < \beta_{\text{G}} = 0.49$. So seven steps are needed, and they give the constant $2^7 \cdot 4.01 + 127\, c_{5.4} = 77{,}681.0327$. Colouring the Gyárfás graphs of the final step with the result of the side step allows a larger increment $\zeta_{\text{f}} = 0.3667$, so one doubling is saved, at the cost of the larger additive term $c_{4.3}(c_{\text{side}})$. Appendix A discusses other schedules.

5.4 Small values of a

When $\log\log a < T_{\text{boot}}$, graphs with no K_a minor and at most N_a vertices are coloured by the following bound.

Lemma 5.6. *Let N_a be as in Proposition 5.5. For every integer $a \geqslant 3$, every graph J with no K_a minor and $v(J) \leqslant N_a$ satisfies*

$$\chi(J) \;\leqslant\; (144.4 + 4\log\log a)\, a.$$

Proof. This is the proof of Theorem 1.5 by Delcourt and Postle [8], with N_a in place of their order bound and with Theorem 3.13 (through Lemma 3.14) in place of the theorem of Duchet and Meyniel. We may assume that J is non-null. Let $p := 2a - 2$, so $p \geqslant a$ since $a \geqslant 3$. Lemma 3.14 gives $\chi(J) \leqslant \big(2 + \max\{\log(v(J)/p), 0\}\big)p$. Since $v(J) \leqslant N_a$ and $a/p \leqslant 1$, we have $v(J)/p \leqslant N_a/p \leqslant 24\tau_{7.1} M^2 \log^2 a$. Since $24\tau_{7.1}M^2 \leqslant 2.7^{70} < e^{70}$, we have $\log(24\tau_{7.1}M^2) \leqslant 70$. Since also $\log\log a > 0$, it follows that $\max\{\log(v(J)/p), 0\} \leqslant 70 + 2\log\log a$. Hence

$$\chi(J) \;\leqslant\; (72 + 2\log\log a) \cdot (2a - 2) \;<\; (144 + 4\log\log a)\, a \;\leqslant\; (144.4 + 4\log\log a)\, a.$$

□

Let $a \geqslant 3$ be an integer with $\log\log a < T_{\text{boot}}$. Then, by Lemma 5.6, every graph J with no K_a minor and $v(J) \leqslant N_a$ satisfies $\chi(J) \leqslant (144.4 + 4T_{\text{boot}})a = 4788.4\, a \leqslant c_{5.5} a$. Together with Proposition 5.5, we obtain the following.

Corollary 5.7. *Let $c_{5.5} = 45{,}024.9873$ and N_a be as in Proposition 5.5. For every integer $a \geqslant 3$, every graph J with no K_a minor and $v(J) \leqslant N_a$ satisfies $\chi(J) \leqslant c_{5.5} a$.*

6 Rooted minors via massed pairs

This section proves Lemma 6.4, which states that, for an integer t with $\log t \geqslant 1.6 \cdot 10^6$, every $9t$-connected graph with at least $70t$ times as many edges as vertices is $(2x, \lfloor 4.004t \rfloor)$-woven, where $x = \lceil t/(M\sqrt{\log t}) \rceil$ and $M = 10^5$. It is used in Section 8, where it shows that the small dense pieces are woven. Lemma 6.2 reduces wovenness to rooted models, in massed pairs, of the disjoint union of a clique of order $2x$ and a matching, Lemma 6.3 provides these rooted models, and Lemma 6.4 combines the two. Only the clique needs a density argument, and the matching is routed by the linkage theorem of Thomas and Wollan. Since $2x$ is tiny compared with t, the clique model lies in a small set of vertices, and deleting this set leaves enough connectivity for the linkage. Recall that $K_s + \ell K_2$ is

the disjoint union of K_s and a matching with ℓ edges. For a graph G and $Y \subseteq V(G)$, let $\rho_G(Y)$ denote the number of edges of G with at least one end in Y. Following Wollan [52] (see also [47]), a pair (G, X) with $X \subseteq V(G)$ is *λ-massed* if:

(M1) $\rho_G(V(G) \setminus X) > \lambda\,|V(G) \setminus X|$, and

(M2) every non-trivial separation (A, B) of G with $X \subseteq A$ and order at most $|X| - 1$ satisfies $\rho_G(B \setminus A) \leqslant \lambda |B \setminus A|$.

Here, as in Section 3.1 and in Wollan's paper, a separation (A, B) is trivial if $A \subseteq B$ or $B \subseteq A$. If $B \subseteq A$, then $\rho_G(B \setminus A) = \rho_G(\emptyset) = 0$. If $X \subseteq A \subseteq B$, then (A, B) has order $|A| \geqslant |X|$. So condition (M2) is the same when it is required for every separation (A, B) of G with $X \subseteq A$ and order at most $|X| - 1$, as in the formalisation. We use (M2) in this last form.

Wollan [52] proves extremal theorems for rooted minors by studying a minimal counterexample. We use the following consequence of his structural results, rather than his final numerical bound. Let F be a graph. A graph G is *F-universal at* a set $Z \subseteq V(G)$ if, for every injection $\pi : Z \to V(F)$, the graph G has a model of $F[\pi(Z)]$ rooted at Z according to π. A graph J is *F-universal* if $v(J) \geqslant |V(F)|$ and J is F-universal at every set of $|V(F)|$ vertices of J. Equivalently, $v(J) \geqslant |V(F)|$ and, for every set $Y \subseteq V(J)$ with $|Y| = |V(F)|$ and every bijection $\pi : Y \to V(F)$, the graph J has a model of F rooted at Y according to π.

Theorem 6.1 (Wollan [52, Theorems 2.1–2.3])**.** *Let F be a graph, and let λ be real with $1 \leqslant |V(F)| \leqslant \lambda$. Suppose that every non-null graph D with $v(D) \leqslant 2\lambda + 1$ and $\delta(D) \geqslant \lambda - |V(F)|$ has an F-universal induced subgraph. Then for every λ-massed pair (G, X) with $|X| \leqslant |V(F)|$ and every injection $\pi : X \to V(F)$, the graph G has a model of $F[\pi(X)]$ rooted at X according to π. That is, G is F-universal at X.*

Wollan does not state Theorem 6.1 in this form, but it is the first step of his proof of [52, Theorem 5.1], and follows from his Theorems 2.2 and 2.3 as below (his Theorem 2.1 is used in their proofs). These theorems concern an *(F, λ)-minimal* pair, that is, a λ-massed pair (G, X) with $|X| \leqslant |V(F)|$ such that G is not F-universal at X, with $v(G)$ as small as possible and, subject to this, with $\rho_G(V(G) \setminus X)$ as small as possible. Wollan's Theorems 2.1–2.3 are formalised in `LinearHadwiger`, following his proofs, and Theorem 6.1 is derived from them in the formalisation of this paper.

Proof. Suppose that the theorem fails, and let (G, X) be an (F, λ)-minimal pair. Then $|X| \geqslant 3$, as Wollan assumes in his proof of [52, Theorem 2.2(2)]. Indeed, let $\pi : X \to V(F)$ be an injection. If $|X| \leqslant 1$, or if $X = \{x_1, x_2\}$ with $\pi(x_1)\pi(x_2) \notin E(F)$, then the sets $\{x\}$ with $x \in X$ form a model of $F[\pi(X)]$ rooted at X according to π. Now let $X = \{x_1, x_2\}$ with $\pi(x_1)\pi(x_2) \in E(F)$. If x_1 and x_2 lie in a common component of G, then splitting an x_1–x_2 path into two subpaths, one containing x_1 and the other containing x_2, gives such a model. Otherwise, let C be the vertex set of the component of G containing x_1. The separations $\big((V(G) \setminus C) \cup \{x_1\}, C\big)$ and $\big(C \cup \{x_2\}, V(G) \setminus C\big)$ of G have order $1 = |X| - 1$, so by (M2), $\rho_G(Y_i) \leqslant \lambda |Y_i|$ for $Y_1 := C \setminus \{x_1\}$ and $Y_2 := V(G) \setminus (C \cup \{x_2\})$. The sets Y_1 and Y_2 partition $V(G) \setminus X$, and no edge joins them. So $\rho_G(V(G) \setminus X) = \rho_G(Y_1) + \rho_G(Y_2) \leqslant \lambda |V(G) \setminus X|$, contrary to (M1). Hence $|X| \leqslant 2$ is impossible for an (F, λ)-minimal pair, and so $|X| \geqslant 3$. No vertex $z \in V(G) \setminus X$ is isolated. Otherwise $(G - z, X)$ is λ-massed, since ρ is unchanged, $|V(G) \setminus X|$ decreases, and a separation (A, B) of $G - z$ as in (M2) gives the separation $(A \cup \{z\}, B)$ of G with the same order and the same set $B \setminus A$. Since $G - z$ is not F-universal at X, this contradicts the minimality of $v(G)$. By [52, Theorem 2.2(2)], some vertex $z \in V(G) \setminus X$ has $\deg_G(z) \leqslant 2\lambda$. Let $D := G[\{z\} \cup N_G(z)]$, so that $v(D) \leqslant 2\lambda + 1$. Let $w \in N_G(z)$. By [52, Theorem 2.2(1)], the vertices z and w have at least $\lfloor \lambda \rfloor$ common neighbours if $w \notin X$, and at least $\lfloor \lambda \rfloor - t_w$

common neighbours if $w \in X$, where $t_w \leqslant |X| - 1$ is the number of neighbours of z in $X \setminus \{w\}$. In both cases they have at least $\lfloor\lambda\rfloor - (|X| - 1)$ common neighbours. These common neighbours and z lie in $N_D(w)$, so $\deg_D(w) \geqslant \lfloor\lambda\rfloor - |X| + 2 > \lambda - |X|$. Since z is not isolated, it has such a neighbour w. As z is adjacent to w and to their common neighbours, $\deg_D(z) = \deg_G(z) > \lambda - |X|$. Hence $\delta(D) \geqslant \lambda - |V(F)|$, and by hypothesis D has an F-universal induced subgraph. This is an F-universal subgraph of G, contrary to [52, Theorem 2.3]. □

Wollan's main theorem [52, Theorem 5.1] combines his structural results with a general argument that works for every graph F. Its consequence [52, Theorem 1.1], which is the form used by Delcourt and Postle [8, Theorem 5.9], states that every $|V(F)|$-connected graph G with at least $(9c + 26{,}833|V(F)|)\, v(G)$ edges has a rooted model of F for every choice of roots, where $c \geqslant 1$ is such that every graph with at least c times as many edges as vertices has an F minor. Lemmas 6.3 and 6.4 replace this theorem for the graphs $F = K_s + \ell K_2$ that we need, where $s \leqslant 2 \cdot 10^{-8} t$ and $\ell = \lfloor 4.004t \rfloor$.

Lemma 6.2. *Let $p \geqslant 1$ and $\ell \geqslant 0$ be integers, let $F := K_p + \ell K_2$, and let $\lambda > 1$ be real with $\lambda \geqslant p + 2\ell$. Suppose that for every λ-massed pair (G_0, X) with $|X| \leqslant p + 2\ell$ and every injection $\pi : X \to V(F)$, the graph G_0 has a model of $F[\pi(X)]$ rooted at X according to π. Then every graph G with $e(G) \geqslant \lambda\, v(G)$ and $\kappa(G) \geqslant 2p + 2\ell$ is (p, ℓ)-woven.*

Proof. The proof combines Wollan's derivation of [52, Theorem 1.1], which checks the massed conditions using connectivity, with the argument of Delcourt and Postle [8, Lemmas 5.11 and 5.13]. As illustrated in Figure 7, the latter uses the graph $K_p + \ell K_2$ and attaches each root r_i that is a terminal to a neighbour r'_i. Let $\ell' \in \{0, \dots, \ell\}$, and let $R = \{r_1, \dots, r_p\}$, $S = \{s_1, \dots, s_{\ell'}\}$ and $T = \{t_1, \dots, t_{\ell'}\}$ be as in the definition of (p, ℓ)-woven. Let $I := \{i \in [p] : r_i \in S \cup T\}$. Since $\delta(G) \geqslant \kappa(G) \geqslant 2p + 2\ell$ and $|R \cup S \cup T| \leqslant p + 2\ell$, each r_i has at least p neighbours outside $R \cup S \cup T$. So, by Hall's theorem, there are pairwise distinct vertices $r'_1, \dots, r'_p \notin R \cup S \cup T$ with r'_i adjacent to r_i for each $i \in [p]$. For $i \in [p]$, let $x_i := r'_i$ if $i \in I$, and $x_i := r_i$ otherwise. Let $X := \{x_1, \dots, x_p\} \cup S \cup T$. The vertices $x_1, \dots, x_p$ are distinct and lie outside $S \cup T$. Moreover $|X| \leqslant p + 2\ell \leqslant \lambda$ and $|X| \leqslant \kappa(G)$.

We claim that (G, X) is λ-massed. For (M2), let (A, B) be a separation of G with $X \subseteq A$ and order at most $|X| - 1 < \kappa(G)$. Since G is $\kappa(G)$-connected, one of $A \setminus B$ and $B \setminus A$ is empty. If $A \setminus B = \emptyset$, then $X \subseteq A \subseteq A \cap B$, contradicting $|A \cap B| < |X|$. So $B \setminus A = \emptyset$, and hence $\rho_G(B \setminus A) = 0$. So (M2) holds. For (M1), note that every edge of G not counted by $\rho_G(V(G) \setminus X)$ has both ends in X. So $e(G) \leqslant \rho_G(V(G) \setminus X) + \binom{|X|}{2} \leqslant \rho_G(V(G) \setminus X) + \binom{|X|+1}{2}$. Since $1 \leqslant p \leqslant |X| \leqslant \lambda$ and $\lambda > 1$, we have $|X| + 1 < 2\lambda$, and hence $\binom{|X|+1}{2} = \frac{(|X|+1)|X|}{2} < \lambda|X|$. Since $e(G) \geqslant \lambda\, v(G)$,

$$\rho_G(V(G) \setminus X) \;\geqslant\; e(G) - \tbinom{|X|+1}{2} \;>\; \lambda v(G) - \lambda|X| \;=\; \lambda|V(G) \setminus X|.$$

Let F have clique vertices $q_1, \dots, q_p$ and matching edges $a_1b_1, \dots, a_\ell b_\ell$. Define $\pi(x_i) := q_i$. If $s_j \neq t_j$, put $\pi(s_j) := a_j$ and $\pi(t_j) := b_j$. If $s_j = t_j$, put $\pi(s_j) := a_j$. Since the sets $\{s_j, t_j\}$ with $j \in [\ell']$ are pairwise disjoint, this defines an injection $\pi : X \to V(F)$. By hypothesis, G has a model $\{B_f : f \in \pi(X)\}$ of $F[\pi(X)]$ rooted at X according to π.

For each j with $s_j \neq t_j$, the sets B_{a_j} and B_{b_j} induce connected subgraphs and are joined by an edge, so their union contains an s_j–t_j path P_j. If $s_j = t_j$, let P_j be the path with the single vertex s_j. The paths P_j are pairwise disjoint and disjoint from every B_{q_i}. For $i \in [p]$, let $M_i := B_{q_i} \cup \{r_i\}$. If $i \notin I$, then $r_i = x_i \in B_{q_i}$, so $M_i = B_{q_i}$. If $i \in I$, then $r_i \in S \cup T$ lies in some set B_{a_j} or B_{b_j}, so not in any $B_{q_{i'}}$, and $G[M_i]$ is connected since $r_i r'_i \in E(G)$ and $r'_i \in B_{q_i}$. So $\mathcal{M} := \{M_1, \dots, M_p\}$ is a K_p model rooted at R, and $\mathcal{P} := \{P_1, \dots, P_{\ell'}\}$ is an

$\{(s_j, t_j)\}$-linkage. Let $w \in M_i \cap V(P_j)$ for some $i \in [p]$ and $j \in [\ell']$. Then $w \in B_{a_j} \cup B_{b_j}$, so $w \notin B_{q_i}$. Hence $w = r_i$ and $i \in I$. Conversely, if $i \in I$, then $r_i \in \{s_j, t_j\}$ for some $j \in [\ell']$, so $r_i \in M_i \cap V(P_j)$. Therefore $V(\mathcal{M}) \cap V(\mathcal{P}) = R \cap (S \cup T)$. □

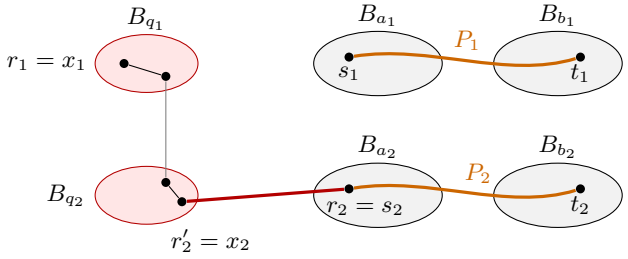


Figure 7: The proof of Lemma 6.2 with $p = \ell = 2$. Since $r_2 = s_2$, the root r_2 is replaced in X by its neighbour r'_2. In the rooted model of $K_2 + 2K_2$, the path P_j joins s_j to t_j inside $B_{a_j} \cup B_{b_j}$. The branch sets B_{q_1} and $B_{q_2} \cup \{r_2\}$ form a K_2 model rooted at $\{r_1, r_2\}$ that meets the linkage only in r_2.

The weaving constant. We call the constant $C_{6.4}$ of Lemma 6.4 below the *weaving constant*. The other parameters of Lemma 6.4 are chosen as follows. The number $\ell = \lfloor 4.004t \rfloor$ is an upper bound on the number of linkage pairs in each application of Lemma 3.8 in the proof of Proposition 8.1. Delcourt and Postle [8, proof of Lemma 6.4] bound the number of pairs by $10t$. They also take $x = \lceil t/\sqrt{\log t} \rceil$, so that the density that Theorem 3.11 requires for a K_{2x} minor alone is about $60t$. The factor $M = 10^5$ makes $s = 2x \leqslant 2 \cdot 10^{-8} t$ when $\log t \geqslant 1.6 \cdot 10^6$. Then the K_s model in the proof of Lemma 6.3 lies in a set Z of at most $0.008t$ vertices of a $(9t+1)$-connected graph J, and $J - Z$ is still $8.99t$-connected, which suffices for the linkage found there. The cost is that Proposition 8.1 has at most $M\sqrt{\log t} + 1$ stages instead of about $\sqrt{\log t}$. This affects only the order bound $N_t = 24\tau_{7.1} M^2 \, t \log^2 t$ and the size thresholds, not the final constant.

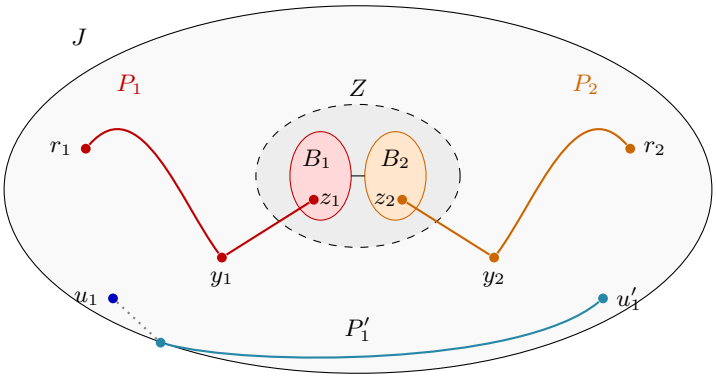


Figure 8: The proof of Lemma 6.3 with $s = 2$ and $\ell = 1$. A set Z of at most $0.008t$ vertices of $J - Y$, chosen by averaging, contains a K_s model $\{B_1, \dots, B_s\}$. A linkage in $J - Z$ joins each root r_i by a path P_i to a neighbour y_i of a vertex $z_i \in B_i$, and joins u_1 to u'_1 by a path P'_1. The branch set for a_1 is $\{u_1\}$, and the branch set for b_1 is the rest of P'_1, joined to u_1 by the first edge of P'_1 (dotted).

Lemma 6.3. *Let t be an integer with $\log t \geqslant 1.6 \cdot 10^6$. Let $s := 2x$, where $x = \lceil t/(M\sqrt{\log t}) \rceil$, let $\ell := \lfloor 4.004t \rfloor$ and $F := K_s + \ell K_2$. Then for every $70t$-massed pair (G, X) with $|X| \leqslant s + 2\ell$ and every injection $\pi : X \to V(F)$, the graph G has a model of $F[\pi(X)]$ rooted at X according to π.*

The formalised form of Lemma 6.3 allows every real t with $\log t \geqslant 1.6 \cdot 10^6$, and so Lemma 6.3 is a special case of it.

Together with Theorem 6.1, the proof of Lemma 6.3 follows Wollan's proof of [52, Theorem 5.1] (in particular Observation 5.2 and Claim 5.3 there). Like Wollan, we take a minimal counterexample and find a dense subgraph in the neighbourhood of a vertex of low degree (both in the proof of Theorem 6.1), then find a minor inside a small set of vertices of this subgraph (Wollan chooses this set at random, and we choose it by averaging), and link the roots to this minor using the theorem of Thomas and Wollan. The differences are that we use Lemma 3.17 to obtain a highly connected dense subgraph J, and that only the clique part of F needs a minor argument, while the matching edges are routed directly by the linkage.

Proof. Let $L := \log t$, $h := s + 2\ell$, $\lambda := 70t$ and $\gamma_s := 30\, s\sqrt{\log s}$. Since $t = e^L \geqslant L^3/6$ and $L \geqslant 1.6 \cdot 10^6$, we have $t \geqslant (1.6 \cdot 10^6)^2 L/6 \geqslant 4 \cdot 10^{11} L \geqslant 4 \cdot 10^{11}\sqrt{L}$ and $\sqrt{L} \geqslant 1264$. In particular $t \geqslant 10^{13}$. Since $t/(M\sqrt{L}) > 1$, we have $x \geqslant 2$ and $s = 2x \geqslant 4$. Also $x < t/(M\sqrt{L}) + 1$, so

$$s\sqrt{L} \;\leqslant\; \frac{2t}{M} + 2\sqrt{L}.$$

Hence $s \leqslant 2t/(1264M) + 2 \leqslant 2 \cdot 10^{-8} t$, since $2 \leqslant 4 \cdot 10^{-9} t$. In particular $s \leqslant t$, so $\log s \leqslant L$, and $\gamma_s \leqslant 30 s\sqrt{L} \leqslant 60t/M + 60\sqrt{L} \leqslant 0.0006001t$, since $60\sqrt{L} \leqslant 10^{-7} t$. Also $\ell \leqslant 4.004t$, so $h \leqslant 8.0081t$, $m := s + \ell \leqslant 4.005t$ and $2s + 2\ell \leqslant 9t$.

Since $|V(F)| = h \geqslant 1$ and $\lambda \geqslant h$, Theorem 6.1 reduces the lemma to the following claim. Every non-null graph D with $v(D) \leqslant 2\lambda + 1$ and $\delta(D) \geqslant \lambda - h$ has an F-universal induced subgraph. Let D be such a graph. Then $v(D) \leqslant 140t + 1 \leqslant 141t$ and $\delta(D) \geqslant 70t - 8.0081t \geqslant 61.99t$, so $\gamma := e(D)/v(D) \geqslant 30.99t$. In particular, $\gamma \geqslant 2k$ with $k = 9t$, as Lemma 3.17 requires.

By Lemma 3.17 with $k = 9t$, D has a $(9t+1)$-connected induced subgraph J with $\delta(J) > \gamma \geqslant 30.99t > 30.5t$ and $e(J) > (\gamma - k)\, v(J) \geqslant 21.99t\, v(J) > 21.5t\, v(J)$. Also $v(J) \leqslant v(D) \leqslant 141t$ and $v(J) > \delta(J) > h$. We show that J is F-universal.

Let $Y \subseteq V(J)$ with $|Y| = h$, and let $\pi : Y \to V(F)$ be a bijection. Let $K_s + \ell K_2$ have clique vertices $q_1, \dots, q_s$ and matching edges $a_1 b_1, \dots, a_\ell b_\ell$. Let $r_i := \pi^{-1}(q_i)$, $u_j := \pi^{-1}(a_j)$ and $u'_j := \pi^{-1}(b_j)$.

As illustrated in Figure 8, we first find a small K_s model in $J - Y$, and then use a linkage that avoids this model to join each root r_i to it and to join u_j to u'_j for each $j \in [\ell]$.

A small clique model avoiding Y. Let $J_0 := J - Y$ and $N := v(J_0)$. Each vertex of Y lies in at most $v(J)$ edges, so $e(J_0) \geqslant e(J) - h\, v(J) > (21.5t - 8.0081t)\, v(J) \geqslant 13.49t\, v(J) \geqslant 13.49t\, N$. Also $N \leqslant v(J) \leqslant 141t$, and $N = v(J) - h > \delta(J) - h > 30.5t - 8.0081t > 22t$. We use the following averaging argument. Let J' be a graph with $n \geqslant 2$ vertices, and let $0 < p \leqslant 1$. Then some set $Z \subseteq V(J')$ satisfies $|Z| \leqslant pn + 2$ and $e(J'[Z]) \geqslant p^2 e(J')$. Indeed, let $n_0 := \min\{n, \lceil pn \rceil + 1\}$, so that $2 \leqslant n_0 \leqslant pn + 2$. Each edge of J' lies in $J'[Z]$ for exactly $\binom{n-2}{n_0-2}$ of the $\binom{n}{n_0}$ sets Z of n_0 vertices, so the average of $e(J'[Z])$ over these sets is $\frac{n_0(n_0-1)}{n(n-1)}\, e(J')$. If $n_0 = n$, this is $e(J') \geqslant p^2 e(J')$. Otherwise $n_0 = \lceil pn \rceil + 1$, so $n_0 - 1 \geqslant pn$ and $n_0(n_0-1) \geqslant (pn+1)pn \geqslant p^2 n(n-1)$. In both cases some set Z of n_0 vertices satisfies $e(J'[Z]) \geqslant p^2 e(J')$.

We apply this with $J' = J_0$, $n = N$, $\nu := 0.0079t - 2$ and $p := \nu/N$. Here $0 < p \leqslant 1$, since $0 < \nu < 22t < N$. This gives $Z \subseteq V(J_0)$ with $|Z| \leqslant \nu + 2 = 0.0079t$ and

$$\begin{aligned} e(J_0[Z]) \;&\geqslant\; \frac{\nu^2}{N^2}\, e(J_0) \;\geqslant\; \frac{13.49t\,\nu^2}{N} \;\geqslant\; \frac{13.49}{141}\,\nu^2 \\ &\geqslant\; 0.0956\,(0.00789t)^2 \;>\; 0.0006001t \cdot 0.0079t \;\geqslant\; \gamma_s |Z|, \end{aligned}$$

using $N \leqslant 141t$, $\nu \geqslant 0.00789t$ (as $t \geqslant 10^{13}$), $\gamma_s \leqslant 0.0006001t$ and $|Z| \leqslant 0.0079t$. In particular $Z \neq \emptyset$. By Theorem 3.11, $J_0[Z]$ has a K_s model $\{B_1, \dots, B_s\}$. Also $Z \cap Y = \emptyset$ and $|Z| \leqslant 0.008t$.

Linking around the clique model. For each $i \in [s]$ choose $z_i \in B_i$ and a neighbour y_i of z_i in J, with $y_i \notin Z \cup Y$ and $y_1, \dots, y_s$ distinct. This is possible by Hall's theorem, since each z_i has more than $30.5t$ neighbours in J, at most $|Z| + |Y| \leqslant 8.0161t$ of them lie in $Z \cup Y$, and $30.5t - 8.0161t \geqslant t \geqslant s$. The graph $J - Z$ is $(9t + 1 - |Z|)$-connected, and $9t + 1 - |Z| \geqslant 8.99t \geqslant 2m$. Also $e(J - Z) \geqslant e(J) - |Z|\, v(J) > (21.5t - 0.008t)\, v(J) \geqslant 5m\, v(J - Z)$. By Theorem 3.5, $J - Z$ is m-linked. The $2m$ vertices r_i, y_i ($i \in [s]$) and u_j, u'_j ($j \in [\ell]$) are distinct and lie in $J - Z$. So $J - Z$ has pairwise disjoint paths P_i from r_i to y_i and P'_j from u_j to u'_j. Let $B'_i := B_i \cup V(P_i)$, which induces a connected subgraph since $y_i z_i \in E(J)$. Let $C_{q_i} := B'_i$ for $i \in [s]$, and let $C_{a_j} := \{u_j\}$ and $C_{b_j} := V(P'_j) \setminus \{u_j\}$ for $j \in [\ell]$. Since $u_j \neq u'_j$, the set C_{b_j} is the vertex set of the non-empty subpath $P'_j - u_j$, which contains u'_j. These sets are pairwise disjoint, because the paths P_i and P'_j are pairwise disjoint and avoid Z. The sets C_{q_i} are pairwise adjacent because the sets B_i are, and C_{a_j} and C_{b_j} are adjacent because the first edge of P'_j joins them. So $\{C_f : f \in V(F)\}$ is a model of F rooted at Y according to π. Hence J is an F-universal induced subgraph of D, which proves the claim. □

Lemma 6.4. *Let $C_{6.4} := 70$. If t is an integer with $\log t \geqslant 1.6 \cdot 10^6$, then every graph G with $e(G) \geqslant C_{6.4} t\, v(G)$ and $\kappa(G) \geqslant 9t$ is $(2x, \lfloor 4.004t \rfloor)$-woven, where $x = \lceil t/(M\sqrt{\log t}) \rceil$ and $M = 10^5$.*

Proof. Let $s := 2x$ and $\ell := \lfloor 4.004t \rfloor$. By the proof of Lemma 6.3, $s + 2\ell \leqslant 8.0081t \leqslant 70t$ and $2s + 2\ell \leqslant 4 \cdot 10^{-8} t + 8.008t \leqslant 9t \leqslant \kappa(G)$. By Lemma 6.3, the hypothesis of Lemma 6.2 holds with $p = s$ and $\lambda = 70t$. So G is $(2x, \lfloor 4.004t \rfloor)$-woven by Lemma 6.2. □

7 A refined form of Lin's theorem

This section proves Theorem 7.1, which supplies the small dense pieces of the chromatic-inseparable case (Section 8). That case needs small highly connected subgraphs in dense graphs with no K_t minor. Delcourt and Postle use their small-dense-subgraph theorem (Section 3.3). Lin [25, Theorem 1.4] proved a variant in which the highly connected subgraph has only $O(t \log t)$ vertices. Let $C_{\mathrm{Lin}} := \max\{30{,}300, 32 \cdot 6400^2\}$, which equals 1,310,720,000. Lin's theorem states that, for all integers $t \geqslant 3$ and $k \geqslant t$, every non-null graph G with no K_t minor and $e(G) \geqslant C_{\mathrm{Lin}} k\, v(G)$ has a k-connected subgraph with at most ${C_{\mathrm{Lin}}}^2\, t \log t$ vertices. Lin states the theorem with an unspecified constant, and C_{Lin} is the value given by Lin's proof [25], where $6400 = C_{3.12}$ is the constant of the Norin–Postle bound (Theorem 3.12). We do not use Lin's theorem. The density constant $C_{\mathrm{Lin}} \approx 1.31 \cdot 10^9$ is quadratic in the Norin–Postle constant, and much larger than the constant $C_{\mathrm{DP}} \approx 3.07 \cdot 10^6$ of the small-dense-subgraph theorem. In our application the density constant determines $C_{8.1}$ (Proposition 8.1). The bound on the number of vertices of the subgraph enters only N_t, and hence only the size thresholds. We therefore rerun Lin's argument with parameters that make the density constant small, at the cost of a larger bound on the number of vertices. We also extract a subgraph that is dense, not only highly connected, since Lemma 6.4 needs density $C_{6.4}t$.

Theorem 7.1. *Let $\omega_{7.1} := 0.0044$ and $\tau_{7.1} := 4.56 \cdot 10^{18}$. Let $t \geqslant 10^{13}$ and $d \geqslant 1.8 \cdot 10^4\, t$ be integers, and let k be an integer with $1 \leqslant k \leqslant \omega_{7.1} d/2$. Every non-null graph G with no K_t minor and $e(G) \geqslant d\, v(G)$ has a set A of at most $\tau_{7.1}\, t \log t$ vertices such that $G[A]$ is $(k+1)$-connected and $e(G[A]) > (\omega_{7.1} d - k)\, |A|$.*

For a graph J and $A \subseteq V(J)$, the *weight* of A in J is $w_J(A) := \sum_{v \in A} \deg_J(v)$. The proof of Theorem 7.1 contracts disjoint connected subgraphs of large weight while few edges are lost (Step 1), discards the vertices with many neighbours among the contracted vertices (Step 2), and finds a set S' of large minimum degree among the remaining vertices (Step 3). A maximal connected set $U \subseteq S'$ of small weight whose contraction loses few edges then has a dense neighbourhood with few vertices (Steps 4–6), and Lemma 3.17 extracts the required set from it (Step 7).

The improvement over Lin's theorem has four sources. First, since $d \geqslant 1.8 \cdot 10^4\, t$, the linear term $(t-2)(|V_1| + |V_2|)$ of the Norin–Postle bound (Theorem 3.12) is at most $dn/(1.8 \cdot 10^4)$ in Claim 2 below, where $n = v(G)$. Second, each contracted subgraph has weight at least Q in G, and the threshold Q below is taken very large. This choice of Q makes the other term of the Norin–Postle bound at most $3.17 \cdot 10^{-8}\, dn$ in Claim 2. The cost is a larger $\tau_{7.1}$, since the final set has fewer than Q vertices. Third, the parameters r and δ_0 of Lemma 7.2 are chosen so that the inequality in Step 3 holds with little room to spare, which makes the density obtained in Step 6 large. Fourth, in Steps 6 and 7 we extract a dense subgraph by a simple averaging argument, and then use the proof of Mader's theorem [30] to obtain a highly connected subgraph whose density is close to that of the dense subgraph (Lemma 3.17). Besides Theorems 3.11 and 3.12 and Lemma 3.17, the proof uses one further tool.

Lemma 7.2 (Delcourt and Postle [8, Lemma 4.2], see also [25, Lemma 2.2])**.** *Let $r > 2$ and $\delta_0 > 0$ be real. Let G be a graph and $S \subseteq V(G)$. If*

$$(r-2)\, e(G[S]) \;>\; (r-1)\, \delta_0\, |S| + e_G(S, V(G) \setminus S),$$

then there is a non-empty set $S' \subseteq S$ such that $\delta(G[S']) \geqslant \delta_0$ and $|N_G(v) \cap S'| \geqslant \deg_G(v)/r$ for every $v \in S'$.

Proof of Theorem 7.1. The proof follows Lin [25, Section 3], which in turn follows the proof of a theorem of Delcourt and Postle [8, Theorem 2.3]. Claims 1–3 below are Lin's claims [25, Claims 3.1–3.3] with our parameters. Let $n := v(G)$. Since $t \geqslant 10^{13}$, we have $d \geqslant 1.8 \cdot 10^{17}$. Since dn is an integer and $e(G) \geqslant dn$, deleting edges gives a spanning subgraph G_0 of G with $e(G_0) = dn$, and G_0 has no K_t minor. If a set A has the required properties in G_0, then it has them in G, since $G_0[A]$ is a spanning subgraph of $G[A]$, adding edges preserves $(k+1)$-connectivity, and $e(G[A]) \geqslant e(G_0[A])$. So we may assume that $e(G) = dn$. Then

$$w_G(V(G)) \;=\; 2e(G) \;=\; 2dn.$$

Since G has no K_t minor and $e(G) \geqslant dn$, Theorem 3.11 gives $d \leqslant 30\, t\sqrt{\log t}$. Define the parameters

$$\varepsilon := 0.083, \quad \theta := 0.003, \quad \varepsilon' := 0.02, \quad r := 2.92, \quad \delta_0 := 0.37\, d,$$

and let

$$Q \;:=\; \frac{2 \cdot 10^{15}\, {C_{3.12}}^2\, t^2 \log t}{d}.$$

Step 1. Contracting heavy subgraphs. Choose pairwise vertex-disjoint non-null connected subgraphs $H_1, \ldots, H_m$ of G with the following two properties, with m as large as possible. First, $w_G(V(H_i)) \geqslant Q$ for each $i \in [m]$. Second, the graph G' obtained by contracting each H_i to a new vertex x_i satisfies

$$e(G) - e(G') \;\leqslant\; \varepsilon \sum_{i=1}^{m} w_G(V(H_i)).$$

The empty family has both properties, so such a family exists. The graph G' is a minor of G, so it has no K_t minor. Let $X := \{x_1, \dots, x_m\}$ and $O := V(G') \setminus X$. Since the sets $V(H_i)$ are disjoint, $|X|Q \leqslant w_G(V(G)) = 2dn$. Hence

$$|X| \;\leqslant\; \frac{2dn}{Q} \;=\; \frac{2d^2 n}{2 \cdot 10^{15}\, {C_{3.12}}^2\, t^2 \log t}.$$

Also $e(G') \geqslant e(G) - 2\varepsilon\, dn = (1 - 2\varepsilon)\, dn$.

Claim 1. There is no non-empty set $A \subseteq O$ such that $G'[A]$ is connected, $w_{G'}(A) \geqslant Q$ and $e(G') - e(G'/A) \leqslant \varepsilon\, w_{G'}(A)$. In particular, $\deg_{G'}(v) < Q$ for every $v \in O$.

Proof of Claim 1. The contractions forming G' do not change adjacencies within O. So $G'[O] = G[O]$, and $\deg_{G'}(v) \leqslant \deg_G(v)$ for $v \in O$. Suppose that A exists. Then $w_G(A) \geqslant w_{G'}(A) \geqslant Q$. Moreover G'/A is obtained from G by contracting $H_1, \dots, H_m$ and $G[A]$. Since

$$e(G) - e(G'/A) \;=\; \big(e(G) - e(G')\big) + \big(e(G') - e(G'/A)\big) \;\leqslant\; \varepsilon \sum_{i=1}^{m} w_G(V(H_i)) + \varepsilon\, w_G(A),$$

the family $H_1, \dots, H_m, G[A]$ contradicts the maximality of m. For the second statement, take $A = \{v\}$, for which $e(G') - e(G'/A) = 0$. ◊

Step 2. Removing the vertices attached to X. Let

$$Y := \big\{v \in O : |N_{G'}(v) \cap X| \geqslant \theta \deg_{G'}(v)\big\},$$

and let $S := V(G') \setminus (X \cup Y)$.

Claim 2. $e(G'[X \cup Y]) + e_{G'}(S, X \cup Y) \leqslant \varepsilon'\, dn$.

Proof of Claim 2. Apply Theorem 3.12 to $G'(X, O)$. Since $|O| \leqslant n$ and $|X| + |O| \leqslant n$,

$$e_{G'}(X, O) \;\leqslant\; C_{3.12}\, t\sqrt{\log t}\, \sqrt{|X|\, n} + tn \;\leqslant\; 3.17 \cdot 10^{-8}\, dn + \frac{t}{d}\, dn \;\leqslant\; \nu'\, dn,$$

where $\nu' := 3.17 \cdot 10^{-8} + 1/(1.8 \cdot 10^4) < 5.559 \cdot 10^{-5}$. Here the second inequality holds since $(C_{3.12}\, t\sqrt{\log t})^2\, |X|\, n \leqslant 10^{-15}(dn)^2$, by the bound on $|X|$. By the definition of Y,

$$\theta \sum_{v \in Y} \deg_{G'}(v) \;\leqslant\; e_{G'}(X, Y) \;\leqslant\; e_{G'}(X, O),$$

so $\sum_{v \in Y} \deg_{G'}(v) \leqslant (\nu'/\theta)\, dn$. Theorem 3.11 and the bound on $|X|$ give

$$e(G'[X]) \;\leqslant\; 30\, t\sqrt{\log t}\, |X| \;\leqslant\; \frac{60}{2 \cdot 10^{15}\, {C_{3.12}}^2} \cdot \frac{d}{t\sqrt{\log t}} \cdot dn \;<\; 10^{-19}\, dn,$$

using $d \leqslant 30\, t\sqrt{\log t}$. We have

$$\sum_{v \in X \cup Y} \deg_{G'}(v) \;=\; 2e(G'[X \cup Y]) + e_{G'}(S, X \cup Y) \qquad \text{and}$$
$$\sum_{v \in X} \deg_{G'}(v) \;=\; 2e(G'[X]) + e_{G'}(X, O).$$

Since X and Y are disjoint, the first sum equals the second sum plus $\sum_{v \in Y} \deg_{G'}(v)$. Hence

$$\begin{aligned} e(G'[X \cup Y]) + e_{G'}(S, X \cup Y) \;&\leqslant\; 2e(G'[X]) + e_{G'}(X, O) + \sum_{v \in Y} \deg_{G'}(v) \\ &\leqslant\; \big(2 \cdot 10^{-19} + \nu'(1 + 1/\theta)\big) dn \;<\; 0.01859\, dn \;<\; \varepsilon'\, dn. \end{aligned}$$

This proves Claim 2. ◊

Step 3. A dense subset. By Claim 2,

$$e(G'[S]) \;=\; e(G') - e(G'[X \cup Y]) - e_{G'}(S, X \cup Y) \;\geqslant\; (1 - 2\varepsilon - \varepsilon')\, dn \;=\; 0.814\, dn.$$

Since $|S| \leqslant n$, Claim 2 also gives

$$\begin{aligned}(r-1)\,\delta_0\,|S| + e_{G'}(S, V(G') \setminus S) \;&\leqslant\; (1.92 \cdot 0.37 + \varepsilon')\, dn \;<\; 0.7305\, dn \\ &<\; 0.92 \cdot 0.814\, dn \;\leqslant\; (r-2)\, e(G'[S]).\end{aligned}$$

By Lemma 7.2, there is a non-empty set $S' \subseteq S$ such that $\delta(G'[S']) \geqslant \delta_0$ and $|N_{G'}(v) \cap S'| \geqslant \deg_{G'}(v)/r$ for every $v \in S'$.

Step 4. A comparison of Q and δ_0. Since $\log x \leqslant x - 1$ for $x > 0$ and $\log 10 < 3$,

$$\log t \;=\; 12 \log 10 + \log(t/10^{12}) \;<\; 35 + t/10^{12} \;\leqslant\; 9.7 \cdot 10^{-12}\, t,$$

where the last inequality holds because $t \geqslant 10^{13}$. Since $d \geqslant 1.8 \cdot 10^4\, t$, it follows that

$$2 \cdot 10^{15}\, {C_{3.12}}^2\, t^2 \log t \;<\; 2 \cdot 10^{15}\, {C_{3.12}}^2 \cdot 9.7 \cdot 10^{-12}\, t^3 \;<\; 0.37^2\, (1.8 \cdot 10^4)^3\, t^3 \;\leqslant\; 0.37^2\, d^3.$$

Dividing by d gives $Q < \delta_0^2$.

Step 5. A maximal light set. Choose a non-empty set $U \subseteq S'$ such that $G'[U]$ is connected, $w_{G'}(U) < Q$ and $e(G') - e(G'/U) \leqslant \varepsilon\, w_{G'}(U)$, with $|U|$ as large as possible. Every singleton in S' has these properties, by Claim 1. Some vertex of $S' \setminus U$ has a neighbour in U. Indeed, otherwise $N_{G'}(u) \cap S' \subseteq U$ for every $u \in U$. Since $\delta(G'[S']) \geqslant \delta_0$, it follows that $|U| \geqslant \delta_0$ and $\deg_{G'}(u) \geqslant \delta_0$ for every $u \in U$. So $w_{G'}(U) \geqslant \delta_0 |U| \geqslant \delta_0^2 > Q$ by Step 4, contrary to the choice of U.

Claim 3. Let $z \in S' \setminus U$ have a neighbour in U. Then $e(G') - e(G'/(U \cup \{z\})) > \varepsilon\, w_{G'}(U \cup \{z\})$.

Proof of Claim 3. Suppose not. The graph $G'[U \cup \{z\}]$ is connected. If $w_{G'}(U \cup \{z\}) < Q$, then $U \cup \{z\}$ contradicts the maximality of $|U|$. Otherwise $w_{G'}(U \cup \{z\}) \geqslant Q$, and since $U \cup \{z\} \subseteq S' \subseteq O$, this contradicts Claim 1. ◊

Step 6. A dense neighbourhood. Let $A \subseteq V(G')$ be non-empty with $G'[A]$ connected. Contracting A deletes the edges of $G'[A]$. For each vertex $y \notin A$ with a neighbour in A, it replaces the $|N_{G'}(y) \cap A|$ edges between y and A by one edge. Hence

$$e(G') - e(G'/A) \;=\; e(G'[A]) + \sum_{\substack{y \in V(G') \setminus A \\ N_{G'}(y) \cap A \neq \emptyset}} \big(|N_{G'}(y) \cap A| - 1\big). \tag{1}$$

Let R be the set of vertices of $V(G') \setminus (U \cup X)$ with a neighbour in U, and let $R' := R \cap S'$. Each vertex of R is a neighbour of a vertex of U, so $|R| \leqslant w_{G'}(U)$. Since $S' \cap X = \emptyset$, the set R' consists of the vertices of $S' \setminus U$ with a neighbour in U. So (1) with $A = U$ gives

$$\begin{aligned}\sum_{u \in U} |N_{G'}(u) \cap S'| \;&=\; 2e(G'[U]) + |R'| + \sum_{y \in R'} \big(|N_{G'}(y) \cap U| - 1\big) \\ &\leqslant\; 2\big(e(G') - e(G'/U)\big) + |R'| \;\leqslant\; 2\varepsilon\, w_{G'}(U) + |R'|.\end{aligned}$$

By the choice of S', the sum on the left is at least $w_{G'}(U)/r$. Hence

$$|R'| \;\geqslant\; \Big(\frac{1}{r} - 2\varepsilon\Big) w_{G'}(U) \;\geqslant\; 0.1764\, w_{G'}(U) \;\geqslant\; 0.1764\, |R|.$$

By Step 5, some vertex $z_0 \in S' \setminus U$ has a neighbour in U, and $z_0 \in R'$. So R', and hence R, is non-empty. Let $z \in R'$ be arbitrary, and let C_z be the set of vertices of $V(G') \setminus (U \cup \{z\})$ that are adjacent to z and have a neighbour in U. The graph $G'[U \cup \{z\}]$ is connected. Compare (1) for $A = U \cup \{z\}$ with (1) for $A = U$. The first term grows by $|N_{G'}(z) \cap U|$, the term $|N_{G'}(z) \cap U| - 1$ of z disappears, the term of each vertex of C_z grows by 1, and every other vertex contributes the same amount, or 0, to both. Hence

$$e(G') - e(G'/(U \cup \{z\})) \;=\; e(G') - e(G'/U) + 1 + |C_z|.$$

By Claim 3 and the choice of U,

$$1 + |C_z| \;>\; \varepsilon\, w_{G'}(U \cup \{z\}) - \varepsilon\, w_{G'}(U) \;=\; \varepsilon \deg_{G'}(z).$$

Since $z \in S$, we have $z \notin Y$, and so $|N_{G'}(z) \cap X| < \theta \deg_{G'}(z)$. Every vertex of C_z lies in $N_{G'}(z) \cap R$ or in $N_{G'}(z) \cap X$. Hence $|N_{G'}(z) \cap R| > (\varepsilon - \theta) \deg_{G'}(z) - 1$. Since $\deg_{G'}(z) \geqslant \delta_0 = 0.37\, d$ and $d \geqslant 1.8 \cdot 10^{17}$,

$$|N_{G'}(z) \cap R| \;>\; 0.08 \cdot 0.37\, d - 1 \;\geqslant\; 0.02959\, d.$$

This holds for every $z \in R'$. Each edge of $G'[R']$ contributes 2 to $\sum_{z \in R'} |N_{G'}(z) \cap R|$, each edge of $G'[R]$ with exactly one end in R' contributes 1, and the other edges of $G'[R]$ contribute 0. Since $R' \subseteq R$, it follows that

$$0.02959\, d\, |R'| \;\leqslant\; \sum_{z \in R'} |N_{G'}(z) \cap R| \;\leqslant\; e(G'[R]) + e(G'[R']).$$

Suppose that $e(G'[R]) < \omega_{7.1} d\, |R|$ and $e(G'[R']) < \omega_{7.1} d\, |R'|$. Since $|R| \leqslant |R'|/0.1764$ and R' is non-empty,

$$e(G'[R]) + e(G'[R']) \;<\; \omega_{7.1} d \Big(\frac{1}{0.1764} + 1 \Big) |R'| \;<\; 0.02935\, d\, |R'| \;<\; 0.02959\, d\, |R'|,$$

a contradiction. So one of $G'[R]$ and $G'[R']$ has at least $\omega_{7.1} d$ times as many edges as vertices. Let R^* be R or R' accordingly, so that $R^* \subseteq R$ is non-empty and $e(G'[R^*]) \geqslant \omega_{7.1} d\, |R^*|$.

Step 7. Conclusion. Let $\gamma := e(G'[R^*])/|R^*|$. By Step 6, $\gamma \geqslant \omega_{7.1} d \geqslant 2k$. By Lemma 3.17, $G'[R^*]$ has a $(k+1)$-connected induced subgraph H with $e(H) > (\gamma - k)\, v(H) \geqslant (\omega_{7.1} d - k)\, v(H)$. Let $A := V(H)$. Then $A \subseteq R \subseteq O$, and $G'[O] = G[O]$ (see the proof of Claim 1). So $G[A] = H$, and A has the required connectivity and density. Finally, since $d \geqslant 1.8 \cdot 10^4\, t$,

$$|A| \;\leqslant\; |R| \;\leqslant\; w_{G'}(U) \;<\; Q \;\leqslant\; \frac{2 \cdot 10^{15}\, {C_{3.12}}^2}{1.8 \cdot 10^4}\, t \log t \;<\; \tau_{7.1}\, t \log t.$$

Here $2 \cdot 10^{15}\, {C_{3.12}}^2 / (1.8 \cdot 10^4) < 4.552 \cdot 10^{18}$. □

Since $d \leqslant 30\, t \sqrt{\log t}$ and $d \geqslant 1.8 \cdot 10^4\, t$, no graph satisfies the hypotheses of Theorem 7.1 unless $\log t \geqslant 3.6 \cdot 10^5$. In Proposition 8.1, Theorem 7.1 is applied with $\log t \geqslant 1.6 \cdot 10^6$.

8 The chromatic-inseparable case

The following proposition is a version of a lemma of Delcourt and Postle [8, Lemma 2.5] in which the bound Γ on the chromatic number of small subgraphs enters additively. It uses Lemma 6.4 and Theorem 7.1.

Proposition 8.1. *Let $\omega_{7.1} = 0.0044$ and $\tau_{7.1} = 4.56 \cdot 10^{18}$ be the constants of Theorem 7.1, and let $C_{6.4} = 70$ and $M = 10^5$ be the constants of Lemma 6.4. Let $C_{8.1} := 18{,}000$, so that $\omega_{7.1} C_{8.1} \geqslant C_{6.4} + 9$. Let t be an integer with $\log t \geqslant 1.6 \cdot 10^6$ (so in particular $t \geqslant 10^{13}$). Let N_t be as in Proposition 5.5, with t in place of a. Let H be a graph with no K_t minor, and let $\Gamma \geqslant 0$ be a real number such that every induced subgraph of H with at most N_t vertices has chromatic number at most Γ. Let*

$$\sigma := 78t + 2\Gamma \qquad \text{and} \qquad \Psi := (32 + 2C_{8.1})t + 2\Gamma + 2.$$

If $\chi(H) \geqslant \Psi$, then H is σ-chromatic-separable. Consequently, if H is s-chromatic-inseparable for some real $s \geqslant 0$, then $s < (2C_{8.1} + 33)\, t + 2\Gamma$.

In the first conclusion, $C_{8.1}$ enters the chromatic threshold Ψ but not the separation cost σ. The lemma of Delcourt and Postle [8, Lemma 2.5] has the shape of the first conclusion with $\Psi = 2\sigma$, so their density constant appears in both. The second conclusion is the single-parameter form, which matches hypothesis $(\mathcal{H}_{\Phi,a_0})$. The final application (Section 9) uses the first conclusion. The paragraph after the proof compares this proof with that of Delcourt and Postle.

The proof of Proposition 8.1 below builds a K_t model in about $M\sqrt{\log t}$ stages, each adding x branch sets, while keeping a k-connected induced subgraph J^* of large chromatic number that is tangent to the model (the statement $(*)_n$ below). At each stage, J^* contains many small dense pieces, since otherwise the chromatic number of J^* would be at most about $2d + \Gamma$ (Step 1). By Lemma 6.4, these pieces are woven. Paths from the old branch sets, from rooted clique models in the pieces and from a new subgraph H_3 of large chromatic number are joined in one piece, the junction, by knitting (Steps 2–4, with Step 5 for the first stage). The branch sets are then shrunk by Lemma 3.10, so that the whole model has chromatic number at most $6t+\Gamma$ (Step 6). Since H is σ-chromatic-inseparable, a k-connected subgraph of large chromatic number outside the model meets H_3 in at least k vertices, and its union with H_3 restores the invariant (Step 7).

Proof. Setup. Let $x := \lceil t/(M\sqrt{\log t})\rceil$ and $r := \lceil M\sqrt{\log t}\rceil$, so that $rx \geqslant t$. Let $k := 12t$ and $d := C_{8.1}t$. Here k is the connectivity of the high-chromatic subgraphs extracted by Corollary 3.3, and d is the density threshold. Theorem 7.1 is applied to subgraphs with at least d times as many edges as vertices. Let $\Theta := 6t+\Gamma+2k$. Since $k = 12t$ and $d = C_{8.1}t$,

$$\sigma = 30t + 2\Gamma + 4k \qquad \text{and} \qquad \Psi = 8t + 2\Gamma + 2k + 2d + 2.$$

For $1 \leqslant n \leqslant r$ we have $n \leqslant M\sqrt{\log t} + 1$, and $(M\sqrt{\log t} + 1)^2 \leqslant 4M^2 \log t$. Hence the numbers $n\tau_{7.1}\, t \log t$ and $6n^2\tau_{7.1}\, t \log t$ are at most N_t.

Each application of Corollary 3.3 below extracts a k-connected induced subgraph from a graph with chromatic number at least $\Psi - \Theta - 24t - \Gamma = 2d - 22t + 2$, as we verify at each application. This is at least $\lceil 3.0625(k-1)\rceil$, because $d = 1.8 \cdot 10^4\, t$ and $k = 12t$. Each application loses at most $2k$ in chromatic number.

Reduction. It suffices to prove the following claim:

> (†) if H is σ-chromatic-inseparable and $\chi(H) \geqslant \Psi$, then H has a K_t minor.

Indeed, since H has no K_t minor, (†) shows that if $\chi(H) \geqslant \Psi$ then H is σ-chromatic-separable. The second conclusion follows from the first by the remark on chromatic thresholds in Section 3, since $\sigma = 78t + 2\Gamma \leqslant \Psi = (32 + 2C_{8.1})t + 2\Gamma + 2 \leqslant (2C_{8.1} + 33)\, t + 2\Gamma$.

Induction. Assume that H is σ-chromatic-inseparable, $\chi(H) \geqslant \Psi$, and H has no K_t minor. We prove by induction on $n \in \{0, 1, \ldots, r\}$ the following statement:

> $(*)_n$: H has a K_{nx} model $\mathcal{A} = \{A_1, \ldots, A_{nx}\}$ and a set $S \subseteq V(H)$ with the following properties:

(I1) S is a core of $\mathcal{A}$, and $|S| \leqslant 2n^2\tau_{7.1}\, t \log t$.
(I2) There is a set $R \subseteq V(H)$ such that $J^* := H[R]$ is k-connected.
(I3) J^* is tangent to $\mathcal{A}$, that is, $|V(J^*) \cap A_j| = 1$ for every $j \in [nx]$.
(I4) $V(J^*) \cap V(\mathcal{A}) \subseteq S$.
(I5) $\chi(J^*) \geqslant \chi(H) - \Theta$.

Since $rx \geqslant t$, the statement $(*)_r$ gives a K_t minor of H, contradicting our assumption and so proving (†). In the inductive step, the pieces and the next high-chromatic subgraph H_3 are found inside J^*. By (I3), each old branch set contains exactly one vertex of J^*, and in Step 4 a path from this vertex extends the branch set.

The case $n = 0$. Since $\chi(H) \geqslant \Psi \geqslant \lceil 3.0625(k-1) \rceil$, Corollary 3.3 shows that H has a k-connected induced subgraph $J^* = H[R]$ with $\chi(J^*) \geqslant \chi(H) - 2k \geqslant \chi(H) - \Theta$. Take $\mathcal{A}$ to be the empty model and $S := \emptyset$. Then (I1)–(I5) hold, (I1), (I3) and (I4) vacuously.

The inductive step. Let $1 \leqslant n \leqslant r$ and assume $(*)_{n-1}$. Let $\mu' := (n-1)x$ and $\mu := nx$. Then $\mu' \leqslant t + M\sqrt{\log t} \leqslant 1.0001t$ and $\mu = \mu' + x \leqslant 3t$. Moreover $r \leqslant M\sqrt{\log t} + 1$ and $x \leqslant t/(M\sqrt{\log t}) + 1$, so

$$\begin{aligned} 3\mu' + \mu \;&=\; (4n-3)x \;\leqslant\; (4M\sqrt{\log t} + 1)\Big(\frac{t}{M\sqrt{\log t}} + 1\Big) \\ &=\; 4t + \frac{t}{M\sqrt{\log t}} + 4M\sqrt{\log t} + 1 \;\leqslant\; 4.004t, \end{aligned}$$

since $\log t \geqslant 1.6 \cdot 10^6$. As $3\mu' + \mu$ is an integer, $3\mu' + \mu \leqslant \lfloor 4.004t \rfloor$. Let $\mathcal{A}' = \{A'_1, \dots, A'_{\mu'}\}$, S' and $H' := J^*$ be given by $(*)_{n-1}$. By (I3), for $j \in [\mu']$ there is a unique vertex w_j in $V(H') \cap A'_j$. Since the sets A'_j are disjoint, the vertices $w_1, \dots, w_{\mu'}$ are distinct, and

$$V(H') \cap V(\mathcal{A}') = \{w_1, \dots, w_{\mu'}\} \subseteq S'. \tag{2}$$

The inclusion holds by (I4). Every subgraph of H' that avoids $\{w_1, \dots, w_{\mu'}\}$ is therefore disjoint from $V(\mathcal{A}')$.

Step 1. Small pieces. Let $H_1 := H' - \{w_1, \dots, w_{\mu'}\}$, an induced subgraph of H. Then $\chi(H_1) \geqslant \chi(H') - \mu' \geqslant \chi(H) - \Theta - \mu'$. Choose a maximal family of pairwise disjoint sets $K \subseteq V(H_1)$ such that $|K| \leqslant \tau_{7.1}\, t \log t$, $H[K]$ is $(9t+1)$-connected and $e(H[K]) > C_{6.4}t\,|K|$.

Suppose for a contradiction that this family has fewer than n members, and let U be the union of its members. Then $|U| \leqslant n\tau_{7.1}\, t \log t \leqslant N_t$, so $\chi(H[U]) \leqslant \Gamma$. Let $T \subseteq V(H_1) \setminus U$ be non-empty. Suppose that $e(H[T]) \geqslant d\,|T|$. Theorem 7.1, applied to $H[T]$ with $9t$ in place of k, gives a set $A \subseteq T$ such that $|A| \leqslant \tau_{7.1}\, t \log t$, $H[A]$ is $(9t+1)$-connected and $e(H[A]) > (\omega_{7.1}d - 9t)\,|A| = 70.2t\,|A| \geqslant C_{6.4}t\,|A|$. Theorem 7.1 applies because $H[T]$ has no K_t minor, $t \geqslant 10^{13}$, $d = C_{8.1}t \geqslant 1.8 \cdot 10^4\, t$ and $9t \leqslant \omega_{7.1}d/2$. The set A could be added to the family, contradicting maximality. So $e(H[T]) < d\,|T|$, and $H[T]$ has a vertex of degree less than $2d$. Hence $H_1 - U$ is $(2d-1)$-degenerate and $\chi(H_1 - U) \leqslant 2d$. Then $\chi(H_1) \leqslant 2d + \Gamma$. But $\chi(H_1) \geqslant \Psi - \Theta - \mu' = 2d + \Gamma + 2t + 2 - \mu' > 2d + \Gamma$, since $\mu' \leqslant 1.0001t$. This is a contradiction.

So the family has at least n members. Choose n of them, $K_0, K_1, \dots, K_{n-1}$, and let $J_i := H[K_i]$ for $0 \leqslant i \leqslant n-1$. Call $J_0, J_1, \dots, J_{n-1}$ the *pieces*, and call J_0 the *junction*. Each piece J is $(9t+1)$-connected, so $v(J) > 9t+1$, and $e(J) > C_{6.4}t\,v(J)$. By Lemma 6.4, J is $(2x, \lfloor 4.004t \rfloor)$-woven. By (2), the pieces are disjoint from $V(\mathcal{A}')$.

Step 2. Paths to the junction J_0. If $n = 1$, let $W := \emptyset$ and $\mathcal{Q}_1 := \emptyset$. Now let $n \geqslant 2$. For $i \in [n-1]$, choose a set X_i of $2x$ vertices of J_i and label them $w_{\mu'+2(i-1)x+1}, \dots, w_{\mu'+2ix}$. Since $2(n-1)x = 2\mu'$, the labels run up to $3\mu'$. Let $W := \{w_1, \dots, w_{3\mu'}\}$. Then $|W| = 3\mu' \leqslant 6t$. The vertices $w_1, \dots, w_{\mu'}$ lie in H' but not in H_1, while each X_i lies in $J_i \subseteq H_1$. So the $3\mu'$

labelled vertices are distinct, and $W \subseteq V(H')$ is disjoint from $V(J_0)$. The graph H' is k-connected, $2|W| = 6\mu' \leqslant 6.0006t \leqslant 12t = k$, and $v(J_0) > 9t+1 \geqslant 2|W|$. Since $H' = H[R]$ for the set R of (I2), Lemma 3.16(a) applied inside R, with H in place of G and $B = V(J_0)$ (see the end of its proof), gives a double fan $\mathcal{Q}_1$ from W to $V(J_0)$ in H with $V(\mathcal{Q}_1) \subseteq V(H')$ and $\chi(H[V(\mathcal{Q}_1)]) \leqslant 4|W| = 12\mu'$. Since $H' = H[R]$ is an induced subgraph of H, $\mathcal{Q}_1$ is also a double fan in H'. For every $n \geqslant 1$, let $\overline{J} := V(J_0) \cup V(J_1) \cup \cdots \cup V(J_{n-1})$. Then $|\overline{J}| \leqslant n\tau_{7.1}\, t \log t \leqslant N_t$, so $\chi(H[\overline{J}]) \leqslant \Gamma$.

Step 3. The next high-chromatic subgraph. Let $H_2 := H_1 - (\overline{J} \cup V(\mathcal{Q}_1))$. So H_2 is obtained from H' by removing the vertices of $\{w_1, \dots, w_{\mu'}\}$, of $\overline{J}$ and of $V(\mathcal{Q}_1)$. Their removal lowers the chromatic number by at most μ', Γ and $4|W| = 12\mu'$, respectively. Since $13\mu' \leqslant 13(t + M\sqrt{\log t}) \leqslant 24t$, we obtain $\chi(H_2) \geqslant \chi(H') - 24t - \Gamma$. By (I5), $\chi(H') \geqslant \chi(H) - \Theta \geqslant \Psi - \Theta$. Hence $\chi(H_2) \geqslant \Psi - \Theta - 24t - \Gamma$, and so Corollary 3.3 applies to H_2. So H_2 has a k-connected induced subgraph H_3 with

$$\chi(H_3) \;\geqslant\; \chi(H) - \Theta - 24t - \Gamma - 2k \;=\; \chi(H) - \sigma.$$

By construction, H_3 is disjoint from $\overline{J}$, from $V(\mathcal{Q}_1)$ and from $W \subseteq \{w_1, \dots, w_{\mu'}\} \cup V(\mathcal{Q}_1)$. By (2), it is also disjoint from $V(\mathcal{A}')$.

Step 4. Building a K_μ model when $n \geqslant 2$.

A linkage to the junction. The graph $H' - W$ is 2μ-connected, as $k - |W| \geqslant 12t - 6t \geqslant 2\mu$. The sets $V(H_3)$ and $V(J_0)$ avoid W. They have more than $k \geqslant 2\mu$ and more than $9t + 1 \geqslant 2\mu$ vertices, respectively. So, by Theorem 3.4, $H' - W$ has 2μ pairwise disjoint paths from $V(H_3)$ to $V(J_0)$. Trimming them gives a family $\mathcal{Q}_2 = \{Q_{i,j} : i \in [\mu],\ j \in \{1,2\}\}$ of 2μ pairwise disjoint $V(H_3)$–$V(J_0)$ paths in $H' - W$. By Step 3, $V(H_3)$ is disjoint from $V(J_0) \cup V(\mathcal{Q}_1)$. So Lemma 3.16(b), with H', $\mathcal{Q}_1$ and $\mathcal{Q}_2$ in place of G, $\mathcal{P}$ and $\mathcal{Q}$, and with $B = V(J_0)$, $U = V(H_3)$ and $m = \mu$, gives $3\mu' + \mu$ pairwise disjoint paths $P_1, \dots, P_{3\mu'+\mu}$ in H' with all their vertices in $V(\mathcal{Q}_1) \cup V(\mathcal{Q}_2)$. Here P_i starts at w_i for $i \leqslant 3\mu'$, and $P_{3\mu'+i}$ starts at a vertex $w_{3\mu'+i} \in V(H_3)$ for $i \in [\mu]$. Let $u_i \in V(J_0)$ be the last vertex of P_i. Each P_i meets $W \cup V(H_3) \cup V(J_0)$ only in its ends. The vertices $w_1, \dots, w_{3\mu'+\mu}$ and $u_1, \dots, u_{3\mu'+\mu}$ are $2(3\mu' + \mu)$ distinct vertices, since the paths P_i are pairwise disjoint and $W \cup V(H_3)$ is disjoint from $V(J_0)$. So $\mathcal{P} = \{P_1, \dots, P_{3\mu'+\mu}\}$ is a $\{(w_i, u_i)\}$-linkage in H'. Also $V(\mathcal{P}) \subseteq V(H')$ and $V(H') \cap V(\mathcal{A}') = \{w_1, \dots, w_{\mu'}\} \subseteq W$. Hence $\mathcal{P}$ has the following three intersection properties:

$$V(\mathcal{P}) \cap V(H_3) = \{w_{3\mu'+1}, \dots, w_{3\mu'+\mu}\}, \qquad V(\mathcal{P}) \cap V(J_0) = \{u_1, \dots, u_{3\mu'+\mu}\},$$
$$V(\mathcal{P}) \cap V(\mathcal{A}') = \{w_1, \dots, w_{\mu'}\}.$$

Rooted models in the pieces. Each J_i is $(2x, \lfloor 4.004t \rfloor)$-woven, by Step 1, and hence $(2x, 3\mu' + \mu)$-woven, since $\lfloor 4.004t \rfloor \geqslant 3\mu' + \mu$. As $3\mu' + \mu$ is the number of pairs of $\mathcal{P}$, Lemma 3.8 applies with $c = 3\mu' + \mu$. Let $\mathcal{P}^{(0)} := \mathcal{P}$. For $i = 1, \dots, n-1$ in turn, apply Lemma 3.8 in the graph H', with the linkage $\mathcal{P}^{(i-1)}$, the subgraph $J_i = H'[K_i]$ in place of H, and the root set X_i. This gives a $\{(w_l, u_l)\}$-linkage $\mathcal{P}^{(i)}$ in H' and a K_{2x} model $\mathcal{M}_i$ in J_i rooted at X_i, with

$$V(\mathcal{P}^{(i)}) \cap V(\mathcal{M}_i) \subseteq X_i \qquad \text{and} \qquad V(\mathcal{P}^{(i)}) \subseteq V(J_i) \cup V(\mathcal{P}^{(i-1)}).$$

Let $\mathcal{P}' := \mathcal{P}^{(n-1)}$. For $i < i'$, the second inclusion gives $V(\mathcal{P}^{(i')}) \subseteq V(\mathcal{P}^{(i)}) \cup V(J_{i+1}) \cup \cdots \cup V(J_{i'})$, and the graphs $J_{i+1}, \dots, J_{i'}$ are disjoint from $\mathcal{M}_i \subseteq J_i$. Hence $V(\mathcal{P}') \cap V(\mathcal{M}_i) \subseteq X_i$. Since every vertex of X_i is the first vertex of a path of every linkage $\mathcal{P}^{(i')}$ with $0 \leqslant i' \leqslant n-1$, in fact $V(\mathcal{P}') \cap V(\mathcal{M}_i) = X_i$. Also $V(\mathcal{P}') \subseteq V(\mathcal{P}) \cup V(J_1) \cup \cdots \cup V(J_{n-1})$. The graphs J_i are disjoint from H_3, from $V(\mathcal{A}')$ and from J_0. So the three displayed intersection

properties of $\mathcal{P}$ also hold for $\mathcal{P}'$. For $i \in [n-1]$ and $w \in X_i$, let $B(w)$ denote the branch set of $\mathcal{M}_i$ containing w. For $l \in [3\mu'+\mu]$, let P'_l denote the path of $\mathcal{P}'$ starting at w_l.

Indices. For $i \in [n-1]$ and $j \in [x]$, let $q_{i,j} := \mu' + 2(i-1)x + x + j$, so that the vertices $w_{q_{i,j}}$ form the second half of X_i. For $j \in [\mu']$, let $i(j) := \lceil j/x \rceil$, let $p_j := \mu' + 2(i(j)-1)x + (j - (i(j)-1)x)$, and let $p'_j := 3\mu' + j$. The vertices w_{p_j} form the first halves of the sets X_i. Define

$$\Pi_j := \{u_j, u_{p_j}, u_{p'_j}\} \quad (j \in [\mu']), \qquad \Pi_{\mu'+j} := \{u_{q_{i,j}} : i \in [n-1]\} \cup \{u_{4\mu'+j}\} \quad (j \in [x]).$$

The index sets $\{j : j \in [\mu']\}$, $\{p_j : j \in [\mu']\}$, $\{p'_j : j \in [\mu']\}$, $\{q_{i,j} : i \in [n-1], j \in [x]\}$ and $\{4\mu'+j : j \in [x]\}$ partition $[3\mu'+\mu]$. So $\Pi_1, \dots, \Pi_\mu$ are pairwise disjoint and contain $3\mu'+\mu$ vertices in total. As illustrated in Figure 9, each branch set A''_l of the model $\mathcal{A}'' = \{A''_1, \dots, A''_\mu\}$ defined below uses some of the paths of $\mathcal{P}'$ to reach the junction J_0. Table 5 lists these paths.

Branch set	Path	First vertex	Lies in
A''_j, $j \in [\mu']$	P'_j	w_j	W, the old branch set A'_j
	P'_{p_j}	w_{p_j}	first half of $X_{i(j)}$, in $J_{i(j)}$
	$P'_{p'_j}$, $p'_j = 3\mu' + j$	$w_{3\mu'+j}$	H_3
$A''_{\mu'+j}$, $j \in [x]$	$P'_{q_{i,j}}$, $i \in [n-1]$	$w_{q_{i,j}}$	second half of X_i, in J_i
	$P'_{4\mu'+j}$	$w_{4\mu'+j}$	H_3

Table 5: The paths of $\mathcal{P}'$ used by each branch set of $\mathcal{A}''$ in Step 4 of the proof of Proposition 8.1. The last vertex of each path lies in Π_j or $\Pi_{\mu'+j}$, respectively, and hence in R_j or $R_{\mu'+j}$.

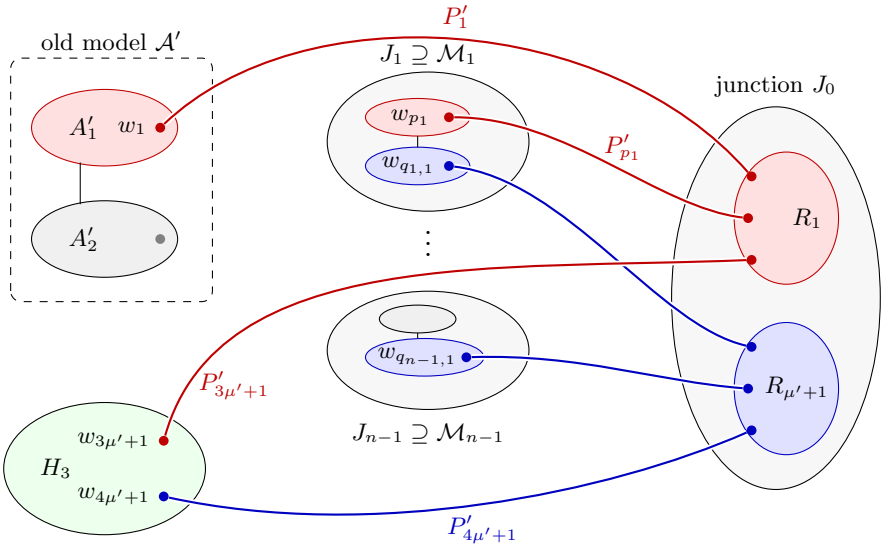


Figure 9: Step 4 of the proof of Proposition 8.1, for the extended branch set A''_1 (red) and the new branch set $A''_{\mu'+1}$ (blue). The set A''_1 is the union of A'_1, the branch set $B(w_{p_1})$ of $\mathcal{M}_1$, the paths P'_1, P'_{p_1} and $P'_{3\mu'+1}$, and R_1. The set $A''_{\mu'+1}$ is the union of the branch sets $B(w_{q_{i,1}})$ of $\mathcal{M}_i$ ($i \in [n-1]$), the paths $P'_{q_{i,1}}$ and $P'_{4\mu'+1}$, and $R_{\mu'+1}$. Each meets H_3 in exactly one vertex. They are adjacent, since they contain distinct branch sets of $\mathcal{M}_1$.

Knitting in the junction. The graph J_0 is $(9t+1)$-connected, and $e(J_0) > C_{6.4}t\, v(J_0) \geqslant 5(3\mu'+\mu)\, v(J_0)$. Moreover, J_0 has the connectivity $2(3\mu'+\mu)-\mu$ required by Lemma 3.6,

since $2(3\mu'+\mu)-\mu = 7\mu'+x \leqslant 7(t+M\sqrt{\log t})+t/(M\sqrt{\log t})+1 \leqslant 7.001t \leqslant 9t+1$. By Lemma 3.6 with $3\mu'+\mu$ and μ in place of k and ℓ, the graph J_0 is $(3\mu'+\mu,\mu)$-knit, so there are pairwise disjoint sets $R_1,\dots,R_\mu \subseteq V(J_0)$ with $\Pi_j \subseteq R_j$ and each $J_0[R_j]$ connected.

The model. For $j \in [\mu']$ let

$$A''_j := A'_j \cup V(P'_j) \cup R_j \cup V(P'_{p_j}) \cup B(w_{p_j}) \cup V(P'_{p'_j}),$$

and for $j \in [x]$ let

$$A''_{\mu'+j} := V(P'_{4\mu'+j}) \cup R_{\mu'+j} \cup \bigcup_{i\in[n-1]} \big(V(P'_{q_{i,j}}) \cup B(w_{q_{i,j}})\big).$$

Each path in these definitions starts in A'_j, in some set $B(\cdot)$, or in H_3, and ends in the set R_j or $R_{\mu'+j}$ of the same definition. Since the graphs $H[A'_j]$, $H[B(\cdot)]$ and $J_0[R_l]$ are connected, $H[A''_l]$ is connected for every $l \in [\mu]$. The sets $A''_1,\dots,A''_\mu$ are pairwise disjoint since:

- Within each of the following families, the members are pairwise disjoint: the sets A'_j, the paths of $\mathcal{P}'$, the sets R_j, and the sets $B(\cdot)$. For the sets $B(\cdot)$, this holds since they are branch sets of a model $\mathcal{M}_i$ in the pairwise disjoint pieces J_i.
- If a path P'_l of $\mathcal{P}'$ meets $V(\mathcal{A}')\cup V(J_0)\cup V(\mathcal{M}_1)\cup\cdots\cup V(\mathcal{M}_{n-1})$ in a vertex v, then v is an end of P'_l, and v lies in the set A''_j whose definition uses P'_l.
- J_0 and the graphs J_i are disjoint from $V(\mathcal{A}')$.
- J_0 is disjoint from the graphs J_i.
- Every element of Π_j belongs to R_j.

The sets are pairwise adjacent. Two sets $A''_j, A''_{j'}$ with distinct $j,j' \leqslant \mu'$ are adjacent because $\mathcal{A}'$ is a model. For $j \leqslant \mu'$ and $j' \in [x]$, the sets A''_j and $A''_{\mu'+j'}$ contain the distinct branch sets $B(w_{p_j})$ and $B(w_{q_{i(j),j'}})$ of $\mathcal{M}_{i(j)}$. For distinct $j,j' \in [x]$, the sets $A''_{\mu'+j}$ and $A''_{\mu'+j'}$ contain the distinct branch sets $B(w_{q_{1,j}})$ and $B(w_{q_{1,j'}})$ of $\mathcal{M}_1$. In the last two cases these branch sets are adjacent, since $\mathcal{M}_{i(j)}$ and $\mathcal{M}_1$ are K_{2x} models. So $\mathcal{A}'' := \{A''_1,\dots,A''_\mu\}$ is a K_μ model.

Tangency and core. The sets A'_j, R_j and $B(\cdot)$ are disjoint from H_3, by Step 3 and since $V(J_0)$ and the sets $V(J_i)$ lie in $\overline{J}$. Let $S := S' \cup \overline{J} \cup \{w_{3\mu'+1},\dots,w_{3\mu'+\mu}\}$. By the intersection properties of $\mathcal{P}'$, the paths of $\mathcal{P}'$ meet $V(H_3)$ exactly in the vertices $w_{3\mu'+1},\dots,w_{3\mu'+\mu}$, and $w_{3\mu'+l}$ is the first vertex of $P'_{3\mu'+l}$ for each $l \in [\mu]$. If $l \leqslant \mu'$, then $3\mu'+l = p'_l$. If $l > \mu'$, then $3\mu'+l = 4\mu'+(l-\mu')$. In both cases the path $P'_{3\mu'+l}$ is used by exactly one set A''_j, namely A''_l. So H_3 meets each A''_j exactly in $w_{3\mu'+j}$, which lies in S. Hence H_3 is tangent to $\mathcal{A}''$. Adjacencies between the sets A''_j with $j \leqslant \mu'$ are witnessed by edges with both ends in S', and all other adjacencies by edges of some J_i, whose ends lie in $V(J_i) \subseteq \overline{J}$. Hence S is a core of $\mathcal{A}''$, and $V(H_3)\cap V(\mathcal{A}'') \subseteq S$. Finally,

$$|S| \;\leqslant\; 2(n-1)^2\tau_{7.1}\,t\log t + n\tau_{7.1}\,t\log t + 3t \;\leqslant\; 2n^2\tau_{7.1}\,t\log t,$$

because $2(n-1)^2+n+1 \leqslant 2n^2$ for $n \geqslant 1$ and $3t \leqslant \tau_{7.1}\,t\log t$.

Step 5. Building a K_μ model when $n = 1$. Here $\mu' = 0$, there are no pieces J_i with $i \geqslant 1$, and $\mu = x$. The construction in Step 4 does not apply, because it uses the model $\mathcal{M}_1$ to make the new branch sets pairwise adjacent. Instead, since H' is x-connected and $V(H_3)$ and $V(J_0)$ have more than $k \geqslant x$ and more than $9t+1 \geqslant x$ vertices, respectively, Theorem 3.4 gives x pairwise disjoint paths from $V(H_3)$ to $V(J_0)$ in H'. Trimming them gives x pairwise disjoint $V(H_3)$–$V(J_0)$ paths $P^\circ_1,\dots,P^\circ_x$ in H', with P°_j joining $y_j \in V(H_3)$ to $z_j \in V(J_0)$. The graph J_0 is $(2x,\lfloor 4.004t\rfloor)$-woven, by Step 1, and in particular $(2x,0)$-woven. Since $v(J_0) > 9t+1 \geqslant 2x$, Lemma 3.1 with $b=0$ shows that J_0 is $(x,0)$-woven. So J_0 has a K_x model $\{B_1,\dots,B_x\}$ with $z_j \in B_j \subseteq V(J_0)$. Let $A''_j := V(P^\circ_j) \cup B_j$ and

$S := V(J_0) \cup \{y_1, \dots, y_x\}$. Since each P_j° meets $V(J_0)$ only in z_j and $V(H_3)$ only in y_j, and each $B_j \subseteq V(J_0)$ is disjoint from H_3 by Step 3, these sets form a K_x model $\mathcal{A}''$ to which H_3 is tangent, and $V(H_3) \cap V(\mathcal{A}'') = \{y_1, \dots, y_x\} \subseteq S$. Also S is a core, since all required edges lie in J_0, and $|S| \leqslant \tau_{7.1}\, t \log t + x \leqslant 2\tau_{7.1}\, t \log t$.

Step 6. Shrinking the branch sets. By Steps 4 and 5, for each $j \in [\mu]$, the graph H_3 meets A_j'' in exactly one vertex h_j, and $h_j \in S$. For each $j \in [\mu]$, apply Lemma 3.10 to the connected graph $H[A_j'']$ and the set $S \cap A_j''$, which is non-empty since it contains h_j. We obtain a set $A_j \subseteq A_j''$ with $H[A_j]$ connected, and a set T_j with $S \cap A_j'' \subseteq T_j \subseteq A_j$, $|T_j| \leqslant 3|S \cap A_j''|$ and $\chi(H[A_j \setminus T_j]) \leqslant 2$. Let $\mathcal{A} := \{A_1, \dots, A_\mu\}$. The sets $A_j \subseteq A_j''$ are pairwise disjoint, and each $H[A_j]$ is connected. Since $S \cap A_j'' \subseteq A_j \subseteq A_j''$, we have $A_j \cap S = A_j'' \cap S$. As S is a core of $\mathcal{A}''$, for all distinct j, j' some edge joins $A_j'' \cap S = A_j \cap S$ to $A_{j'}'' \cap S = A_{j'} \cap S$. So $\mathcal{A}$ is a K_μ model with core S. Next, $V(H_3) \cap A_j'' = \{h_j\}$ and $h_j \in S \cap A_j'' \subseteq A_j$. Hence $|V(H_3) \cap A_j| = 1$ for every $j \in [\mu]$, that is, H_3 is tangent to $\mathcal{A}$. Also $V(H_3) \cap V(\mathcal{A}) \subseteq V(H_3) \cap V(\mathcal{A}'') \subseteq S$. Let $T := \bigcup_j T_j$. Then $|T| \leqslant 3|S| \leqslant 6n^2\tau_{7.1}\, t \log t \leqslant N_t$, so $\chi(H[T]) \leqslant \Gamma$. Also $V(\mathcal{A}) \setminus T$ is the disjoint union of the sets $A_j \setminus T_j$, and each $H[A_j \setminus T_j]$ can be coloured with two colours of its own. So $\chi(H[V(\mathcal{A}) \setminus T]) \leqslant 2\mu \leqslant 6t$. Hence $\chi(H[V(\mathcal{A})]) \leqslant \chi(H[T]) + \chi(H[V(\mathcal{A}) \setminus T]) \leqslant 6t + \Gamma$.

Step 7. Restoring the invariant. Let $H_4 := H - V(\mathcal{A})$. Then H_4 is an induced subgraph of H, and

$$\chi(H_4) \;\geqslant\; \chi(H) - \chi(H[V(\mathcal{A})]) \;\geqslant\; \chi(H) - 6t - \Gamma \;\geqslant\; \Psi - 6t - \Gamma \;\geqslant\; \lceil 3.0625(k-1) \rceil.$$

By Corollary 3.3, H_4 has a k-connected induced subgraph H_5 with $\chi(H_5) \geqslant \chi(H) - 6t - \Gamma - 2k = \chi(H) - \Theta$. Since H_4 is an induced subgraph of H, so is H_5. Also $V(H_5) \cap V(\mathcal{A}) = \emptyset$. As illustrated in Figure 10, there are two cases, according to whether H_5 and H_3 share fewer than k vertices.

Case 1. $|V(H_5) \cap V(H_3)| < k$. Let $Y_1 := V(H_5) \setminus V(H_3)$ and $Y_2 := V(H_3)$. These sets are disjoint, so $H[Y_1]$ and $H[Y_2]$ are vertex-disjoint induced subgraphs of H. Deleting a vertex lowers the chromatic number by at most 1, and $H[Y_1] = H_5 - (V(H_5) \cap V(H_3))$ since H_5 is induced. Hence

$$\chi(H[Y_1]) \;\geqslant\; \chi(H_5) - (k-1) \;\geqslant\; \chi(H) - \Theta - k \;\geqslant\; \chi(H) - \sigma,$$

since $\sigma - \Theta = 24t + \Gamma + 2k \geqslant k$. Moreover $H[Y_2] = H_3$, since H_3 is an induced subgraph of H, so $\chi(H[Y_2]) = \chi(H_3) \geqslant \chi(H) - \sigma$ by Step 3. Thus H is σ-chromatic-separable, a contradiction.

Case 2. $|V(H_5) \cap V(H_3)| \geqslant k$. Let $R := V(H_3) \cup V(H_5)$ and $J^* := H[R]$. We claim that J^* is k-connected. Indeed, $v(J^*) \geqslant v(H_5) > k$. Let $Z \subseteq R$ with $|Z| < k$. Then $H_3 - Z$ and $H_5 - Z$ are connected, because H_3 and H_5 are k-connected. They share a vertex, because $|V(H_3) \cap V(H_5)| \geqslant k > |Z|$. Their union is a connected spanning subgraph of $J^* - Z$. So $J^* - Z$ is connected. This proves the claim.

We now verify $(*)_n$ for $\mathcal{A}$, S and J^*. Property (I1) holds by Step 6, and by the bound on $|S|$ in Step 4 or Step 5. Property (I2) was just shown. For (I3) and (I4), recall that $V(H_5) \cap V(\mathcal{A}) = \emptyset$. So $V(J^*) \cap A_j = V(H_3) \cap A_j$ for every $j \in [\mu]$. This set has exactly one vertex, by Step 6. Also $V(J^*) \cap V(\mathcal{A}) = V(H_3) \cap V(\mathcal{A}) \subseteq S$, by Step 6. Finally, $\chi(J^*) \geqslant \chi(H_5) \geqslant \chi(H) - \Theta$, which is (I5). This proves $(*)_n$. □

Comparison with the proof of Delcourt and Postle. The proof of Proposition 8.1 follows the proof of Lemma 6.4 of Delcourt and Postle [8], and $(*)_n$ is their inductive statement. The main differences are as follows. We use Corollary 3.3, Theorem 7.1 and

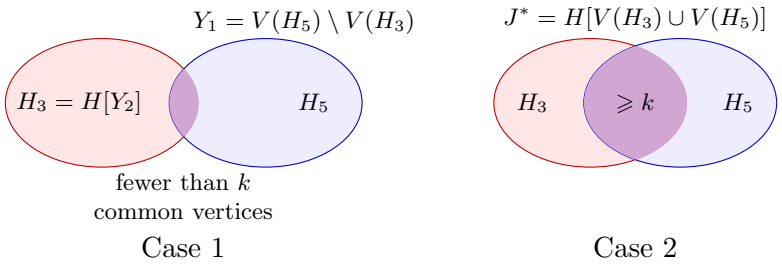


Figure 10: Step 7 of the proof of Proposition 8.1. In Case 1, the k-connected graphs H_3 and H_5 share fewer than k vertices, and $Y_1 = V(H_5) \setminus V(H_3)$ and $Y_2 = V(H_3)$ give a chromatic separation of H. In Case 2, they share at least k vertices, so $J^* = H[V(H_3) \cup V(H_5)]$ is k-connected.

Lemmas 3.6 and 6.4 in place of [8, Theorems 5.1, 2.3 and 5.8 and Lemma 5.13]. We use M times as many stages, each adding $x = \lceil t/(M\sqrt{\log t})\rceil$ branch sets instead of $\lceil t/\sqrt{\log t}\rceil$. We record the chromatic number lost at each step, so that Γ enters σ and Ψ only with coefficient 2, and $C_{8.1}$ enters only through the term $2d$ of Ψ. Step 5 is new, since for $n = 1$ their construction gives no adjacency between the new branch sets.

9 Proof of Theorem 1.2

This section combines Proposition 8.1, Corollary 5.7 and an adaptation of Lemma 4.2 in a final separation step, and then completes the proof.

Corollary 9.1. *Let b be an integer with $\log b \geqslant 1.6 \cdot 10^6$, and let*

$$\sigma_b := (78 + 2c_{5.5})\, b \qquad \text{and} \qquad \Psi_b := (32 + 2C_{8.1} + 2c_{5.5})\, b + 2.$$

Every graph F with no K_b minor and $\chi(F) \geqslant \Psi_b$ is σ_b-chromatic-separable.

Proof. Since $\log b \geqslant 1.6 \cdot 10^6$, we have $b \geqslant 3$. Let $U \subseteq V(F)$ with $|U| \leqslant N_b$. Then $F[U]$ has no K_b minor, so Corollary 5.7 gives $\chi(F[U]) \leqslant c_{5.5}b$. So Proposition 8.1 applies with b, F and $c_{5.5}b$ in place of t, H and Γ, and its first conclusion is the claim. □

In the final application, we use Corollary 9.1 in Step 7b of the proof of Lemma 4.2, in place of $(\mathcal{H}_{\Phi,a_0})$, with $\Lambda_{9.2}$ and $\Lambda'_{9.2}$ as defined in Lemma 9.2 below. The separation cost $\sigma_b = (78 + 2c_{5.5})b$ does not involve $C_{8.1}$. Each separation in Step 7b has cost σ_b with $b = \min\{2a, t\} \leqslant 2a$, and these costs are covered by the term $\Lambda_{9.2}a$, as in Step 8. The chromatic threshold Ψ_b does involve $C_{8.1}$. Since $b \leqslant t$, the requirement $\chi(G_4) \geqslant \Psi_b + \sigma_b$, which allows two successive separations, is met by the term $\Lambda'_{9.2}t$ in the hypothesis of Lemma 9.2. This term is the same at every level of the induction on a. So $C_{8.1}$ does not enter $\Lambda_{9.2}$, and enters $\Lambda'_{9.2}$, and hence the final constant, only through the summand $2C_{8.1}$. By contrast, using $(\mathcal{H}_{\Phi,a_0})$ with the second conclusion of Proposition 8.1 would put $2C_{8.1}$ into Φ, and the coefficient 10 of Φ in Λ_{U} would give the summand $20C_{8.1}$ (see Appendix A).

Lemma 9.2. *Let $T_{\mathrm{fin}} := 18.4$. Let t be an integer with $\log\log t \geqslant T_{\mathrm{fin}}$, and let $a_0 := t/\log t$ and $\eta := 5/(a_0 - 4)$. Let*

$$\Lambda_{9.2} := (1+\eta)\big(897.5 + 10(78 + 2c_{5.5})\big) \qquad \text{and} \qquad \Lambda'_{9.2} := 110 + 2C_{8.1} + 4c_{5.5} + \frac{2}{t}.$$

Let $a \geqslant 1$ be an integer, and let G be a $21a$-connected graph with no K_t minor and

$$\chi(G) \;\geqslant\; D\big(\tfrac{46}{3}a_0 + 4\big) + 1 + \Lambda_{9.2}a + \Lambda'_{9.2}t.$$

Then G is $(a, \lceil 10a/3 \rceil)$-woven.

Proof. Since $a_0 = t/\log t \geqslant 9$ when $\log\log t \geqslant T_{\mathrm{fin}}$, we have $\eta \in (0,1]$ and $a_0 = 4 + 5/\eta$. The proof is the proof of Lemma 4.2 for the uneven scheme, with the following changes. Throughout, t is fixed, and we use induction on a. Every graph to which the induction hypothesis is applied is an induced subgraph of G, and so has no K_t minor. Steps 1–6, 7a and 9 are unchanged, and Step 2 gives $a > a_0$ as before, since $\chi(G) \geqslant D(\frac{46}{3}a_0 + 4) + 1$. Steps 3 and 5 use only $\chi(G) \geqslant 1 + 897.5a$, which holds since $\Lambda_{9.2} > 897.5$. In Step 7b, the choice of R' and S', the definition of G_4 and the bound $\chi(G_4) \geqslant \chi(G_3) - 3a$ are unchanged. The rest of Step 7b up to the definition of J_1, J_2, J_3 is replaced by the following, and Step 8 is modified as described below.

Let $b := \min\{2a, t\}$, which is an integer. The graph G_4 has no K_{2a} minor by Step 7a, and no K_t minor since G has none. So G_4 has no K_b minor. Also $b \geqslant \min\{a, t\} \geqslant a_0$. Let $L := \log t$, so that $L \geqslant e^{T_{\mathrm{fin}}} \geqslant 9.7 \cdot 10^7$. Then $\log L \leqslant L/2$, since $e^{L/2} \geqslant (L/2)^2/2 = L^2/8 \geqslant L$. So $\log b \geqslant \log a_0 = L - \log L \geqslant L/2 \geqslant 4.85 \cdot 10^7 \geqslant 1.6 \cdot 10^6$. Since $b \leqslant t$,

$$\Psi_b + \sigma_b \;=\; (110 + 2C_{8.1} + 4c_{5.5})\, b + 2 \;\leqslant\; \Lambda'_{9.2} t.$$

By Steps 3–5 and the bound $\chi(G_4) \geqslant \chi(G_3) - 3a$, we have $\chi(G_4) \geqslant \chi(G) - (137.5a + 17)$. Since $\Lambda_{9.2} a \geqslant 137.5a + 17$, it follows that $\chi(G_4) \geqslant \Lambda'_{9.2} t \geqslant \Psi_b + \sigma_b$. By Corollary 9.1, the graph G_4 is σ_b-chromatic-separable. So it has vertex-disjoint induced subgraphs F_1, F_2 with $\chi(F_i) \geqslant \chi(G_4) - \sigma_b \geqslant \Psi_b$. Applying Corollary 9.1 to F_1, which also has no K_b minor, gives vertex-disjoint induced subgraphs $F_{1,1}, F_{1,2}$ of F_1 with $\chi(F_{1,j}) \geqslant \chi(F_1) - \sigma_b$. Let $J_1 := F_{1,1}$, $J_2 := F_{1,2}$ and $J_3 := F_2$. Then $\sigma_b = (78 + 2c_{5.5})b \leqslant 2(78 + 2c_{5.5})a$. So the rest of Step 7b, and Step 8, go through as in the proof of Lemma 4.2 with σ_b in place of σ, with Φ replaced by $78 + 2c_{5.5}$, with $\Lambda_{9.2}$ in place of Λ_{U}, and with $\Lambda'_{9.2} t$ added to the lower bound on $\chi(G)$ in the hypothesis and to the resulting lower bounds on $\chi(J'_i)$ in Step 8. In particular, each J'_i is a $21a'_i$-connected graph with no K_t minor and

$$\chi(J'_i) \;\geqslant\; D\big(\tfrac{46}{3}a_0 + 4\big) + 1 + \Lambda_{9.2} a'_i + \Lambda'_{9.2} t,$$

so the induction hypothesis applies to it. □

The even scheme would give $(1+\eta)\big(517.5 + 12(78 + 2c_{5.5})\big)$ in place of $\Lambda_{9.2}$, which is larger, since $78 + 2c_{5.5} > 190$.

We now prove the main theorem.

Proof of Theorem 1.2. Let G be a graph, and let $h := \mathrm{had}(G)$. If $h \leqslant 1$, then G has no edge, since an edge gives a K_2 minor. So $\chi(G) = 0 = h$ if G is null, and $\chi(G) = 1 = h$ otherwise. Now let $h \geqslant 2$ and $t := h + 1$, so that G has no K_t minor. Since $t \geqslant 3$, Theorem 3.11 gives $\chi(G) < 60\, t\sqrt{\log t} + 1$.

If $h < 10^6$, then $\log t \leqslant t - 1 = h < 10^6 \leqslant 1001^2$, so $\sqrt{\log t} \leqslant 1001$. Since $h \geqslant 2$,

$$\chi(G) \;<\; 60{,}060\, t + 1 \;=\; 60{,}060\, h + 60{,}061 \;\leqslant\; 1{,}118{,}430\, h.$$

Now let $h \geqslant 10^6$.

Case 1. $\log\log t < T_{\mathrm{fin}}$. Then $\log t < e^{T_{\mathrm{fin}}}$, so $\sqrt{\log t} < e^{T_{\mathrm{fin}}/2} = e^{9.2} < 10^4$. Since $h \geqslant 10^6$,

$$\chi(G) \;<\; 600{,}000\, t + 1 \;=\; 600{,}000\, h + 600{,}001 \;\leqslant\; 1{,}118{,}430\, h.$$

Case 2. $\log\log t \geqslant T_{\mathrm{fin}}$. Let a_0, η, $\Lambda_{9.2}$ and $\Lambda'_{9.2}$ be as in Lemma 9.2, and let $L := \log t$. Then $L \geqslant e^{T_{\mathrm{fin}}} \geqslant 9.7 \cdot 10^7$. Since $t = e^L \geqslant L^3/6$, we have $t \geqslant a_0 = t/L \geqslant L^2/6 \geqslant 2.5 \cdot 10^9$.

We first show that $D(\frac{46}{3}a_0 + 4) + 1 \leqslant t/10$. Let $s_0 := \lfloor \frac{46}{3}a_0 + 4 \rfloor$, so that $D(\frac{46}{3}a_0 + 4) = D(s_0)$. Since $a_0 \geqslant 6$, we have $2 \leqslant s_0 \leqslant 16a_0 \leqslant 16t$, and so $\log s_0 \leqslant \log 16 + L < L + 3$.

Since $L \geqslant 9.7{\cdot}10^7 \geqslant 9650^2(1{+}3/L)$, we have $L{+}3 \leqslant (L/9650)^2$, that is, $\sqrt{L+3} \leqslant L/9650$. By Theorem 3.11, and since $a_0 L = t$,

$$D(s_0) \;\leqslant\; 60\, s_0 \sqrt{\log s_0} + 1 \;\leqslant\; 60 \cdot 16 a_0 \cdot \frac{L}{9650} + 1 \;=\; \frac{960}{9650}\, t + 1 \;=\; \frac{t}{10} - \frac{t}{1930} + 1.$$

Since $t \geqslant 3860$, it follows that $D(s_0) \leqslant t/10 - 1$.

Next, every $21t$-connected graph G' with no K_t minor satisfies

$$\chi(G') \;<\; \frac{t}{10} + \Lambda_{9.2} t + \Lambda'_{9.2} t.$$

Otherwise $\chi(G') \geqslant D(\frac{46}{3}a_0 + 4) + 1 + \Lambda_{9.2}t + \Lambda'_{9.2}t$, and Lemma 9.2 with $a = t$ shows that G' is $(t, \lceil 10t/3 \rceil)$-woven. Since G' has more than $21t$ vertices, it has a K_t minor, a contradiction.

We now claim that

$$\chi(G) \;<\; \frac{t}{10} + 42t + \Lambda_{9.2} t + \Lambda'_{9.2} t.$$

Suppose not. Then $\chi(G) \geqslant 42t + \Lambda_{9.2}t \geqslant 3.0625(21t - 1)$, since $\Lambda_{9.2} \geqslant 64$. As $\chi(G)$ is an integer, $\chi(G) \geqslant \lceil 3.0625(21t{-}1) \rceil$. So by Corollary 3.3 with $k = 21t$, G has a $21t$-connected induced subgraph G' with $\chi(G') \geqslant \chi(G) - 42t \geqslant t/10 + \Lambda_{9.2}t + \Lambda'_{9.2}t$. Since G' has no K_t minor, this contradicts the previous paragraph. This proves the claim.

Finally, we bound the right-hand side of the claim. Let $\Lambda_0 := 897.5 + 10(78 + 2c_{5.5}) = 902{,}177.246$, so that $\Lambda_{9.2} = (1{+}\eta)\Lambda_0$. Since $a_0{-}4 \geqslant 2500\,\Lambda_0$, we have $\eta\Lambda_0 = 5\Lambda_0/(a_0{-}4) \leqslant 0.002$. Also $\Lambda'_{9.2}t = (110 + 2C_{8.1} + 4c_{5.5})t + 2$. Since $10(78 + 2c_{5.5}) + 4c_{5.5} = 780 + 24c_{5.5}$ and $0.1 + 42 + 897.5 + 780 + 110 = 1829.6$,

$$\frac{t}{10} + 42t + \Lambda_0 t + (110 + 2C_{8.1} + 4c_{5.5})t \;=\; (1829.6 + 2C_{8.1} + 24c_{5.5})\, t \;=\; c_{1.2} t.$$

Hence, by the claim,

$$\chi(G) \;<\; (c_{1.2} + 0.002)\, t + 2 \;\leqslant\; 1{,}118{,}430\,(t-1) \;=\; 1{,}118{,}430\, h.$$

Here the second inequality holds because $c_{1.2} = 1{,}118{,}429.2952$, so it is equivalent to $0.7028\, t \geqslant 1{,}118{,}432$, which holds as $t \geqslant 2.5 \cdot 10^9$.

In all cases $\chi(G) \leqslant 1{,}118{,}430$ had(G). □

The margin $1{,}118{,}430 - c_{1.2} = 0.7048$ absorbs the error term $\eta\Lambda_0 t \leqslant 0.002t$ from $\Lambda_{9.2}t$, the summand 2 of $\Lambda'_{9.2}t$, and the change from t to $h = t - 1$. The value $T_{\text{fin}} = 18.4$ is, to one decimal place, the least value for which the argument of Case 2 gives $D(\frac{46}{3}a_0{+}4){+}1 \leqslant t/10$. That argument bounds $D(s_0)$ by $960\, t\sqrt{L+3}/L + 1$, which is less than $t/10$ only if $960\sqrt{L+3} < L/10$, and this fails for $L = e^{18.3}$. A larger value of T_{fin} up to 19.6 would give the same constant, since $60\, e^{19.6/2} < 1.09{\cdot}10^6$, so Case 1 would still give $\chi(G) \leqslant 1{,}118{,}430\, h$ (as $h \geqslant 10^6$).

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A Postscript

This paper was first submitted to the arXiv on the morning of 7th October 2026 (Melbourne time). A few hours later, OpenAI released its disproof of Hadwiger's Conjecture [40].[1] OpenAI's construction gives infinitely many graphs G with $\alpha(G) \leqslant 2$ and $\mathrm{had}(G) < \frac{26}{75}v(G) + \frac{2}{3}$, implying $\chi(G) \geqslant \frac{1}{2}v(G) > \frac{75}{52}\,\mathrm{had}(G) - \frac{25}{26}$ (and $\mathrm{had}(G) \to \infty$ along this family). Two natural questions arise. Let c_1 be the infimum of the real numbers c such that $\chi(G) \leqslant c\,\mathrm{had}(G)$ for every graph G. Let c_2 be the infimum of the real numbers c such that $\chi(G) \leqslant (c + o(1))\,\mathrm{had}(G)$ for every graph G, where the $o(1)$ term tends to 0 as $\mathrm{had}(G) \to \infty$. What are c_1 and c_2? We know that $\frac{75}{52} \leqslant c_2 \leqslant c_1 \leqslant 1{,}118{,}430$. With the goal of closing this gap, we now identify the parts of the argument that determine the upper bound in Theorem 1.2, and indicate what would be needed to reduce it further.

The dominant term. The constant $c_{1.2} = 1829.6 + 2C_{8.1} + 24c_{5.5}$ is dominated by $24c_{5.5} \approx 1.081 \cdot 10^6$, whose origin is explained in Section 2.4. The term $2C_{8.1} = 36{,}000$ contributes about 3.2%. So $c_{1.2}$ is determined mainly by the bootstrap constant $c_{5.5}$ and by the coefficient 10 of Φ in Λ_{U}, which turns $78 + 2c_{5.5}$ into the term $20c_{5.5}$ of $24c_{5.5}$. In particular, further improvements to the weaving constant $C_{6.4}$ or to the proportion $\omega_{7.1}$ affect only $C_{8.1}$, and so would change $c_{1.2}$ by less than 3.2%. Using Proposition 8.1 only through its second conclusion, that is, Lemma 4.2 with $(\mathcal{H}_{\Phi,a_0})$ in place of Lemma 9.2, gives a constant of about $939.6 + 10(2C_{8.1} + 33) + 20c_{5.5} \approx 1.262 \cdot 10^6$, which is about 13% more. Several estimates below are values of the formula $1829.6 + 2C_{8.1} + 24c_{5.5}$ or of a variant of it. A value of $c_{1.2}$ below $60\,e^{T_{\mathrm{fin}}/2}$, which exceeds 593,827, would also require a smaller size threshold T_{fin}, since Case 1 of the proof of Theorem 1.2 uses only $\chi(G) \leqslant 60\,t\sqrt{\log t} + 1$, which is close to $60\,e^{T_{\mathrm{fin}}/2}\,t$ when $\log\log t$ is just below T_{fin}.

[1] OpenAI also proved that $\mathrm{ch}(G) \in O(\mathrm{had}(G))$ [41], where $\mathrm{ch}(G)$ is the list chromatic number of G. Since $\chi(G) \leqslant \mathrm{ch}(G)$, this implies the Linear Hadwiger Theorem of Norin and Steiner [38]. No explicit constant is given for the bound $\mathrm{ch}(G) \in O(\mathrm{had}(G))$.

The chromatic-inseparable case. This case affects the main constant mainly through Γ, which enters each of σ and Ψ twice (see the paragraph after the proof of Proposition 8.1). In each of σ and Ψ, one copy of Γ enters through Θ. This copy of Γ bounds the chromatic number of the set T of Step 6 of the proof of Proposition 8.1, which lies in the new branch sets. The other copy bounds the chromatic number of a union of pieces, namely the set $\overline{J}$ of Step 2 (used in Step 3) for σ, and the set U of Step 1 for Ψ. In the final separation step (Corollary 9.1), $\Gamma = c_{5.5}b$, and the two copies of Γ produce the term $24c_{5.5}$ of $c_{1.2}$ (Section 2.4). An argument that loses Γ only once would lower this term to $12c_{5.5}$, a reduction of $12c_{5.5} \approx 5.4 \cdot 10^5$, almost half of $c_{1.2}$. The other contributions of this case are small. Better connectivity results would change only the terms $2C_{8.1} = 36{,}000$ and 1829.6, which together are less than 4% of $c_{1.2}$. Lemma 6.4 needs pieces that are $9t$-connected and have density $C_{6.4}t = 70t$, and Lemma 3.17, used in the proof of Theorem 7.1, gives both. By Lemma 3.17, average degree $4k$ forces a $(k+1)$-connected subgraph with more than $2k$ vertices. Bernshteyn and Kostochka [3] showed that average degree $(3+\frac{1}{6})k$ forces a $(k+1)$-connected subgraph, and Carmesin [7] showed that average degree $(3+\frac{1}{3})k$ forces a $(k+1)$-connected subgraph with more than $2k$ vertices, and that this is sharp. These results give connectivity without the density, and a result that gives only one of the two properties does not help. Kawarabayashi and Yu [21] prove that every 8ℓ-connected graph is *ℓ-knitted*, that is, $(\ell, 1)$-knit in our terminology, correcting a gap in their earlier argument for 9ℓ [20]. The knitting in the proof of Proposition 8.1 needs connectivity at most $7.001t$, which is below the $9t$ already required by Lemma 6.4, so a better knitting result would not help.

The bootstrap. Within the framework of Section 5, the schedule of Proposition 5.5 is optimal up to the error terms 73ξ. A dynamic program over all schedules (script `bootstrap_dp.py`) takes $\xi = 0$ and rounds each exponent reached up to the right end of a short interval, which can only help a schedule. Its value 45,024.98 is therefore a lower bound on the constant given by any schedule, and $c_{5.5} = 45{,}024.98 + 73\xi$. So a reduction of $c_{5.5}$ by more than 73ξ needs a change to the framework itself. There are three places to look. First, each step doubles the previous constant, because G_1 is coloured through its contraction (step (i) of the proof of Lemma 5.4). A factor smaller than 2 would reduce each of the first three terms of $c_{5.5} = 2^6 \cdot 4.01 + 62c_{5.4} + c_{4.3}(c_{\text{side}}) + \xi$. Second, each step raises the exponent by less than half the exponent β_{G} at which the Gyárfás graphs are coloured, since these graphs have order up to $2b'k(k-1)$ with $k = \lceil (\log t)^{\zeta} \rceil$ (by the condition $(\beta_{\text{G}} - 2\zeta)T \geqslant 1$ of Lemma 5.4). Third, the final exponent must exceed 2, because the order bound N_t of Proposition 8.1 satisfies $N_t \in O(t \log^2 t)$ (see the remark after Proposition 5.5). A larger increment per step, or a smaller order bound, would save steps. For example, one fewer main step would replace $c_{5.5}$ by $2\bar{c}_4 + c_{4.3}(c_{\text{side}}) + \xi \approx 25{,}453$, and $c_{1.2}$ by about $6.49 \cdot 10^5$. The additive term $c_{4.3}(\Phi)$ of each step and the base case are discussed below.

The base case. The base case enters the main constant through its constant and through its exponent. In $c_{5.5} = 2^6 \cdot 4.01 + 62c_{5.4} + c_{4.3}(c_{\text{side}}) + \xi$, the base constant 4.01 appears with coefficient $2^6 = 64$ in the first term, with coefficient 12 in each of the 62 copies of $c_{5.4} = c_{4.3}(4.01) + \xi$, and with coefficient $10 \cdot 14 = 140$ in $c_{4.3}(c_{\text{side}})$, since $c_{\text{side}} = 2 \cdot 4.01 + c_{5.4}$. So each unit of the base constant contributes $64 + 744 + 140 = 948$ to $c_{5.5}$, and hence 22,752 to $c_{1.2}$. The constant 4 of the small-graph bound of Norin and Steiner and of Theorem 5.1 is the factor 2 of Reed and Seymour times the factor 2 from a bipartition step in the proof of [38, Lemma 8]. Removing the second factor would lower $c_{1.2}$ by about 45,500. The exponent matters more. The base case stops at exponent $\frac{1}{2}$ because the additive term $(100/\varepsilon)^{2000/\varepsilon^2}$ of Theorem 5.1 is at most εa for ε of order $(\log a)^{-\beta}$ only if $\beta < \frac{1}{2}$. In the

argument of Norin and Steiner, this reflects the error term $(d+1)^{4m}$ of [38, Lemma 8], with $d \in \exp(O(m \log m))$, which is in $\exp(O(m^2 \log m))$ [38, Section 2.1]. An error term in $\exp(O(m \log m))$ would extend the base case to exponent $1 - o(1)$. Then ζ could be close to $\frac{1}{2}$ in Lemma 5.4, and three bootstrapping steps with the base case for the Gyárfás graphs would suffice. These three steps would give the constant $2^3 \cdot 4.01 + 7c_{5.4} \approx 4285.4$. However, Corollary 5.7 also uses the bound $(144.4 + 4T_{\text{boot}})a = 4788.4\,a$ of Section 5.4. So $c_{5.5}$ would be replaced by 4788.4, and $c_{1.2}$ by about $1.53 \cdot 10^5$. If T_{boot} were also lowered to at most 1035.25, so that $144.4 + 4T_{\text{boot}} \leqslant 4285.4$, then $c_{5.5}$ would be replaced by about 4285.4, and $c_{1.2}$ by about $1.41 \cdot 10^5$.

The chromatic-separable reduction. The cost of this reduction is $c_{4.3}(\Phi) = 42 + \min\{517.5 + 12\Phi,\ 897.5 + 10\Phi\}$. In the uneven-scheme term $897.5 + 10\Phi$ of $c_{4.3}(\Phi)$, the coefficient 10 of Φ produces the term $20c_{5.5}$ of $c_{1.2}$, since $78 + 2c_{5.5}$ plays the role of Φ in the final step (see above). The coefficient 12 enters $c_{5.5}$ through the side step and the five main steps, and the coefficient 10 enters $c_{5.5}$ through the final bootstrapping step. The coefficient $10 = 5 \cdot 2$ has two factors. The factor 2 arises because Lemma 3.9 gives rooted K_a models only from a K_{2a} minor, so Step 7b of the proof of Lemma 4.2 separates graphs with no K_{2a} minor. A lemma that finds rooted K_a models from a $K_{(1+o(1))a}$ minor in $21a$-connected graphs would halve both coefficients, to 6 and 5. With all other terms unchanged, this would give $c_{5.5} \approx 40{,}335$ and $c_{1.2} \approx 6.0 \cdot 10^5$. The factor 5 arises because two chromatic separations give three subgraphs, and each branch set uses two of them (Step 9). Among the ways of assigning two of the three subgraphs to each branch set, the uneven scheme is optimal (see the discussion after Table 3), so a smaller factor needs a different way to combine the subgraphs. The additive terms $517.5 = 3 \cdot 172.5$ and $897.5 = 5 \cdot 179.5$ come from the losses listed in Table 3, mainly from the set Z, the hub and the two extractions of $21a$-connected subgraphs. Lowering each of 517.5 and 897.5 by 1 would lower $c_{5.5}$ by 73 and $c_{1.2}$ by about 1750. The connectivity $21a$ is needed because Step 9 needs $G_3 - R'$ to be $2a$-linked, and the connectivity form of the Thomas–Wollan theorem gives this when $G_3 - R'$ is $20a$-connected. A lower connectivity would lower the losses $42a$ of the two extractions of $21a$-connected subgraphs. Theorem 3.5 might allow lower connectivity, but it needs density and connectivity in the same subgraph, which Corollary 3.3 does not provide.

B Computer verification

The scripts. The scripts check the numerical inequalities in three ways. First, rational constants such as $c_{5.4}$, $c_{5.5}$ and $c_{1.2}$ are checked in exact rational arithmetic (`precise_check.py`). Second, inequalities involving logarithms, square roots and exponentials at specific values are checked in interval arithmetic with outward rounding at 60 digits (`precise_check.py`, using the `mpmath` library), so these checks are certified despite rounding errors. Third, since numbers such as $\exp(e^{1161})$ are far too large to handle directly, each inequality required for all t with $\log\log t$ at least a size threshold is rewritten in terms of $y := \log\log t$, so that $\log t = e^y$. It is then reduced to the case where y equals the size threshold. This case is checked in interval arithmetic. This reduction uses that, in each case, the inequality can be written as $f(y) \geqslant 0$ with f non-decreasing in y, so that the inequality at the size threshold implies it for all larger y. For most inequalities this is clear, because the terms that must be small decrease with y. Where it is not clear, the text proves it. The script `explicit_thresholds.py` locates the size thresholds, and the script `refined_lin_check.py` summarises the computations of Sections 6 and 7. Both use floating-point arithmetic and are not needed for the proof. Table 6 lists where each numerical claim of the paper is checked.

Claim	Where stated	Script
Constants $c_{5.4}$, c_{side}, $\bar{c}_j$, $c_{5.5}$, $c_{1.2}$, and the parts $897.5 + 10(78 + 2c_{5.5})$ of $\Lambda_{9.2}$ and $110 + 2C_{8.1} + 4c_{5.5}$ of $\Lambda'_{9.2}$	Secs. 2.4, 5, 9	`precise_check.py` (exact)
The remarks after Theorem 3.11 and Lemma 4.1	Secs. 3, 4	`precise_check.py` (exact, interval)
Linear bounds in Lemmas 4.1, 4.2 and Theorem 4.3, Table 3, the crossover $\Phi = 190$, the coefficients 10 and 12	Sec. 4	`precise_check.py` (exact)
The bounds $2.7182818283^{23} \cdot 1.22 > 1.188 \cdot 10^{10} > 578.4K + 10$ and $1.188 \cdot 10^{10} > 24.5K$, and the other bounds in Lemmas 5.2 and 5.3	Sec. 5.1	`precise_check.py` (interval)
The bounds for $y \geqslant 100$ in the proof of Lemma 5.4	Sec. 5.2	`precise_check.py` (exact, interval)
The conditions of Lemma 5.4 on T, ζ, β_{G} and Φ at the size thresholds, the bounds $24\tau_{7.1}M^2 \leqslant 2.7^{79}$ and $0.0792\,T_{\mathrm{boot}} \geqslant 79$, and the bounds $79/0.0792 < 998$ and 873.4 in the remark after Table 4	Sec. 5.3	`precise_check.py` (exact, interval)
The bounds $24\tau_{7.1}M^2 \leqslant 2.7^{70}$, $\log(24\tau_{7.1}M^2) \leqslant 70$ and 4788.4 of Section 5.4	Sec. 5.4	`precise_check.py` (interval)
Bounds in Lemmas 6.3 and 6.4	Sec. 6	`precise_check.py` (exact, interval)
Constants in Section 7	Sec. 7	`precise_check.py` (exact, interval)
Bounds in the proof of Proposition 8.1 ($4.004t$, $7.001t$, $24t$, orders)	Sec. 8	`precise_check.py` (exact, interval)
The bounds in the proof of Lemma 9.2, the bounds $t/10$, 9650, $2.5 \cdot 10^9$, 0.002, 600,000 and 60,060 in the proof of Theorem 1.2, and the remarks on T_{fin} after it	Sec. 9	`precise_check.py` (interval)
Estimates in Section 2 (including the argument of Norin and Steiner) and in Appendix A	Sec. 2, App. A	`precise_check.py` (exact, interval)
Optimality of the schedule within the framework	App. A	`bootstrap_dp.py`
Location of the size thresholds (not needed for the proof)	Secs. 5, 9	`explicit_thresholds.py`
Floating-point summary of Sections 6 and 7 (not needed)	Secs. 6, 7	`refined_lin_check.py`

Table 6: The numerical claims of the paper and the scripts that check them. Here "exact" means exact rational arithmetic, and "interval" means interval arithmetic, after the reduction to the size threshold described above.

The Lean formalisation. The formalisation of Theorem 1.2 builds on the Lean library `LinearHadwiger`, which formalises the proof of Norin and Steiner and its external inputs (`https://github.com/snorin239/LinearHadwiger`, commit `9ef0db5`, Lean and Mathlib v4.32.1). This library accompanies their paper [38], and was produced by OpenAI Codex following guidance from Norin and Steiner. On top of this, Claude Opus 5.5 formalised the arguments of the present paper, and the following external results not in `LinearHadwiger`:

- the linkage theorem of Thomas and Wollan [47, Corollary 1.2] (Theorem 3.5),
- Theorem 1.6 of Thomas and Wollan [47], which states that every graph with min-

imum degree at least $5k$ and at most $10k$ vertices has a k-linked subgraph with at least $2k$ vertices, and on which their proof of Theorem 3.5 relies,

- the degree condition on which Theorem 1.6 of Thomas and Wollan relies, due to Egawa et al. [12] and to Kawarabayashi, Kostochka and Yu [19, Theorem 1.1] (formalised in the minimum-degree form stated by Thomas and Wollan [47, Theorem 4.1], by adapting the proof of Kawarabayashi, Kostochka and Yu),
- the sharpened form of Mader's theorem (Lemma 3.17, proved in Section 3), since `LinearHadwiger` contains only the basic form, which gives a k-connected induced subgraph,
- Nguyen's theorem [34, Theorem 1.1], and
- the greedy colouring bound of Delcourt and Postle (Lemma 3.14).

The final Lean statement, displayed in Section 1, uses Mathlib's chromatic number `G.chromaticNumber` and the Hadwiger number `hadwigerNumber G`, which is defined from Mathlib notions in `MainMathlib.lean`. The library `LinearHadwiger` defines `chromatic G` as `G.chromaticNumber` converted to a natural number, which loses nothing since G is finite. In `LinearHadwiger`, `cliqueMinorNumber G` is the largest n such that G has a K_n model. The file `BSMain.lean` proves the bound of Theorem 1.2 for `chromatic G` and `cliqueMinorNumber G`. The Lean files also prove that `hadwigerNumber G` equals `cliqueMinorNumber G`, and derive the final statement from this bound. Every numbered result of the paper is formalised, and every statement and proof in the paper follows the Lean files. Results of `LinearHadwiger` are stated in the paper in the forms formalised there. Published results that the paper cites without proof, such as Wollan's Theorems 2.1–2.3 [52], are formalised in `LinearHadwiger` or in the list above.